Two-channel transmission mechanism of a global inflation shock across emerging economies*

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Abstract

This paper examines the two-channel transmission of a global inflation shock to CPI inflation in emerging economies. A high frequency external instrument identification strategy, following Känzig (2021), is employed to empirically test the impact of the global inflation shock on the relevant nominal exchange rate. Thereafter A three bloc open economy dynamic stochastic general equilibrium (DSGE) model is then constructed to theoretically examine the multi-regional spillover impact of a global inflation shock. The paper finds opposing price pressures emanating from the factors of production, with the cost of composite intermediate goods exerting upward price pressure while downward price pressure is exerted owing to a fall in wages. Evidence is also found for the existence of an indirect channel of the global inflation shock, which operates through changes in the relevant nominal exchange rate following the shock. The indirect channel either amplifies or dampens the direct channel, depending on the direction of the exchange rate change induced by the global inflation shock. The paper further identifies the importance of currency invoicing, the degree of exchange rate pass-through, and trade structure as key channels through which global inflation shocks are transmitted to domestic prices in Emerging Economies.

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1 Introduction

In recent years, inflation across the world has undergone a notable shift, marked by a sharp decline during the COVID-19 pandemic, followed by a surge in the post-pandemic period. This period also coincides with a substantial increase in the share of variance in headline and core inflation explained by a global inflation factor across different regions of the world (Otrok and Strackman, 2024). This is consistent with the idea of global inflation synchronization, which refers to the comovement of national inflation rates across countries driven by common global factors.

Several studies examining the impact of global factors on inflation have found an increasingly important role for global factors in explaining domestic inflation in recent years, particularly for advanced economies (Hakkio, 2009; Ciccarelli and Mojon, 2010; Auer et al., 2024). The estimated proportion varies in the literature from 12% - 75% across different groups of countries and time periods (see Table 1). The influence of the global inflation factor on domestic inflation rates has become more broad-based in recent years, as documented by Ha et al. (2019).

Table 1: Domestic Inflation Explained by Global Factors

Studies	**Percentage Explained**	**Countries Analyzed**	**Time Period**
Ciccarelli and Mojon (2010)	70%	22 OECD Countries	1960-2008
Neely and Rapach (2011)	35%	64 Countries	1951-2009
Ha et al. (2019)	12%	99 Countries	1970-2017
Binici et al. (2022)	40%	30 European Countries	2002-2022
Auer et al. (2024)	58%	56 Countries	1960-2023
Otrok and Strackman (2024)	75%	-	2020-2024

Several factors can account for the existence of global inflation synchronization, namely similar policy responses, the propagation of common shocks, and greater openness to international trade and financial flows (Clarida et al., 2002; Bianchi and Civelli, 2015; Auer et al., 2017). This paper focuses on the role of exchange rate pass-through, currency invoicing, trade openness, and intermediate goods in the transmission of global inflation across emerging economies. The rationale for focusing on emerging economies stems from their high degree of exchange rate pass-through (Brandão-Marques et al., 2023), their high dependence on the US dollar as the currency of choice for trade invoicing (Georgiadis and Mösle, 2019), and the increase in their share of trade openness in recent years (see Figure 1).

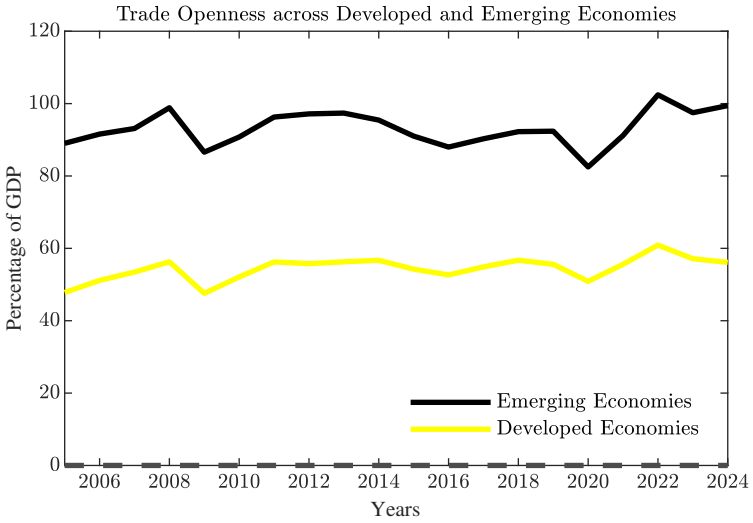


Figure 1: Trade Openness across Developed and Emerging Economies. The data for this figure is from UNCTAD Goods and Services (BPM6): Trade Openness Indicators. Sum of imports and exports is taken as the measure of trade openness. The value of the indicator for the developed economies is provided in the dataset, while the data for the emerging economy is calculated by averaging the trade openness to GDP ratio of emerging economies which forms part of the MSCI Emerging Markets Index. The data is available for all the 24 economies except for Taiwan.

A global inflation shock leads to changes in the relevant nominal exchange rates of emerging economies and propagates through two channels. The direct channel contributes to price pressure through the factors of production affected by the global inflation shock. The indirect channel, however, either amplifies or dampens the direct channel, depending on the change in nominal exchange rates brought about by the global inflation shock.

Figure 2 illustrates the two-channel transmission of the global inflation shock for the retail price of final goods produced in the United States (US)/Rest of the World (ROW) for the Emerging Economy (EE) under Dominant Currency Pricing (DCP). The middle arrow highlights the direct channel of the global inflation shock, while the indirect channel is represented by the side arrows in the figure. If global inflation is associated with an appreciation of the US dollar against the currencies of emerging economies, this leads to an amplification of the global inflation shock as represented by the left-hand side arrow channel. However, if global inflation is associated with a depreciation of the US dollar against the currencies of emerging economies, this dampens the impact of the global inflation shock.

An empirical analysis is conducted to shed light on this relationship. The correlation

between global inflation and the growth rate of the US dollar index against the top 19 emerging-economy trading partners of the US for the period from Q1 1999 to Q4 2024, is -0.58. This shows that global inflation is negatively associated with the growth rate of the US dollar index against emerging economies. Thus, an increase in global inflation is associated with a depreciation of the US dollar against a basket of emerging economies' currencies. To further investigate the causality and contemporaneous relationship between global inflation and the US dollar index against emerging economies, a structural VAR model and the local projection method are employed. The oil supply news shock as in Känzig (2021), is used as a proxy for the global inflation shock. The results indicate that the global inflation shock leads to an appreciation of the EE currency against the US dollar; thus, the indirect channel dampens the impact of the global inflation shock on the retail price of final goods produced in the US/ROW for the EE under DCP.

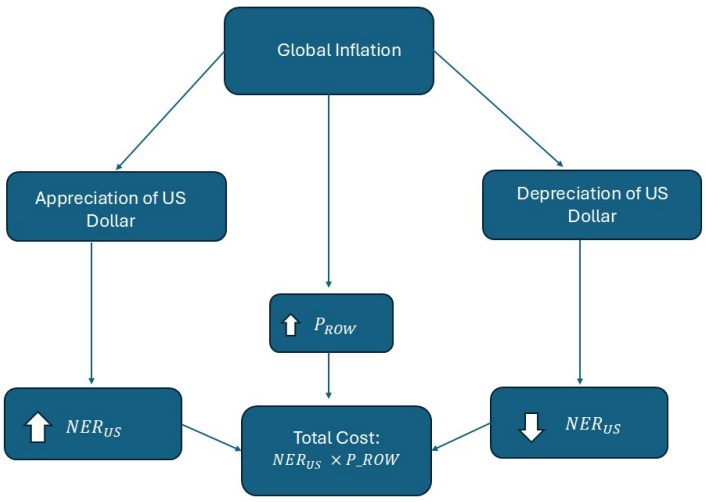


Figure 2: Two-channel transmission of global inflation shock for emerging economy

To better understand the two-channel transmission of the global inflation shock, a three-bloc open-economy dynamic stochastic general equilibrium (DSGE) model featuring the emerging economy, the US, and the rest of the world is constructed. A three-bloc structural model provides nuanced insights into the impact of the global inflation shock on CPI inflation in emerging economies and its sub-components. It enables one to disentangle the indirect channel by examining the impact of relevant nominal exchange rates on the

dock and retail prices of goods imported into the EE. The model is analyzed under three different currency invoicing regimes, namely dominant currency pricing (DCP), producer currency pricing (PCP), and local currency pricing (LCP). Under PCP, exporters invoice in their own currency. Under LCP, exporters choose the consumer country's currency as the invoicing currency. Under DCP, the invoicing currency is neither that of the producer nor that of the consumer; instead, a widely accepted and easily marketable currency, such as the US dollar, is chosen.

The model is also analyzed under different degrees of exchange rate pass through and trade openness, along with an examination of the differential impact of a global inflation shock across final and intermediate goods. The rationale for investigating the model across exchange rate pass-through, currency invoicing regimes, trade openness, and intermediate goods stems from the importance of these factors in the transmission of the global inflation shock. Exchange rate pass-through is critical in regulating the extent to which changes are passed on to domestic prices, while the remaining factors affect the degree of ERPT. The focus on only these factors is due to the scope of this study, which is restricted to examining the impact of the trade sector.

Table 2: Price Pressure from Nominal Exchange Rate Channel

Amplification of Global Inflation	**Dampening of Global Inflation**
DP of goods produced in ROW (DCP)	RP of goods produced in ROW (DCP)
RP of goods produced in ROW (PCP)	RP of goods produced in US (DCP)
RP of goods produced in US (PCP)	DP of goods produced in US (LCP)
DP of goods produced in ROW (LCP)	

The two major results that hold across all model scenarios and currency invoicing regimes are the opposing price pressure stemming from the factors of production following a global inflation shock and the existence of a relevant nominal exchange rate channel (indirect channel) of the global inflation shock. The first result finds that the cost of composite intermediate goods exerts upward price pressure, while downward price pressure is exerted owing to a fall in wages. The second result relates to the existence of the indirect channel of global inflation shock. It operates through changes in the relevant nominal exchange rate - the exchange rate between the invoicing currency and consumer or producer country's currency - following a global inflation shock. When the invoicing currency differs from either the currency of the producer country or the

consumer country, any changes in the exchange rate between the invoicing currency and either of the two currencies affect the transmission of a global inflation shock. It either amplifies or dampens the direct channel of the global inflation shock, depending on the currency invoicing regime and movements in the exchange rates (see Table 2).

In addition to the two overarching results outlined above, the paper also identifies several section-specific findings. In the exchange rate pass-through subsection, differential ERPT is found to generate heterogeneous effect on CPI inflation in EE and its subcomponents under DCP and PCP. The paper further finds that ERPT is pivotal to the existence of the indirect channel of the global inflation shock. When ERPT is assumed to be zero, the two overarching themes identified in the baseline scenario no longer hold. Trade structure further emerges as a key channel through which the global inflation shock is transmitted in EE, with both the type of good affected by the shock and the degree of trade openness shaping this transmission.

The paper proceeds as follows: Section 2 presents the relevant literature on the topic. Section 3 provides an empirical analysis of the relationship between global inflation and the growth rate of the US dollar index against the currency of emerging economies. Section 4 outlines the structure of the three-bloc DSGE model, whose results are analyzed in Section 5. Conclusions are drawn in Section 6.

2 Literature Review

This chapter contributes to the literature on the two-channel transmission of a global inflation shock in emerging economies, specifically examining how an unexpected increase in global inflation affects CPI inflation in these economies. To answer this question comprehensively, it is necessary first to understand how global inflation affects the nominal exchange rate between the invoicing currency and the consumer or producer country's currency, and thereafter to examine the transmission of the change in the nominal exchange rate into domestic prices, a mechanism referred to as exchange rate pass through (ERPT).

This chapter draws on the following strands of literature: first the impact of global factors, primarily global inflation, on the co-movement of national inflation rates across countries. Second, the impact of global inflation on nominal exchange rates. Third,

the importance of ERPT and its relevant features. Fourth, the role of different factors, namely, currency invoicing, trade openness, and intermediate goods, in contributing to either movements in exchange rates or their pass through into domestic prices. Fifth, the use of high frequency identification of external instrument in estimating VAR and local projections. Sixth, the application of a three-bloc structural macroeconomic model to theoretically examine the dynamics of the two-channel transmission. The model is further used to examine the impact of trade structure on the CPI inflation of emerging economies.

In recent years, the literature has increasingly focused on the impact of global factors on inflation, with a large number of studies finding a growing role of global factors in explaining domestic inflation, particularly for advanced economies. Initial research examining the influence of a common global factor on national inflation rates was conducted by Hakkio (2009) and Ciccarelli and Mojon (2010). Both studies find that a substantial proportion of domestic inflation can be attributed to a common global factor, with the proportion varying in the literature from 12% - 72% across different groups of countries and time periods.

In a recent study by Auer et al. (2024), 58% of inflation was explained by a global factor for a sample of 31 countries from 1960-2023, with the corresponding figure for OECD countries pegged at 72% for the same period. The global factors employed in this study include, but are not restricted to, the VIX, the US dollar index, a commodity price index, and the US federal funds rate. The influence of global inflation factor on national inflation rates has become more broad-based in recent years, as documented by Ha et al. (2019). They find that the proportion of countries where global inflation factor contributes more than 10% to inflation variation has increased to two-thirds for the period 2001-2017.

The literature provides nuanced evidence on the impact of global inflation on the US dollar. Känzig (2021) find depreciation of US dollar against other currencies after an oil supply news shock which increases the oil price. However, other studies argue that an increase in global inflation leads to strengthening of the US dollar due to the monetary policy differentials and its safe-heaven asset status via the global financial cycle (Davis and Sagnanert (2024); Yoldas (2024); Obstfeld and Zhou (2023)). The impact of global inflation on exchange rates depends on the spillover channels of the underlying shock

driving global inflation, as illustrated by Bäurle et al. (2021) for Switzerland.

The transmission of global inflation shock happens occurs via two channels, one of which operates through movements in the exchange rates i.e., the relevant nominal exchange rate channel. This makes ERPT an integral part of the transmission mechanism of the global inflation shock as it is the factor which dictates how much of the movement in exchange rates gets passed on into the domestic prices. Krugman (1987) was one of the foundational papers on exchange rate pass through, in which the focus was on studying the impact of real appreciation of the US dollar since the 1980s on the price of US imports. He finds that 35 to 40 percent of real appreciation of the US currency was absorbed by foreign exporters by raising their US prices relative to those in other countries. The idea of incomplete exchange rate pass through was further validated by Goldberg and Knetter (1996), who find incomplete and heterogeneous pass through of exchanges rate due to third-degree price discrimination. Burstein and Gopinath (2014) find evidence of both heterogeneous and incomplete pass through of exchange rates, and also observe ERPT to be time-varying.

Exchange rate pass through varies across different currency invoicing regimes. Gopinath (2015) finds that the degree of exchange rate pass through differs across currencies of invoicing. Countries with a higher share of imports invoiced in a foreign currency, with the exception of the US dollar, have higher exchange rate pass through. This is particularly relevant for emerging economies, as a large proportion of their goods are invoiced in a foreign currency, mostly the US dollar (Georgiadis and Mösle, 2019). Another factor that plays a significant role in affecting exchange rate pass through is trade openness. Gust et al. (2010) find low ERPT due to greater openness on account of greater competition from abroad, which leads exporters to vary their markups instead of passing the changes in exchange rates into prices. Meanwhile, Iorngurum (2024) identifies trade openness as one of several country-specific characteristics that causes ERPT to vary across different studies and countries. In contrast, Benigno and Faia (2016) finds increase in the degree of pass-through with increase in the number of foreign competitors.

Global value chain participation and large import-intensive exporters are two other factors that affect ERPT. Georgiadis et al. (2025) find lower exchange rate pass through to import prices with greater global value chain participation by a country's trading

partners. Meanwhile, Amiti et al. (2014) find that large import-intensive exporters have a pass through of around 50%, as against nearly complete pass through for small non-importing firms. Intermediate goods also affect movements in exchange rates, Craighead (2019) finds that the introduction of intermediate goods trade in their models results in higher real exchange rate volatility relative to output. In addition, ERPT is also affected by the inflation regime and monetary policy credibility, the type of shock affecting the exchange rate, the state of the economy, and the size and type of firms (Taylor (2000); Carrière-Swallow et al. (2016); Forbes et al. (2018); Carrière-Swallow et al. (2024); Amiti et al. (2014)).

On the methodological front, I employ Känzig (2021)'s methodology of high frequency identification of an external instrument to study the impact of oil supply news shock which acts as a proxy for the global inflation shock, on the US dollar index against the currencies of emerging economies. The high frequency identification of external instruments has been previously employed for identification of monetary policy shock by Gertler and Karadi (2015), central bank information shock by Jarociński and Karadi (2020), and the global financial cycle shock by Miranda-Agrippino and Rey (2020).

For the purpose of theoretical analysis, I construct a three-bloc open economy model to study the multi regional spillover impact of a global inflation shock on CPI inflation and its sub-components. Among the studies that have employed this framework are Eichenbaum et al. (2020), Casas et al. (2017), and Kollmann et al. (2016).

The five different strands of literature identified in this section collectively form the foundational framework on which this study is built. The first three strands help identifying the gap in the literature pertaining to the transmission of a global inflation shock. They highlight the importance of exchange rate pass-through in the transmission of any shock that is dependent on the movements in exchange rates. ERPT is incomplete, heterogeneous, time-varying, and affected by numerous factors. With the focus of this study on the trade sector, the different types of currency invoicing regimes, trade openness, and intermediate goods become significant factors in determining ERPT. Therefore, all possible scenarios with differential degrees of ERPT across different currency invoicing regimes, varying degrees of trade openness, and trade in intermediate goods need to be examined. The final two strands provide the methodological framework for analyzing the two-channel transmission of a global inflation shock, with the choice of currency in-

voicing, the degree of ERPT, and trade structure serving as the channels through which the shock is transmitted.

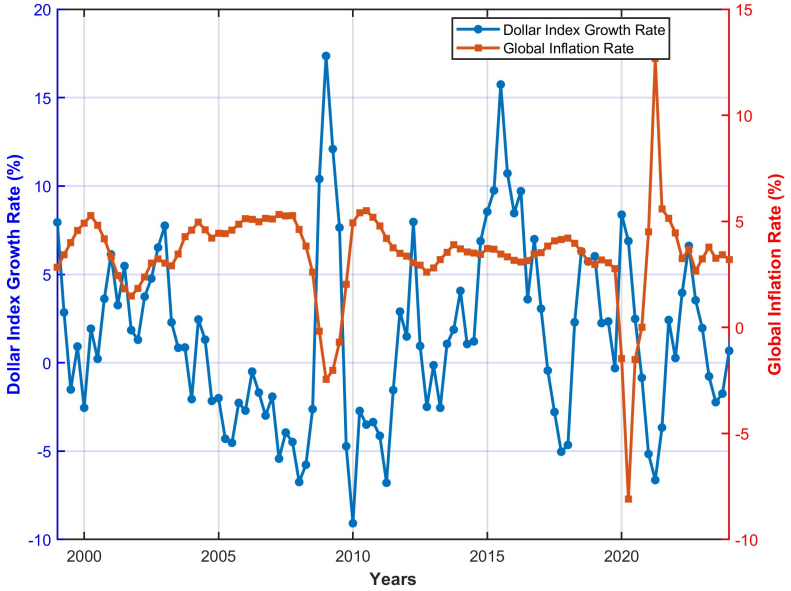


Figure 3: Correlation between global inflation and growth rate of US dollar index against the currency of emerging economies. The global inflation is derived as the percentage change of the quarterly GDP based chain linked volume price index of G20 countries from OECD dataset. The US dollar index against the emerging economies is taken from the Board of Governors of the Federal Reserve System. The time period considered is from Q1 1999 to Q4 2024

3 Empirical Analysis

This section undertakes empirical analysis on the impact of global inflation on US dollar index against emerging economies. Global inflation is based on the quarterly GDP based chain linked volume price index of G20 countries. G20 countries cover 80% of global GDP and 75% of international trade. The US dollar index vis a vis emerging economies is taken from Board of Governors of the Federal Reserve System (US) and covers top 19 bilateral trading emerging economy partners of the US. An increase in the index represent US dollar appreciation against the emerging economies currency.

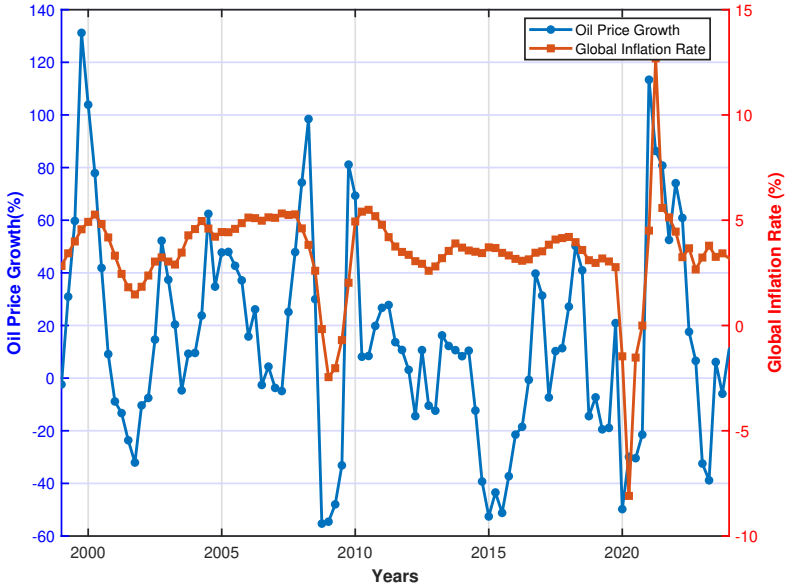


Figure 4: Correlation between global inflation and growth in Oil Price. The global inflation is derived as the percentage change of the quarterly GDP based chain linked volume price index of G20 countries from OECD dataset. The oil price is the WTI spot crude oil price and the time period considered is from Q1 1999 to Q4 2024.

Figure 3 shows the quarterly global inflation rate and the US dollar growth rate against emerging economies from Q1 1999 to Q4 2024. The correlation between the two variables for the same period is -0.58. Even after removing the period of volatile movement in global inflation rate, and focusing on the period from Q1 2010 to Q4 2019, the correlation between global inflation and the US dollar growth rate against emerging economies is -0.57. Thus, global inflation is associated with an appreciation of emerging economies currency vis a vis US dollar. This result that the total impact of global inflation is dampened for emerging economies that invoice their goods in US dollar. However, correlation only captures the strength of the comovement between the two variables. To further investigate the causality and contemporaneous relationship between global inflation and US dollar index against the emerging economies, a more rigorous and flexible empirical approach needs to be employed.

3.1 Empirical Model Framework

This subsection analyze the oil supply new shock as in Känzig (2021) as a proxy for global inflation shock to study its impact on the nominal exchange rate between US and

the emerging economies. The rationale for using the aforementioned shock as a proxy for the global inflation is the high positive correlation of 0.53 between the growth rate of oil price and global inflation as depicted in figure 4. In addition, the oil price shock contributes nearly 40 percent to the variation in global inflation (Ha et al., 2019).

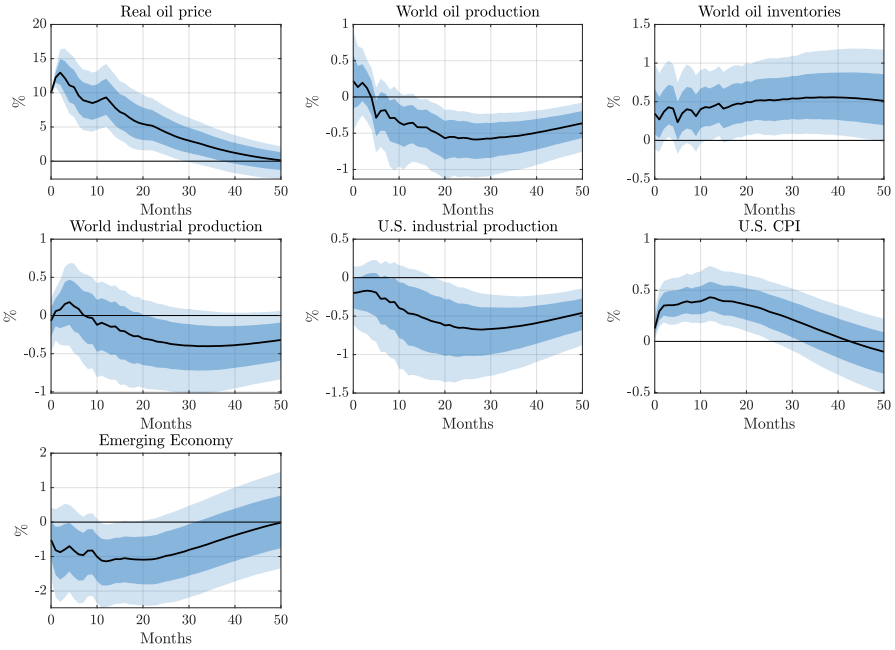


Figure 5: Impulse Responses to a Global Inflation Shock based on VAR analysis. The shock corresponds to a 10% increase in the real oil price. The panel in the third row is the exchange rate between US dollar and basket of emerging economies, where decline in the exchange rate means depreciation of US dollar against the emerging economies' currency

The structural shock is constructed using the external instrument approach, which are based on the oil supply surprise series, that are constructed using the OPEC announcements and high frequency future oil prices. The empirical model framework comprises of the baseline VAR model and Local projections as described in Känzig (2021), with an addition of nominal exchange rate between US and emerging economies. The VAR methodology allows to employ restrictions on the VAR residual covariance with the moment condition for the identification of the structural shock using the external instrument. However, the assumption of invertibility inherent within a VAR structure makes the identification empirically restrictive.

3.1.1 VAR

The reduced form VAR(p) model is described as follows:

$$y_t = b + B_1 y_{t-1} + ... + B_p y_{t-p} + u_t \tag{3.1}$$

where y_t is a n $\times$ 1 vector of endogenous variables, b is an n $\times$ 1 vector of constants, p is the lag order, $B_1....B_p$ are n $\times$ n coefficient matrices, and u_t is an n $\times$ 1 vector of reduced form innovations with covariance matrix $var(u_t) = \Sigma$.

The structural shocks are related to the reduced form innovations based on the following linear equation:

$$u_t = S\epsilon_t$$

where the non-singular structural impact matrix of n $\times$ n dimensions defined as S and ϵ_t is an n $\times$ 1 vector of structural shocks. The variance of the structural shocks is specified as $var(\epsilon_t) = \Omega$, which is a diagonal matrix because structural shocks are mutually uncorrelated. The variance of structural shocks and reduced form innovations are related based on the following identity.

$$\Sigma = S\Omega S' \tag{3.2}$$

The VAR identification focuses on finding the structural impact vector associated with the oil supply news shock, which is assumed as s_1, the first column of the structural impact matrix S. This is identified using the external instruments approach where, for an instrument z_t to be considered a valid instrument, it needs to satisfy the following two conditions:

$$E[z_t \epsilon_{1,t}] = \alpha \neq 0$$
$$E[z_t \epsilon_{2:n,t}] = 0$$

The two condition states that the external instrument (z_t) is correlated with the oil supply news shock ($\epsilon_{1,t}$) and uncorrelated with all the other shocks ($\epsilon_{2:n,t}$). The external instrument for the purpose of this exercise is the oil supply surprise series. Based on the aforementioned conditions, s_1 is identified as:

$$\tilde{s}_{2:n,1} \equiv \frac{s_{2:n,1}}{s_{1,1}} = \frac{E[z_t u_{2:n,t}]}{E[z_t u_{1,t}]}$$

This holds only when $E[z_t u_{1,t}] \neq 0$. The structural impact vector associated with the shock of interest is given by $s_1 = (s_{1,1}, \tilde{s}_{2:n,1} s_{1,1})'$. The scale $s_{1,1}$ is set by normalization, which is designed to incorporate a 10 percent increase in the price of oil in the aftermath of the shock.

3.2 Local Projections

Local projections provides a convenient way of empirically testing the relationship between two variables without providing the long run dynamic relationship between the variables. It employs single equation methods, to estimate the impact of an exogenous shock on the outcome without specifying the full structural model (Jorda and Taylor, 2025). However, the impulse responses generated are more erratic, less persistent, and less precisely estimated at longer horizons due to the absence of any dynamic restrictions across impulse horizons (Känzig, 2021). A linear local projection model can be estimated as follows:

$$y_{i,t+h} = \beta_0^i + \psi_h^i Shock_t + \beta_h^{i'} x_{t-1} + \xi_{i,t,h} \tag{3.3}$$

where, $y_{i,t+h}$ is the outcome variable of interest, x_{t-1} is the vector of controls, $Shock_t =$

$\hat{\epsilon}_{1,t}$ denotes the oil supply news shock identified from external instrument VAR, ψ_h^i is the impulse response to the oil supply news shock of variable i at horizon h, and $\xi_{i,t,h}$ is potentially serially correlated error term.

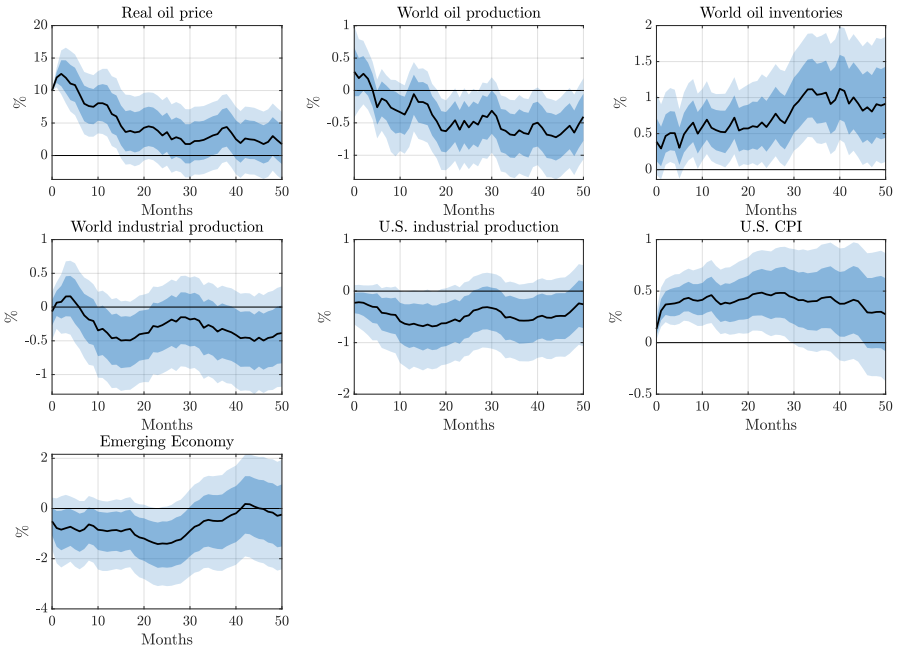


Figure 6: Impulse Responses to a Global Inflation Shock based on Local Projections. The shock corresponds to a 10% increase in the real oil price. The panel in the third row is the exchange rate between US dollar and basket of emerging economies, where decline in the exchange rate means depreciation of US dollar against the emerging economies' currency

3.3 Results

Similar to the results obtained in Känzig (2021), the price of oil and the world oil inventories increase in the aftermath of the shock. While the world oil production and world industrial production displays no immediate impact but both of them starts to fall significantly after the shock. The fall in the US industrial production materializes more quickly than the world industrial production, where the CPI inflation in the US rises substantially on impact and continues to rise for a year before it starts to fall.

The US dollar depreciates against the basket of emerging economies on impact and keep on depreciating for a year and starts to appreciate after a period of nearly two years from the oil supply news shock. The observed trajectory of the impulse response function resembles both the impact of the shock on the US dollar against the narrow currency

index and broad currency index in the additional propagation mechanism analyzed in the original paper. Against the narrow currency index, the US dollar starts to appreciate after reaching trough while the depreciation of the US dollar is persistent against the broad currency index.

Comparable results are yielded by the impulse responses under local projections with the caveat of impulse responses being more erratic due to lack of dynamic restrictions. Over long horizons, the results across local projections depicts lower degree of persistence and are less precisely estimated.

The depreciation of the US dollar against the currencies of emerging economies yields two notable insights about the transmission of a global inflation shock to CPI inflation in emerging economies. First, the relevant nominal exchange rate channel affects the transmission of a global inflation shock due to the high reliance on US dollar trade invoicing in emerging economies and the movement of the US dollar exchange rate in the aftermath of a global inflation shock (or its proxy). Invoicing goods in US dollar creates a difference between the consumer and invoicing currency, and any movement in the exchange rate between these two currencies affects the retail price of goods. This is referred to as the relevant nominal exchange rate channel. Second, with the US dollar depreciating against the currencies of the emerging economies, downward pressure is exerted on the retail price of imported goods in EE, as a smaller amount is required to be paid in EE's currency to acquire the imported goods.

4 Model

The model used for the analysis is a standard New Keynesian small open-economy model and streamlined from Eichenbaum et al. (2020). It comprises of three blocs, i.e. the emerging economy, the US economy, and the rest of the world. The baseline scenario assumes dominant currency invoicing, where goods are invoiced in a widely accepted and easily marketable currency, which is then compared with producer currency invoicing and local currency invoicing. The model consists broadly of two agents, households and firms, with incomplete asset markets, while the remaining features follow the standard New Keynesian model framework. It abstracts from capital and the financial sector in order to isolate the impact of the global inflation shock on CPI inflation in the EE via the goods sector. This section provides summarized details about the agents problems

and market clearing conditions.

4.1 Households

The utility of a household depends on its per capita consumption - which includes consumption of both domestically produced and imported final goods-, leisure, and the real stock of one-period domestic and foreign bonds it holds. The bonds are invoiced in the currency of the issuing country, and there exist two distinct types of foreign bonds for each country. The utility function for household 'k' is defined as follows:

$$U_{k,i} = E_t \sum_{t=0}^{\infty} \beta_E^t [log(C_{i,t}) + \varrho_i log(1 - H_{i,t}) + \sum_{j=[E,\$,R]} log(\eta_{i,j,t}) V(\frac{B_{i,j,t}}{P_{i.t}} NER_{i,j,t})]$$

$C_{i,t}$ is per capita consumption in country 'i', and $H_{i,t}$ is hours worked by the household in country i. $B_{i,j,t}$ is the end-of-period t nominal bond holdings of country j bonds held by households in country i. $P_{i,t}$ is the consumer price index in country i and $NER_{i,j,t}$ is the nominal exchange rate between country i and country j. An increase in the value of $NER_{i,j,t}$ translates into a depreciation of country i's currency against country j's currency, and vice versa. The Emerging Economy (E), the US Economy ($), and the Rest of the World (R) are the three possible types of countries, and thus i = [E,$,R].

As in Eichenbaum et al. (2020), V is the utility function that determines the utility households derive from holding domestic and foreign bonds. V is assumed to be increasing and strictly concave, and $V'(0)$ is set equal to the steady-state value of $\lambda_{i,t}$. $\eta_{i,j,t}$ is a spread shock that affects the utility in country i from holding country j bonds.

The budget constraint for the households is given by the following equation:

$$\begin{aligned} \sum_{j=[E,\$,R]} RER_{i,j,t} b_{i,j,t} + C_{i,t} = & \sum_{j=[E,\$,R]} RER_{i,j,t} b_{i,j,t-1} \frac{1}{\pi_{j,t-1,t}} + w_{i,t} H_{i,t} \\ & - [\sum_{j=[E,\$,R] \forall i \neq j} \frac{\kappa_{i,j}}{2} RER_{i,j,t} (b_{i,j,t} - b_{i,j}^{ss})^2] \end{aligned} \tag{4.1}$$

$RER_{i,j,t} = NER_{i,j,t} \frac{P_{j,t}}{P_{i,t}}$ is the real exchange rate between country i and country j, and

follows the same principle of appreciation/depreciation as the nominal exchange rate. $\pi_{j,t-1,t}$ is CPI-based inflation in country j between period t-1 and period t. The real wage rate is denoted as $w_{i,t}$, while bracketed term on the RHS is the portfolio adjustment cost as in Schmitt-Grohe and Uribe (2003), used to induce stationarity.

4.1.1 Final Consumption Goods

Households consume both domestic and imported final goods, making the final consumption good a combination of the two. Households purchase domestic final goods from the final goods firms and imported final goods from retailers in their own countries. Households purchase goods from final goods firms and retailers in their own currency. The final consumption good is subject to home bias and is defined as follows:

$$C_{i,t} = [\sum_{j=[E,\$,R]} (\omega_{i,j})^{\frac{1}{\rho_i}} (C_{i,j,t})^{\frac{\rho_i - 1}{\rho_i}}]^{\frac{\rho_i}{\rho_i - 1}}$$

where,

$$\sum_{j=[E,\$,R]} \omega_{i,j} = 1$$

$C_{i,j,t}$ is the final consumption good produced in country j and consumed in country i. $\omega_{i,j}$ denotes the proportion of final goods consumed in country i that originate from country j. The elasticity of substitution between domestic and imported final goods in country i is governed by ρ_i. The Dixit-Stiglitz price index corresponding to the aggregate consumption function is given by:

$$1 = [\sum_{j=[E,\$,R]} \omega_{i,j} (\frac{P_{i,j,t}}{P_{i,t}})^{1-\rho_i}]^{\frac{\rho_i}{\rho_i - 1}} \tag{4.2}$$

$p_{i,j,t} = \frac{P_{i,j,t}}{P_{i,t}}$ is the retail price of the final good produced in country j for country i.

4.2 Firms

At a granular level of functionality, firms can be classified into six different categories: wholesalers, production firms, intermediate good firms, final goods firms, exporters, and retailers. Among these firms, wholesalers and retailers operate under a monopolistic market structure. The rationale for assuming wholesalers are monopolistic is to incorporate nominal rigidity into the model. Retailers are assumed to be monopolistic, as they control the exchange rate pass-through. Exporters are assumed to be perfectly competitive in order to uphold the law of one price for the dock price of intermediate and final goods.

This chain of production can be described as follows: wholesalers in country i produce goods using factors of production, and sell them to the final goods firm and exporters in country i. Both the final goods firm and exporters in country i assemble these goods, with the final goods firm selling its assembled goods to consumers in country i, while the exporters sell their assembled goods to retailers in country j, which in turn sell them to consumers in country j. Similar chain of production exist for intermediate goods.

Table 3: Notations for different types of Goods

Notation	Description
$X^{M}_{i,j,t}$	Intermediate goods in country j to be utilized in country i
$M_{i,t}$	Composite intermediate good used for production in country i
$C_{i,j,t}$	Final goods produced in country j and consumed in country i
$X^{imp}_{i,j,t}$	Final Goods exported by country j to country i

4.2.1 Wholesalers

Wholesale firms are monopolistic and update the price of their goods based on Calvo pricing. The goods are produced using a Cobb-Douglas production function, with labour and composite intermediate goods as factors of production. It produces goods for the production firm, intermediate goods firm, final goods firm and exporters. The production function of wholesale firm q is as follows:

$$\sum_{j=[E,\$,R]} X_{j,i,t}(q) + \sum_{j=[E,\$,R]} X^{M}_{j,i,t}(q) = A_{i,t} H^{1-\alpha_i}_{i,t}(q) M^{\alpha_i}_{i,t}(q)$$

$X_{i,j,t}(q)$ is a good used to produce the final good in country j to be consumed in country i. $X^M_{i,j,t}(q)$, on the other hand, is used to produce intermediate goods, which are required as part of the production process.

Wholesale firms determine the price of their goods by maximizing their profits, which involves solving the following Calvo problem:

$$E_t \sum_{k=0}^{\infty} \Lambda_{i,t,t+k} (\xi^{imp}_{j,i})^k [\frac{P_{j,i,t}(i)}{P_{i,t+k}} - MC_{i,t+k}] X_{j,i,t+k}(q)$$

$\Lambda_{j,t,t+k}$ is the stochastic discount factor in the production country between period t and t+k. $\xi^{imp}_{i,j}$ is the Calvo parameter for goods produced in country j and consumed in country i. This changes to $\xi_{i,i}$ for domestically produced and consumed goods. The real marginal cost is denoted by $MC_{j,t+k}$. A similar profit-maximizing problem exists for intermediate goods.

4.2.2 Production Firm

Production firms combine domestically produced intermediate goods with imported intermediate goods to form a composite intermediate good, which is used as part of the production function of the wholesale firms. They combine different types of intermediate goods using a Dixit-Stiglitz aggregator, which is defined as follows:

$$M_{i,t} = [\sum_{j=[E,\$,R]} (m_{i,j})^{\frac{1}{z_i}} (X^M_{i,j,t})^{\frac{z_i - 1}{z_i}}]^{\frac{z_i}{z_i - 1}}$$

$M_{i,t}$ is the composite intermediate good used for production in country i. $X^M_{i,j,t}$ is the intermediate good produced in country j and utilized in country i. $m_{i,j}$ denotes the proportion of intermediate goods utilized in country i that originate from country j. The elasticity of substitution between domestic and imported intermediate goods in country i is governed by z_i. The Dixit-Stiglitz price index corresponding to the function is given by:

$$\frac{P^M_{i,t}}{P_{i,t}} = [\sum_{j=[E,\$,R]} m_{i,j}(\frac{P^M_{i,j,t}}{P_{i,t}})^{1-z_i}]^{\frac{1}{1-z_i}} \tag{4.3}$$

$p^M_{i,t} = \frac{P^M_{i,t}}{P_{i,t}}$ is the price of the composite intermediate good in country i, deflated by the consumer price index in country i. The real price of intermediate goods produced in country j for country i is denoted by $p^M_{i,j,t} = \frac{P^M_{i,j,t}}{P_{i,t}}$.

4.2.3 Intermediate Good Firm

These firms combine the goods produced by the wholesale firm into domestic intermediate goods, which are then utilized by the production firm to produce the composite intermediate good. They use the following aggregator to combine the goods:

$$X_{i,i,t} = [\int_0^1 (X^M_{i,i,t}(q))^{\frac{\theta^M_{i,i}-1}{\theta^M_{i,i}}}]^{\frac{\theta^M_{i,i}}{\theta^M_{i,i}-1}}$$

4.2.4 Final Goods Firm

There exists perfect competition among final good firms. They combine goods produced by wholesale firms to create the domestic final good, which is then consumed by households. The final goods firms combine goods in the following way:

$$C_{i,i,t} = [\int_0^1 (X_{i,i,t}(q))^{\frac{\theta_{i,i}-1}{\theta_{i,i}}}]^{\frac{\theta_{i,i}}{\theta_{i,i}-1}}$$

4.2.5 Exporters

Exporters operate under a perfectly competitive market structure and combine goods produced by wholesalers to produce intermediate and final goods, which are imported by other economies. They assemble the final goods using the following aggregator:

$$X_{i,j,t}^{imp} = [\int_0^1 (X_{i,j,t}(q))^{\frac{\theta_{i,j}^{imp}-1}{\theta_{i,j}^{imp}}}]^{\frac{\theta_{i,j}^{imp}}{\theta_{i,j}^{imp}-1}}$$

Similar aggregation occurs for intermediate goods:

$$X_{i,j,t}^{M} = [\int_0^1 (X_{i,j,t}^{M}(q))^{\frac{\theta_{i,j}^{Mi}-1}{\theta_{i,j}^{Mi}}}]^{\frac{\theta_{i,j}^{Mi}}{\theta_{i,j}^{Mi}-1}}$$

Under dominant currency pricing, export firms set the dock price of goods in US currency. With export firms operating under perfect competition, the law of one price holds for the dock price of intermediate and final goods.

4.2.6 Retailers

Retail firms are monopolistic and update the price of their goods using Calvo pricing. They import intermediate goods from exporters in other countries, which are then purchased by production firms to make the composite intermediate good. They also import final goods from other economies, which are then consumed by households. They assemble the final goods using the following aggregator:

$$C_{i,j,t} = [\int_0^1 (X_{i,j,t}^{imp}(r))^{\frac{\theta_{i,j}-1}{\theta_{i,j}}}]^{\frac{\theta_{i,j}}{\theta_{i,j}-1}}$$

Similar aggregation occurs for intermediate goods:

$$X_{i,j,t} = [\int_0^1 (X_{i,j,t}^{M}(r))^{\frac{\theta_{i,j}^{M}-1}{\theta_{i,j}^{M}}}]^{\frac{\theta_{i,j}^{M}}{\theta_{i,j}^{M}-1}}$$

Retail firms control the exchange rate pass-through for imported intermediate and final goods by solving the following dynamic problem:

$$E_t \sum_{k=0}^{\infty} \Lambda_{i,t,t+k} (\xi_{i,j})^k [\frac{P_{i,j,t}(i)}{P_{i,t+k}} - \frac{NER_{i,j,t+k} P^{imp}_{i,j,t+k}}{P_{i,t+k}}] X^{imp}_{i,j,t+k}(r)$$

$P^{imp}_{i,j,t+k}$ is the dock price of final goods produced in country j and consumed in country i. A similar dynamic problem exists for imported intermediate goods.

4.3 Monetary Policy

The nominal interest rate in country i is determined using the following Taylor type rule:

$$r_{i,t} = (r_{i,t-1})^{\sigma_i} (r^{ss}_i (\frac{\pi_{i,t}}{\pi^{ss}_i})^{\theta_{\pi_i}} (\frac{Y_{i,t}}{Y^{ss}_i})^{\theta_{y_i}})^{1-\sigma_i} e^{\epsilon_{M,i}} \tag{4.4}$$

$R_{i,t}$ is the nominal interest in country i, with its steady state counterpart denoted as R^{ss}_i. The monetary policy shock is denoted as $\epsilon_{M,i}$ and π^{ss}_i is the target level of CPI inflation. $Y_{i,t}$ is the total amount of goods produced in an economy and Y^{ss}_i is its steady state counterpart. It is incorporated within the model to study the indirect channel of global inflation. An increase in a country's CPI inflation leads to an increase in its nominal interest rate via the Taylor rule, which causes a realignment of the saving decisions of households across countries, and thus impacts the nominal exchange rates.

4.4 Market Clearing Conditions

The goods market clearing condition is defined as follows:

$$Y_{i,t} = C_{i,i,t} + \sum_{j=[E,\$,R] \forall i \neq j} X^{imp}_{j,i,t} + I_{i,t} + [\sum_{j=[\$,R] \forall i \neq j} \frac{\kappa_{i,j}}{2} RER_{i,j,t} (b_{i,j,t} - b^{ss}_{i,j})^2] \tag{4.5}$$

where, $Y_{i,t}$ is defined as the total output of country i in time period t.

Aggregation across firms provides the following equation:

$$A_{i,t}(H_{i,t})^{1-\alpha_i}(K_{i,t-1})^{\alpha_i} = C_{i,i,t}PD_{i,i,t} + \sum_{j=[E,\$,R]\forall i\neq j} PD^{imp}_{j,i,t}X^{imp}_{j,i,t} \tag{4.6}$$

Bond market equilibrium is given by the following identity:

$$\sum_{j=[\$,R]} b_{j,i,t} = 0 \tag{4.7}$$

4.5 Stochastic Process

The main stochastic process considered for this exercise is the markup/cost-push shock in the ROW, which acts as a global inflation shock and is denoted by $MS_{R,t}$. It allows for an examination of the two-channel mechanism through which global inflation affects CPI inflation in the EE. It leads to an increases in the dock price of the final goods produced in the ROW for the EE, along with the price of the intermediate goods produced in the ROW for its own production and for imports to the US and the EE. This causes CPI inflation to increase in all three blocs, thus causing global inflation. The markup shock, along with the technology variable ($A_{i,t}$) and the components of the spread shock ($\tilde{\eta}_{i,j,t} and \eta_{j,t}$), evolve as follows:

$$log(x_{i,t}) = \rho_{x,i} log(x_{i,t-1}) + \epsilon_{x,i,t} \tag{4.8}$$

where x is the variable of interest and $|\rho_{x,i}| < 1$ is the persistence parameter. $\epsilon_{x,i,t}$ are iid normal random variables, which is also true for the monetary policy shock.

4.6 Calibration

The calibrated values of the majority of parameters are based on the relevant literature, except for the parameters responsible for the exchange rate pass-through (ERPT), the share of domestic and imported intermediate and final goods from each country, the steady-state level of debt, and the portfolio adjustment cost.

Table 4: Value of Selected Calibrated Parameters

Variables	**Description**	**Value**
$\xi_{\$,R}, \xi_{\$,E}$	ERPT parameter for Final Goods imported in US	0.75
$\xi_{E,\$}, \xi_{R,\$}$	ERPT parameter for Final Goods produced in US	0.25
$\xi_{E,R}$	ERPT parameter for Final Goods produced in ROW for EE	0.5
$\xi_{R,E}$	ERPT parameter for Final Goods produced in EE for ROW	0.5
$\xi^M_{\$,R}, \xi^M_{\$,E}$	ERPT parameter for Int Goods imported in US	0.75
$\xi^M_{E,\$}, \xi^M_{R,\$}$	ERPT parameter for Int Goods produced in US	0.25
$\xi^M_{E,R}$	ERPT parameter for Int Goods produced in ROW for EE	0.5
$\xi^M_{R,E}$	ERPT parameter for Int Goods produced in EE for ROW	0.5
$\omega_{E,\$}$	Proportion of Final Goods produced in US for EE	0.02
$\omega_{E,R}$	Proportion of Final Goods produced in ROW for EE	0.22
$\omega_{\$,\$}$	Proportion of Final Goods produced and consumed in US	0.786
$\omega_{\$,E}$	Proportion of Final Goods produced in EE for US	0.007
$\omega_{\$,R}$	Proportion of Final Goods produced in ROW for US	0.207
$\omega_{R,R}$	Proportion of Final Goods produced and consumed in ROW	0.977
$\omega_{R,E}$	Proportion of Final Goods produced in EE for ROW	0.0055
$\omega_{R,\$}$	Proportion of Final Goods produced in US for ROW	0.0175
$m_{E,E}$	Proportion of Int Goods produced and consumed in EE	0.77
$m_{E,\$}$	Proportion of Int Goods produced in US for EE	0.02
$m_{E,R}$	Proportion of Int Goods produced in ROW for EE	0.21
$m_{\$,\$}$	Proportion of Int Goods produced and consumed in US	0.886
$m_{\$,E}$	Proportion of Int Goods produced in EE for US	0.002
$m_{\$,R}$	Proportion of Int Goods produced in ROW for US	0.112
$m_{R,R}$	Proportion of Int Goods produced and consumed in ROW	0.97
$m_{R,E}$	Proportion of Int Goods produced in EE for ROW	0.01
$m_{R,\$}$	Proportion of Intermediate Goods produced in US for ROW	0.02
$\kappa_{E,\$}, \kappa_{R,\$}$	Portfolio Adjustment Costs US Bonds	0.001
$\kappa_{E,R}, \kappa_{\$,R}$	Portfolio Adjustment Costs ROW Bonds	0.01
$\kappa_{\$,E}, \kappa_{R,E}$	Portfolio Adjustment Costs EE Bonds	0.01

In the baseline scenario, the value of ERPT is assumed to be high under producer currency pricing, low under local currency pricing, and intermediate under dominant currency pricing. This is based on the empirical evidence found in Gopinath (2015). The share of domestic and imported intermediate and final goods from each country in the baseline scenario is calculated using the Asian Development Bank's multiregional input-output table. The final value used is an annual average of the data available since 2007 for 62 countries and the rest of the world across 35 sectors. The selection of the group of emerging economies is based on the availability of data for the economies included in the MSCI Emerging Markets Index.

The steady-state level of debt in the baseline scenario is based on the trade balance to output ratio based calculation, as illustrated in Uribe and Schmitt-Grohe (2017). The values for the trade balance to output ratio are derived from the quarterly log-quadratic time-trend data available in the aforementioned reference. The mean value across all countries is assumed to be the trade balance to output ratio of the ROW, while the data for the emerging economy and the US are provided explicitly in the dataset. In the baseline scenario, the portfolio cost associated with US bonds purchased in the emerging economy and the ROW is assumed to be close to zero, in order to highlight the easy availability and marketability of US bonds. The table containing value of all the parameters in the model can be accessed in the appendix.

5 Results and Analysis

This section analyzes the three-bloc open-economy model defined in the previous section. The first sub-section focuses on the baseline scenario in which the US dollar is the dominant currency, with the analysis undertaken across all three currency invoicing regimes. The second sub-section examines differential degrees of ERPT, first considering an unequal degree of pass-through across currency invoicing regimes and subsequently investigating an equal degree of pass-through across currency invoicing regimes. The third sub-section shifts focus to different types of goods and analyzing the impact of a global inflation shock across various types of goods. The last sub-section evaluates the impact of varying trade openness on the transmission of a global inflation shock.

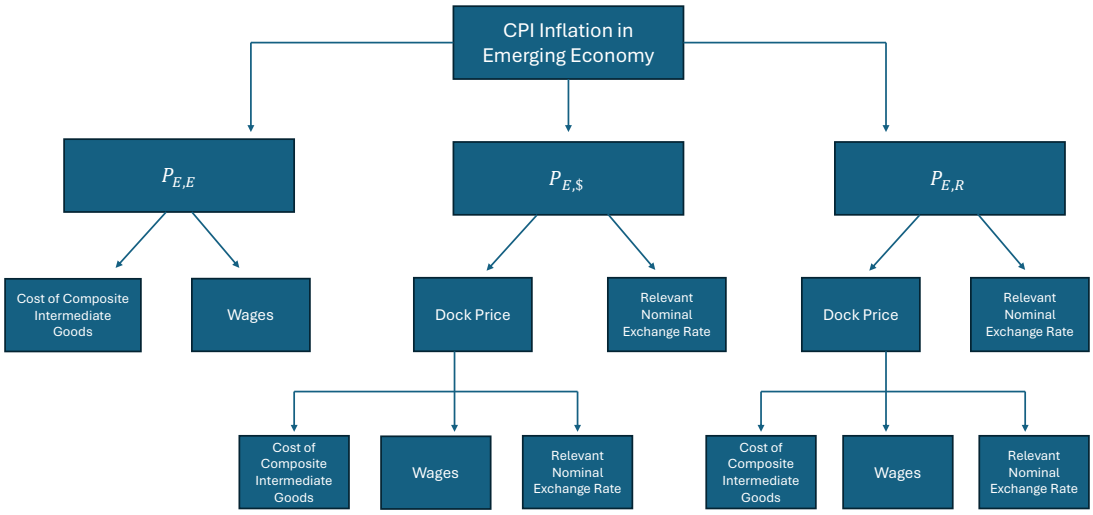


Figure 7: Variables impacting CPI inflation and its subcomponents

5.1 Baseline Scenario

The analysis focuses on the impact of a global inflation shock on the CPI inflation of the emerging economy (EE). For the purpose of this study, a cost-push/mark-up shock with a magnitude of one percent in the ROW is considered the global inflation shock. It affects both intermediate and final goods firms in the ROW and evolves according to Equation 4.8. The impact of the global inflation shock is examined for the variables of interest identified in Figure 7.

The CPI inflation of the emerging economy is based on the retail price of the final goods produced domestically ($P_{E,E}$), in the US ($P_{E,\$}$), and in the ROW ($P_{E,R}$). The retail price is the price paid by the consumer in their domestic currency, to purchase the final good. $P_{E,E}$ is influenced by the cost of composite intermediate goods and wages in the emerging economy. The cost of the composite intermediate goods reflects the total price of the intermediate goods used in the production of final goods that are produced and consumed domestically. The composite intermediate good used in production is made from intermediate goods produced domestically and imported from the US and the ROW.

Two broad themes of similarity exist across the different currency invoicing regimes. The first relates to the opposing price pressures emanating from the cost of composite intermediate goods and wages. Following a global inflation shock, the cost of composite intermediate goods increases while wages fall across all blocs and under all currency invoicing regimes. This holds true for all blocs across all currency invoicing regimes (see Figure 9 (a), (b), (e), (f) and 10 (e), (f)). For the final goods produced and consumed in EE, the upward price pressure emanating from the cost of composite intermediate goods overpowers the downward price pressure stemming from wages. This can be seen in Figure 8 (b), where the growth rate of $P_{E,E}$ increases on impact and stays above the steady-state level across all currency invoicing regimes. The growth rate of $P_{E,E}$ is the highest under PCP, followed by DCP and LCP.

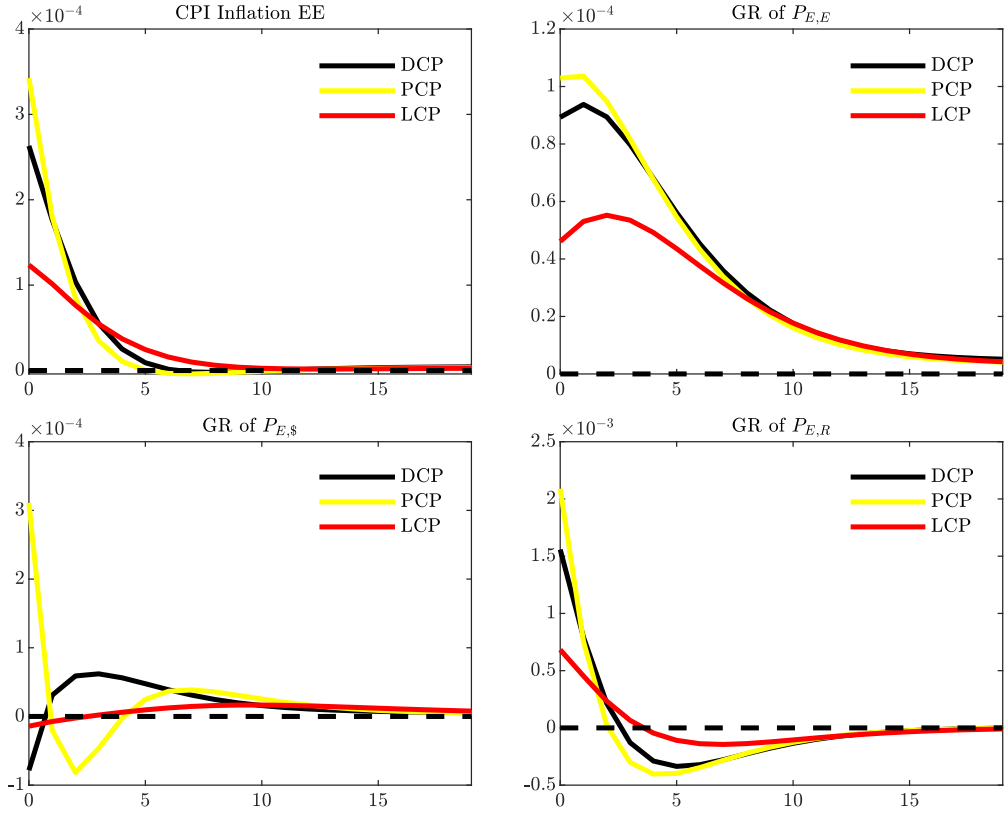


Figure 8: Impulse Responses of CPI Inflation and its sub-components to 1 s.d. Global Inflation Shock under different currency invoicing regimes. The panels contain the following variables (a) CPI Inflation in EE; (b) Growth Rate of retail price of domestically produced and consumed final goods in EE ($P_{E,E}$); (c) Growth Rate of retail price of final goods produced in the US for the EE ($P_{E,\$}$); (d) Growth Rate of retail price of final goods produced in the ROW for the EE ($P_{E,\$}$)

The second theme relates to the impact of the change in the relevant nominal exchange rate following the global inflation shock, on the dock and retail prices of the goods imported into the EE. The relevant nominal exchange rate is the exchange rate that impacts either the dock or retail price due to the difference between the invoicing currency and the currency of the producer or consumer country. It differs across blocs and currency invoicing regimes. For instance, under DCP and PCP, $NER_{E,\$}$ becomes the relevant nominal exchange rate at the retail price level, whereas under LCP, $NER_{E,\$}$ is the relevant nominal exchange rate for the dock price of goods produced in the US for the EE. This is illustrated below through examples at different price levels across all currency invoicing regimes.

The relevant nominal exchange rate channel is evident under LCP for the dock price of the final goods produced in the US for the EE. Since the invoicing currency (the EE's currency) differs from the currency of the producer (the US's currency), the nominal exchange rate between the EE and the US becomes relevant for this dock price. The EE's nominal exchange rate appreciates against the US currency, allowing producers in the US to earn more US currency for every unit of EE currency (see Figure 9 (d)). This allows them to reduce their prices without a decline in relative profits in order to increase demand for their products, thus putting downward pressure on the dock price. The downward pressure stemming from $NER_{E,\$}$, combined with that stemming from the fall in wages, overpowers the upward pressure stemming from the increase in the cost of composite intermediate goods in the US, leading to a fall in this dock price (see Figure 9 (c)).

A similar impact of the relevant nominal exchange rate can be seen at the retail price for the final goods produced in the US for the EE under PCP. The invoicing currency is different from the currency of the consumer country, thus the nominal exchange rate channel becomes active at the retail price level. The increase in $P_{E,\$}$ under PCP can be attributed to the upward pressure emanating from the dock price of the good, along with the depreciation of the EE's currency against the US currency. The depreciation of the EE's currency against the US currency puts upward pressure on the retail price, as it becomes more expensive for the retailer in the EE to import goods from the US. This leads to an even greater increase in $P_{E,\$}$ when combined with the upward pressure stemming from the dock price, as can be seen in Figure **??** (c). The depreciation of the EE's currency against the US currency is due to the relatively greater slowdown in global

demand under PCP, the comparatively higher import dependence of emerging economies on the US, and the differential degree of pass-through across currency invoicing regimes and prices.

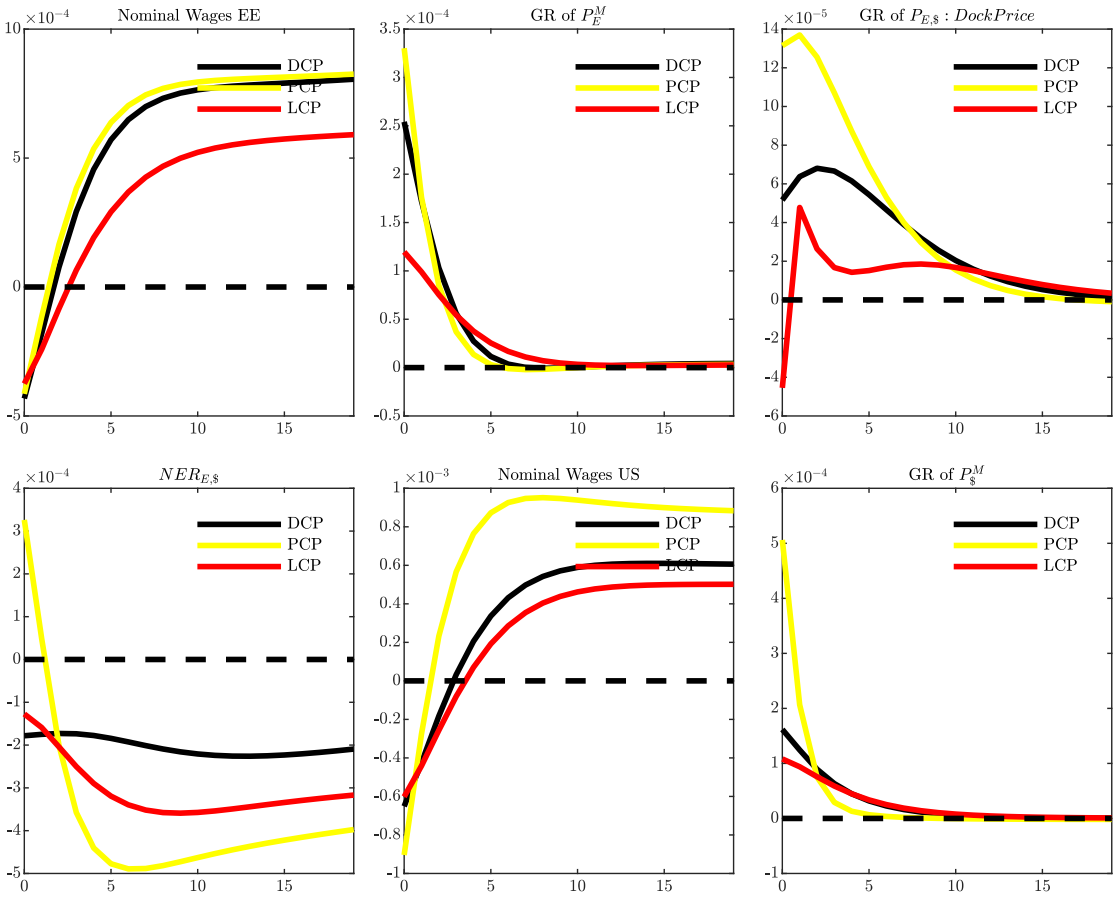


Figure 9: Impulse Responses of CPI Inflation and its sub-components to 1 s.d. Global Inflation Shock under different currency invoicing regimes. The panels contain the following variables (a) Nominal Wages in EE; (b) Growth Rate of cost of composite intermediate goods in EE (P_E^M); (c) Growth rate of dock price of final goods produced in the US for EE ($P_{E,\$}$: $DockPrice$) (d) Nominal Exchange Rate between the US and the EE ($NER_{E,\$}$); (e) Nominal Wages in US (f) Growth Rate of cost of composite intermediate goods in US ($P_\$^M$)

The relevant nominal exchange rates can also impact both the dock and retail prices of the same good in opposite directions. Under DCP, the goods produced in the ROW for the EE are invoiced in US dollars, making the invoicing currency different from the currency of the producer. The ROW's nominal exchange rate appreciates against the US dollar, causing producers in the ROW to earn less ROW currency for every unit of

US currency. This puts upward pressure on the dock price of the goods produced in the ROW for the EE. On the other hand, since the currency of the consumer country differs from the invoicing currency, $NER_{E,\$}$ becomes the relevant nominal exchange rate at the retail price level. The EE's nominal exchange rate appreciates against the US currency, which puts downward pressure on the retail price of the final goods produced in the ROW for the EE. This is due to the lower cost in EE currency, for retailers in the EE to import these goods. More such illustrations covering different price levels across all currency invoicing regimes, can be found in the appendix.

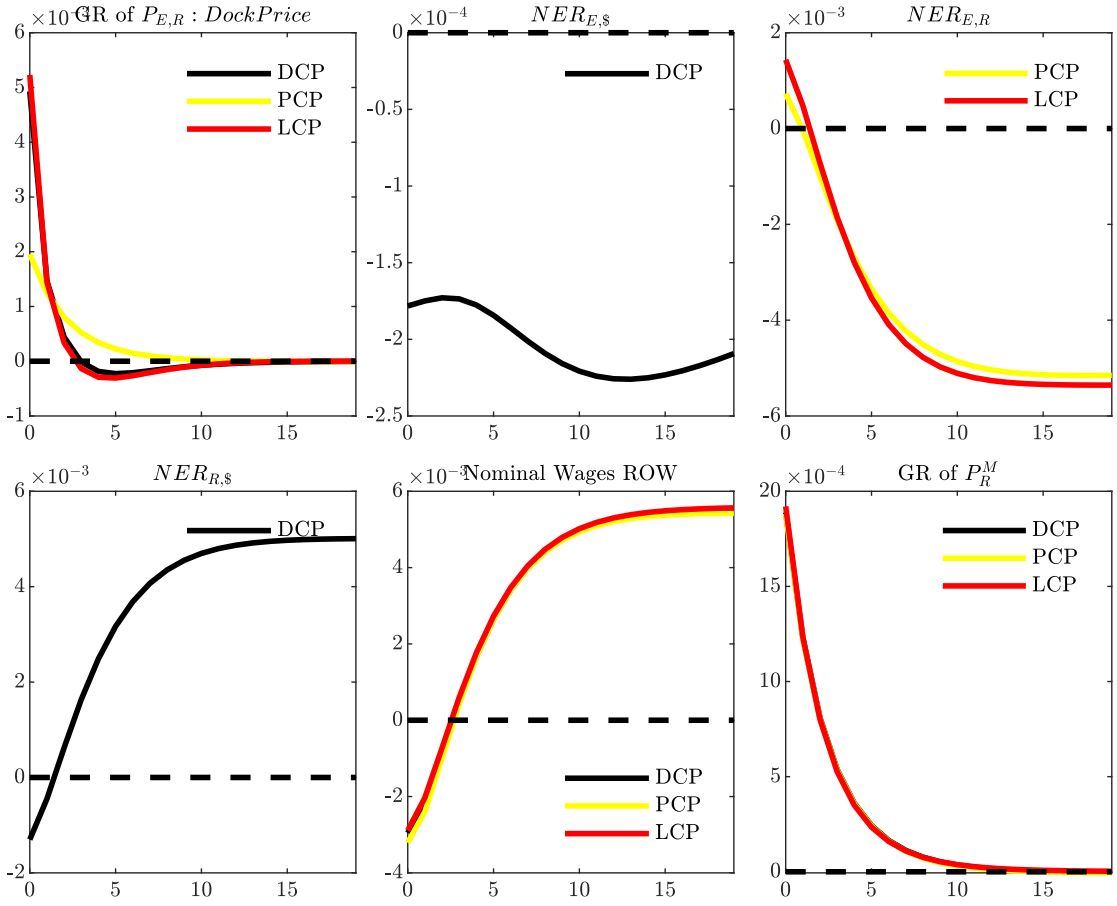


Figure 10: Impulse Responses of CPI Inflation and its sub-components to 1 s.d. Global Inflation Shock under different currency invoicing regimes. The panels contain the following variables (a)Growth rate of dock price of final goods produced in the ROW for EE ($P_{E,R}$: $DockPrice$); (b) Nominal Exchange Rate between the US and the EE ($NER_{E,\$}$); (c) Nominal Exchange Rate between the ROW and the EE ($NER_{E,R}$) (d) Nominal Exchange Rate between the US and the ROW ($NER_{R,\$}$) ; (e) Nominal Wages in ROW (f) Growth Rate of cost of composite intermediate goods in ROW (P_R^M)

The above analysis highlights the importance of the relevant nominal exchange rate channel as a catalyst for the differential transmission of a global inflation shock across different currency invoicing regimes. For the first of the two main themes identified, for the first set of results, the direction of change for the factors of production is the same across all currency invoicing regimes, even though the magnitudes of change differ. For the second, the more contrasting results across the currency invoicing regimes are due to the relevant nominal exchange rate channel. The workings of the relevant nominal exchange rate channel are due not only to the different relevant exchange rates but also to different movements in the same relevant exchange rates across currency invoicing regimes. Thus, it is not just the assumed features of the currency invoicing regimes but also the overall model dynamics that play a critical part in the transmission of a global inflation shock via the relevant nominal exchange rate channel.

Table 5: Model Scenarios Unequal ERPT

Scenario	**High**	**Middle**	**Low**
Baseline	PCP	DCP	LCP
Scenario 1	DCP	PCP	LCP
Scenario 2	DCP	LCP	PCP
Scenario 3	PCP	LCP	DCP
Scenario 4	LCP	PCP	DCP
Scenario 5	LCP	DCP	PCP

5.2 Exchange Rate Pass Through

The review of the literature highlights the importance of ERPT in regulating the transmission of a global inflation shock by controlling the degree to which movements in exchange rates are passed through to domestic prices. This is a key mechanism through which the relevant nominal exchange rate channel to function. The literature also identifies that ERPT is time-varying and heterogeneous, differing across different currency invoicing regimes. To better understand these features of ERPT better, this section focuses on analyzing the different model scenarios of ERPT to gauge the importance of heterogeneity in its degree across different currency invoicing regimes. This also allows for an investigation into the time-varying nature of ERPT. All possible relative ERPT model scenarios are identified in Table 5.

The second part of this subsection examines the role of different degrees of ERPT, re-

moving the heterogeneity across currency invoicing regimes by assuming an equal degree of ERPT for all of them. This provides insights into the effect of different degrees of ERPT in isolation and also highlights the importance of heterogeneity in the differential impact of a global inflation shock across currency invoicing regimes.

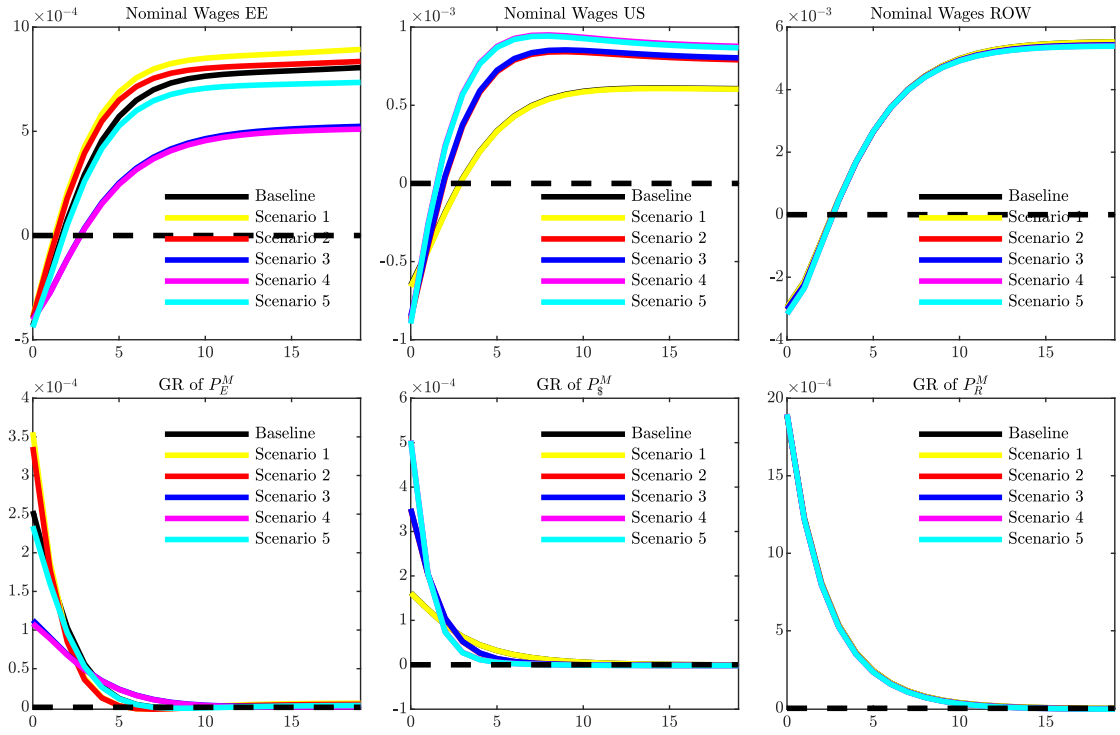


Figure 11: Impulse Responses of Factors of Production across all blocs under DCP to 1 s.d. Global Inflation Shock across unequal exchange rate pass through model scenarios. The panels contain the following variables (a) Nominal Wages in EE; (b) Nominal Wages in US; (c) Nominal Wages in ROW; (d) Growth Rate of cost of composite intermediate goods in EE (P_E^M); (e) Growth Rate of cost of composite intermediate goods in US ($P_\$^M$); (f) Growth Rate of cost of composite intermediate goods in EE (P_R^M);

5.2.1 Unequal Exchange Rate Pass Through

The six model scenarios in this subsection are defined in Table 5, with each scenario specifying a distinct relative ERPT for each currency invoicing regime. In each model scenario, one currency invoicing regime carries a high ERPT, another carries a middle ERPT, and a third carries a low ERPT. The high ERPT parameter is assumed to be 0.25, which implies that exchange rate changes are fully passed through to retail prices after 1.33 quarters. The middle ERPT parameter is assumed to be 0.5 (2 quarters), while the low ERPT parameter is assumed to be 0.75 (4 quarters). The analysis in this

subsection focuses on comparison across all model scenarios with the baseline scenario, especially with regard to the two overarching results identified previously.

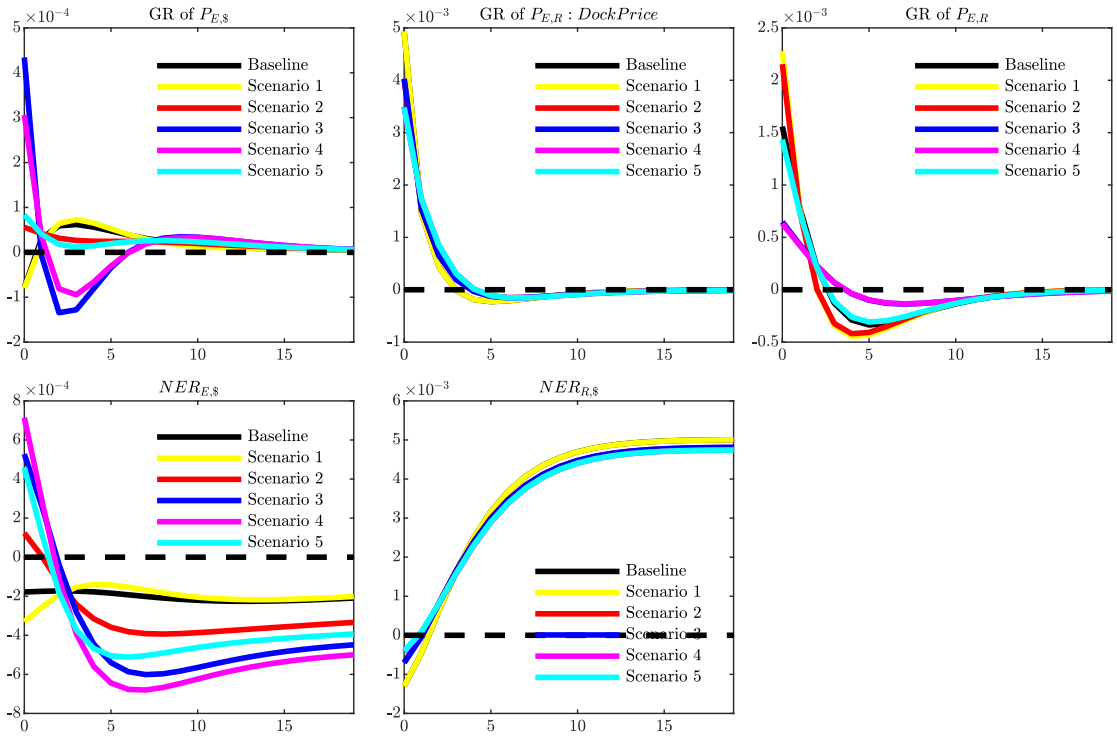


Figure 12: Impulse Responses of sub-components of CPI Inflation in EE and Relevant Nominal Exchange Rates under DCP to 1 s.d. Global Inflation Shock across unequal exchange rate pass through model scenarios. The panels contain the following variables (a) Growth Rate of retail price of final goods produced in the US for EE ($P_{E,\$}$); (b) Growth Rate of dock price of final goods produced in the ROW for EE ($P^{imp}_{E,R}$); (c) Growth Rate of retail price of final goods produced in the ROW for EE ($P_{E,R}$); (d) Nominal Exchange Rate between EE and US ($NER_{E,\$}$); (e) Nominal Exchange Rate between ROW and US ($NER_{R,\$}$);

The two broad themes of results discussed in the baseline scenario exist across all five additional scenarios assumed in this section. Similar to the baseline scenario, the cost of composite intermediate goods and wages exert opposing price pressures across all blocs in all five scenarios (see Figure 11). Wages exert downward price pressure while the cost of composite intermediate goods exerts upward price pressure. Across all six scenarios, the upward price pressure stemming from the cost of composite intermediate goods overpowers the downward price pressure stemming from wages.

The changes in the relevant nominal exchange rate impact the dock and retail prices of the goods imported into the EE following a global inflation shock, although this impact

differs across scenarios relative to the baseline scenario. Under DCP in the baseline scenario, the relevant nominal exchange rate impacts the dock and retail prices of the goods produced in the ROW for the EE in opposite directions. However, under Scenarios 2, 3, 4, and 5, the relevant nominal exchange rate exerts upward pressure on both the dock and retail prices of the goods produced in the ROW for the EE. The depreciation of the US currency against the ROW currency leads to upward pressure on $P^{imp}_{E,R}$ while, the depreciation of EE's currency against the US currency exerts upward pressure on $P_{E,R}$.

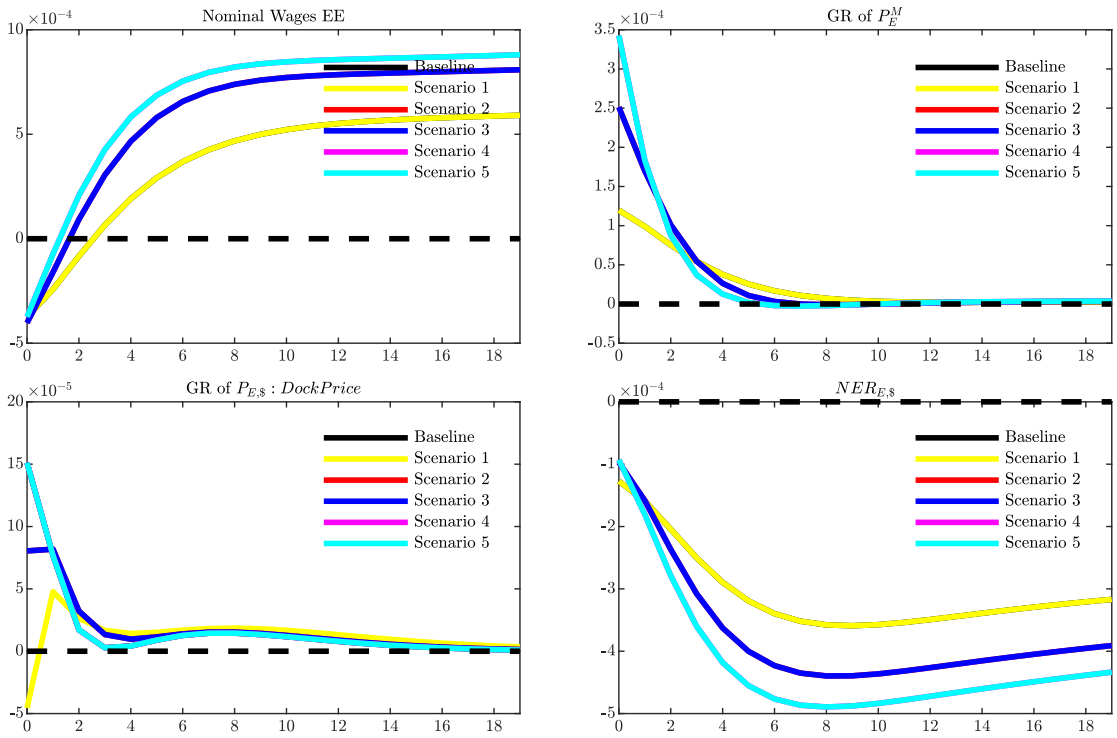


Figure 13: Impulse Responses of Factors of Production in EE, Dock Price, and Nominal Exchange Rate under LCP to 1 s.d. Global Inflation Shock across unequal exchange rate pass through model scenarios. The panels contain the following variables (a) Nominal Wages in EE; (b) Growth Rate of cost of composite intermediate goods in EE (P^M_E); (c) Growth Rate of dock price of final goods produced in the US for EE ($P^{imp}_{E,\$}$); (d) Nominal Exchange Rate between EE and US ($NER_{E,\$}$);

The depreciation of the EE's currency against the US currency under Scenarios 2, 3, 4, and 5 also impacts the retail price of the final goods produced in the US for the EE ($P_{E,\$}$) under DCP and the dock price of the final goods produced in the US for the EE ($P^{imp}_{E,\$}$) under LCP. Under the baseline scenario, the appreciation of the EE's currency

against the US currency ($NER_{E,\$}$) translates into a decline in $P_{E,\$}$ under DCP and $P^{imp}_{E,\$}$ under LCP, due to the downward pressure stemming from $NER_{E,\$}$. In contrast, with the depreciation of the EE's currency against the US currency for the aforementioned scenarios, instead of a downward price pressure, $NER_{E,\$}$ exerts upward price pressure, causing both $P_{E,\$}$ under DCP and $P^{imp}_{E,\$}$ under LCP to increase on impact (see Figure 12 (a), and (d), Figure 13 (d)).

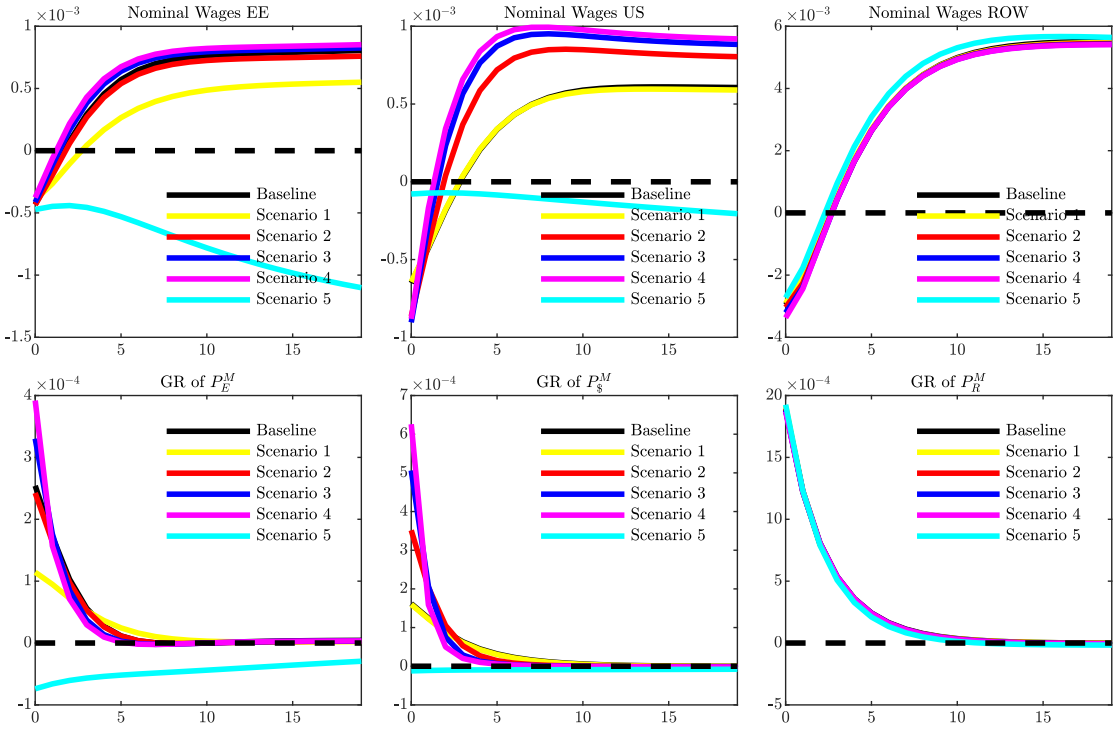


Figure 14: Impulse Responses of Factors of Production across all blocs under DCP to 1 s.d. Global Inflation Shock across equal exchange rate pass through model scenarios. The panels contain the following variables (a) Nominal Wages in EE; (b) Nominal Wages in US; (c) Nominal Wages in ROW; (d) Growth Rate of cost of composite intermediate goods in EE (P^M_E); (e) Growth Rate of cost of composite intermediate goods in US ($P^M_\$$); (f) Growth Rate of cost of composite intermediate goods in EE (P^M_R);

5.2.2 Equal Exchange Rate Pass Through

This subsection comprises five model scenarios in which ERPT is assumed to be equal across all currency invoicing regimes. In addition to the previously considered values of ERPT, two additional scenarios are included to represent near-zero and near-complete exchange rate pass-through. The analysis in this subsection is focused on comparisons across model scenarios for each currency invoicing regime, as well as on the two overar-

ching themes identified in the baseline scenario.

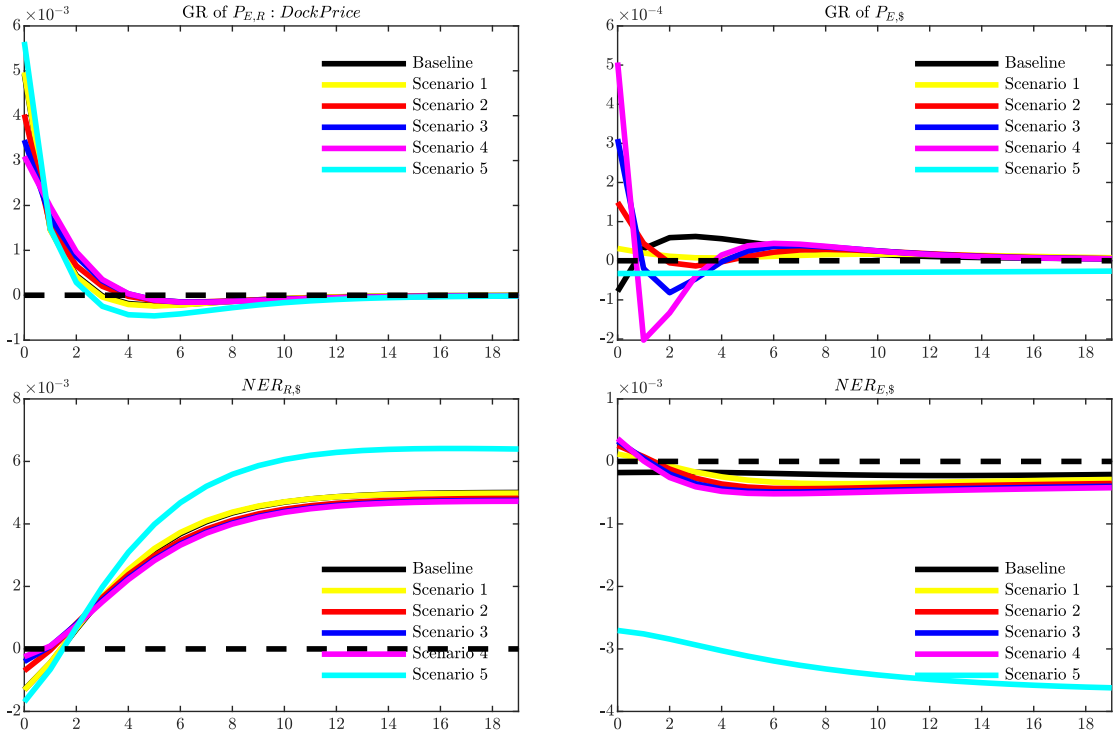


Figure 15: Impulse Responses of sub-components of CPI Inflation in EE and Relevant Nominal Exchange Rates under DCP to 1 s.d. Global Inflation Shock across equal exchange rate pass through model scenarios. The panels contain the following variables (a) Growth Rate of dock price of final goods produced in the ROW for EE ($P^{imp}_{E,R}$); (b) Growth Rate of retail price of final goods produced in the US for EE ($P_{E,\$}$); (d) Nominal Exchange Rate between ROW and US ($NER_{E,\$}$); (e) Nominal Exchange Rate between EE and US ($NER_{E,\$}$);

With the exception of Scenario 5, where ERPT is near-zero, the two overarching themes identified in the baseline scenario persist across Scenarios 1 to 4. This identifies ERPT as the factor responsible for the existence of the two-channel transmission (i.e., the two overarching results) following a global inflation shock. The opposing price pressures emanating from the factors of production are prevalent across all blocs under all four scenarios (see Figure 14). The upward price pressure stemming from the cost of composite intermediate goods overpowers the downward price pressure stemming from wages for goods produced and consumed in the EE, across all three currency invoicing regimes. This also holds for the dock price of the goods produced in the US and the ROW for the EE under DCP and PCP. Under LCP, this effect cannot be readily isolated, as the dock price of the goods produced in the US and the ROW for the EE is also affected by

the relevant nominal exchange rate.

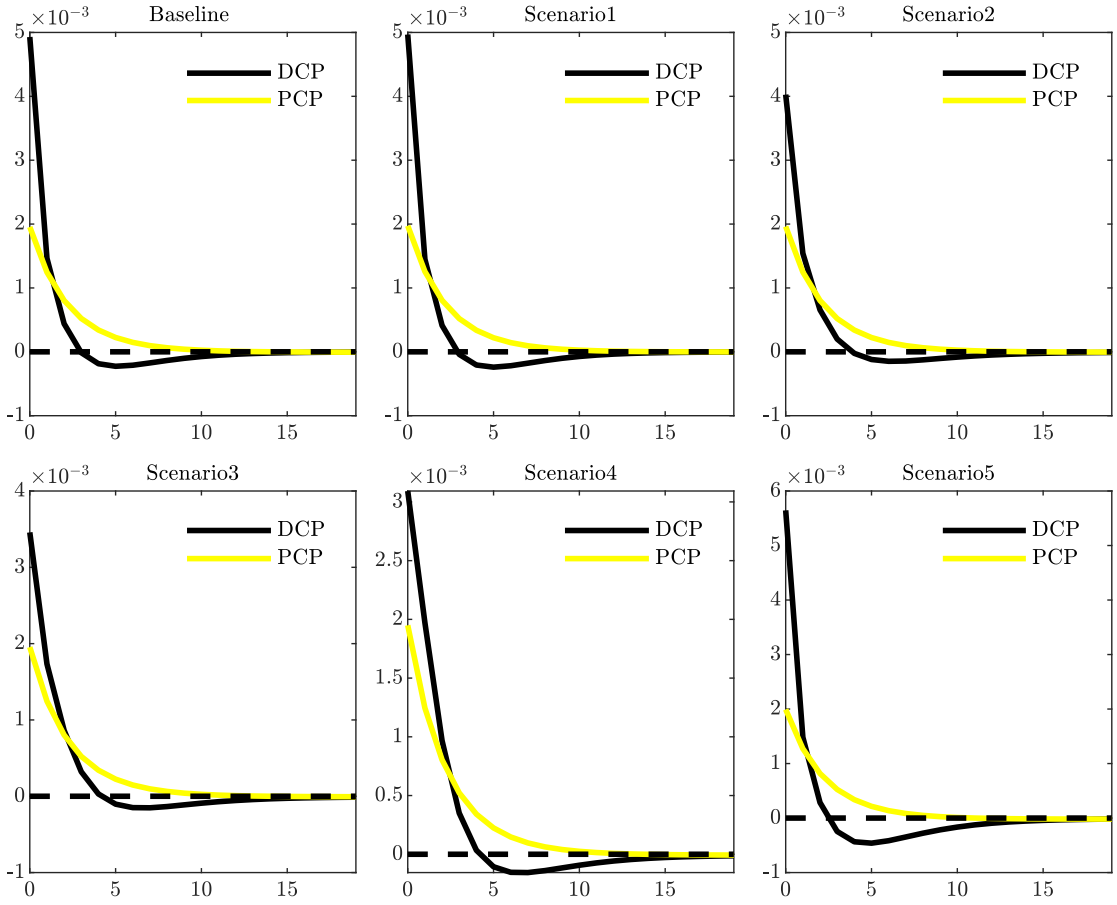


Figure 16: Impulse Responses of the dock price of final goods produced in the ROW for EE ($P_{E,R}^{imp}$ under Dominant and Producer Currency Pricing to 1 s.d. Global Inflation Shock across different scenarios of equal exchange rate pass-through. The panels contain the following variables (a) Baseline Scenario (b) Scenario 1 (c) Scenario 2 (d) Scenario 3 (e) Scenario 4 (f) Scenario 5

The impact of the indirect channel - - - the relevant nominal exchange rate - - - is evident in the dock price of the goods produced in the ROW for the EE under DCP. This is due to the impact of the relevant nominal exchange rate channel. Under DCP, the exchange rate between the ROW and the US is the relevant nominal exchange rate for the dock price of the goods produced in the ROW for the EE. The appreciation of the ROW currency against the US currency exerts upward price pressure on the dock price, as exporters receive fewer units of ROW currency for every unit of US currency (see Figure 15 (a) and (c)) . Consequently, $P_{E,R}^{imp}$ rises under DCP, whereas under PCP, $P_{E,R}^{imp}$ is unaffected by the relevant nominal exchange rate channel, as the invoicing currency is

identical to the currency of the producer country.

A further impact of the indirect channel is observed in $P_{E,\$}$ under DCP. Under the baseline scenario, the retail price of the goods produced in the US for the EE falls due to the downward price pressure stemming from the appreciation of the EE's currency against the US currency. However, with the EE's currency depreciating against the US currency across Scenarios 1 to 4, $P_{E,\$}$ rises on impact rather than falling. Another key result originating from the analysis in this subsection is that the impulse response functions across all variables of interest, except for $P^{imp}_{E,R}$, follow identical trajectories under DCP and PCP for all model scenarios (see Figure 16). This highlights the importance of differential ERPT in shaping the impact of a global inflation shock across DCP and PCP.

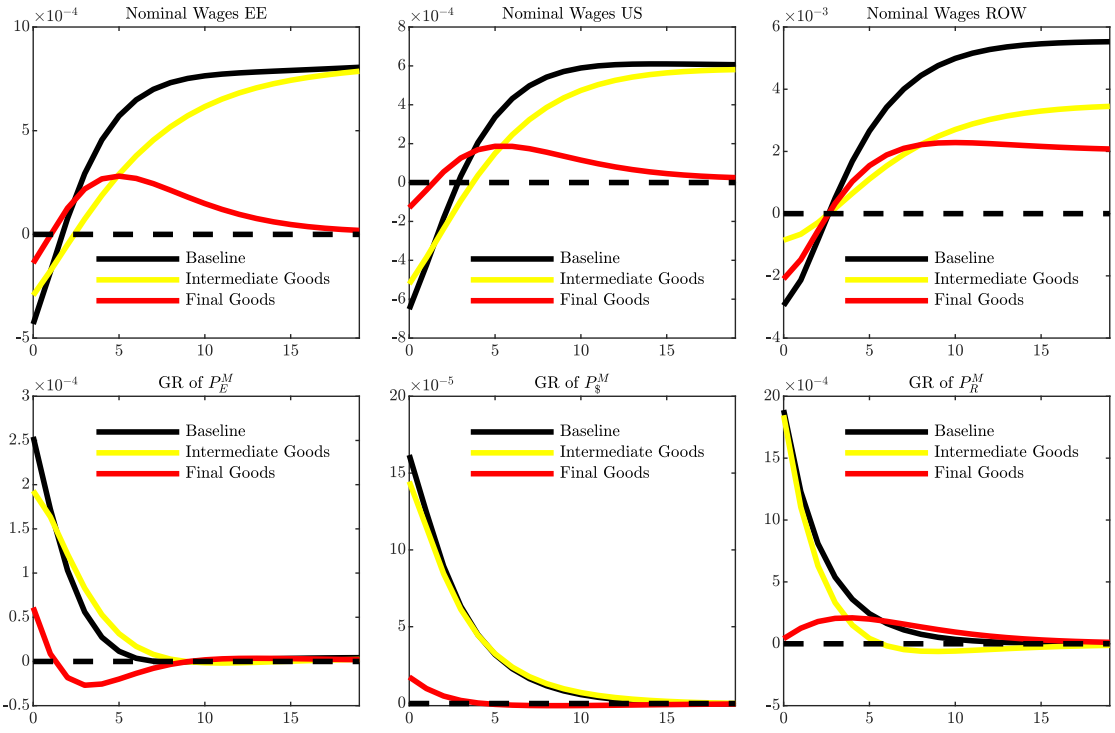


Figure 17: Impulse Responses of Factors of Production across all blocs under DCP to 1 s.d. Global Inflation Shock across different types of goods. The panels contain the following variables (a) Nominal Wages in EE; (b) Nominal Wages in US; (c) Nominal Wages in ROW; (d) Growth Rate of cost of composite intermediate goods in EE (P^M_E); (e) Growth Rate of cost of composite intermediate goods in US ($P^M_\$$); (f) Growth Rate of cost of composite intermediate goods in EE (P^M_R);

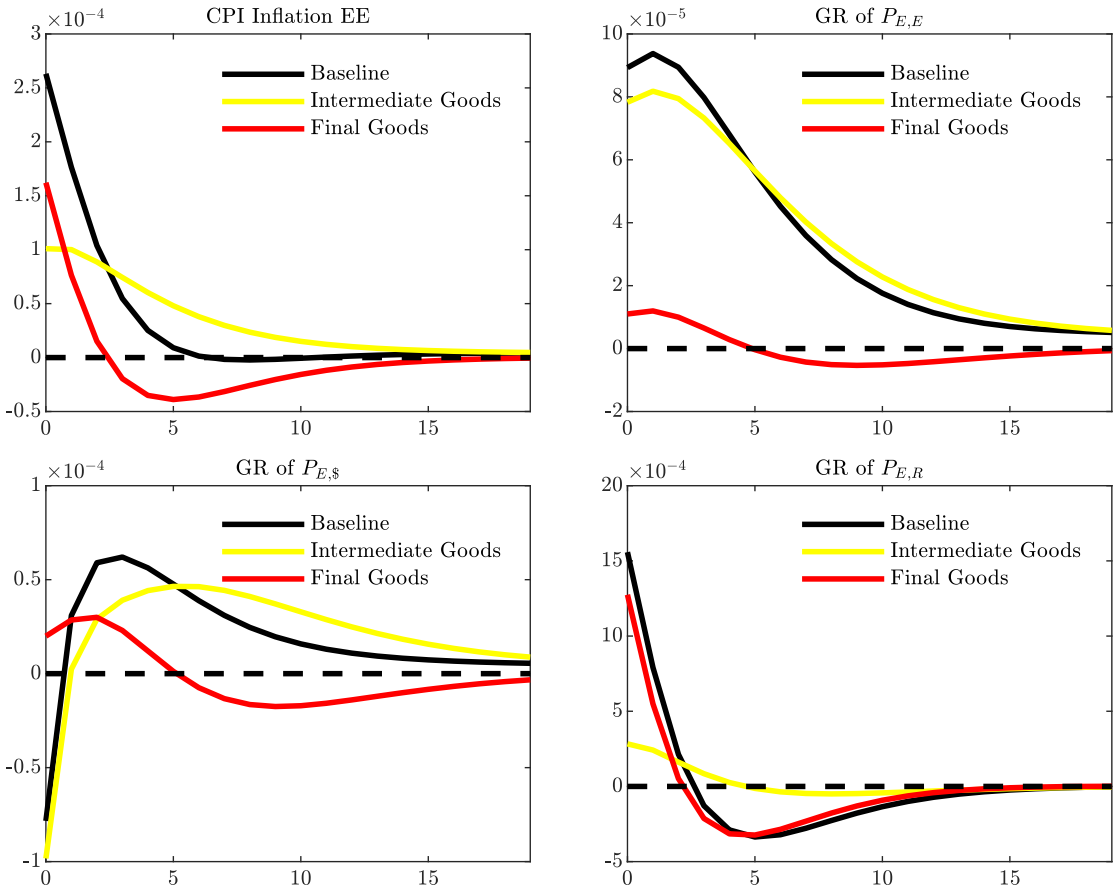


Figure 18: Impulse Responses of CPI Inflation and its sub-components under Dominant Currency Pricing to 1 s.d. Global Inflation Shock across different types of goods. The panels contain the following variables (a) CPI Inflation in EE (b) Growth rate of retail price of final goods produced and consumed in the EE (c) Growth rate of retail price of final goods produced in the US for the EE (d) Growth rate of retail price of final goods produced in the ROW for the EE.

5.3 Intermediate and Final Goods

This section examines the contributions of intermediate and final goods to CPI inflation in EE. In the baseline scenario, the global inflation shock affects both intermediate and final goods simultaneously. To isolate the impact across each type of good, two additional model scenarios are constructed and analyzed. Under the final good model scenario, the global inflation shock affects only the firms that assemble the final goods, while the intermediate goods model scenario captures the impact of the global inflation shock on firms involved in the production of intermediate goods.

The rationale for studying the transmission of the global inflation shock in isolation

for different types of goods and comparing it with the baseline scenario stems from the different impact of the same shock on intermediate goods and final goods (Nakamura and Steinsson (2010); Bems et al. (2011)). Intermediate goods have been identified in the literature as a factor that affects both ERPT and movements in exchange rates. Therefore, this exercise provides insights into the dynamics of a global inflation shock that affects only the intermediate goods firms and compares it with model scenarios in which the shock affects only the final goods firms, and in which the shock affects both types of firms.

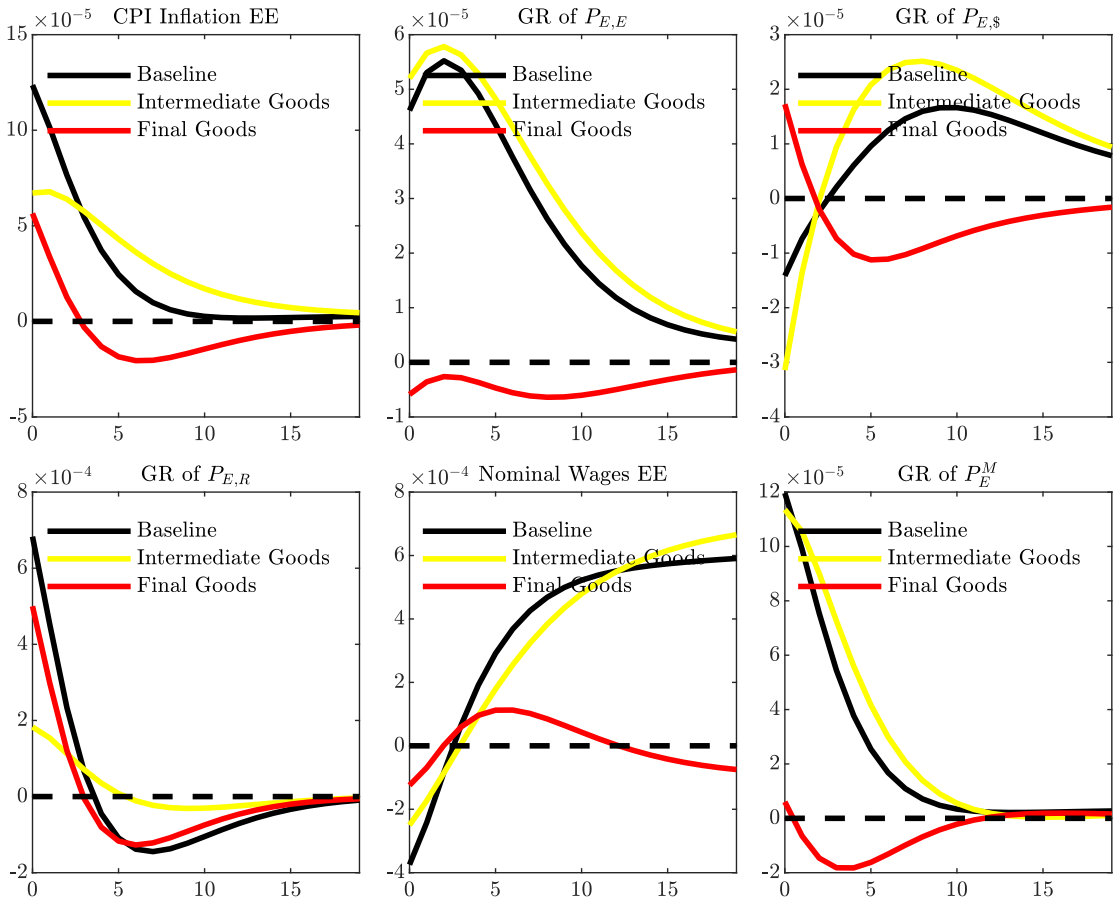


Figure 19: Impulse Responses of CPI Inflation and its sub-components under Local Currency Pricing to 1 s.d. Global Inflation Shock across different types of goods. The panels contain the following variables (a) CPI Inflation in EE (b) Growth rate of retail price of final goods produced and consumed in the EE (c) Growth rate of retail price of final goods produced in the US for the EE (d) Growth rate of retail price of final goods produced in the ROW for the EE (e) Nominal Wages in EE. (f) Growth rate of cost of composite intermediate goods in EE

The two overarching results observed in the baseline scenario hold true across both the intermediate goods and final goods models. The opposing price pressures emanating from the factors of production hold across all blocs and all currency invoicing regimes for the additional model scenarios introduced in this subsection (see Figure 17). The relevant nominal exchange rate channel plays an important role in the divergent transmission of the global inflation shock under the intermediate goods model relative to the final goods and baseline models.

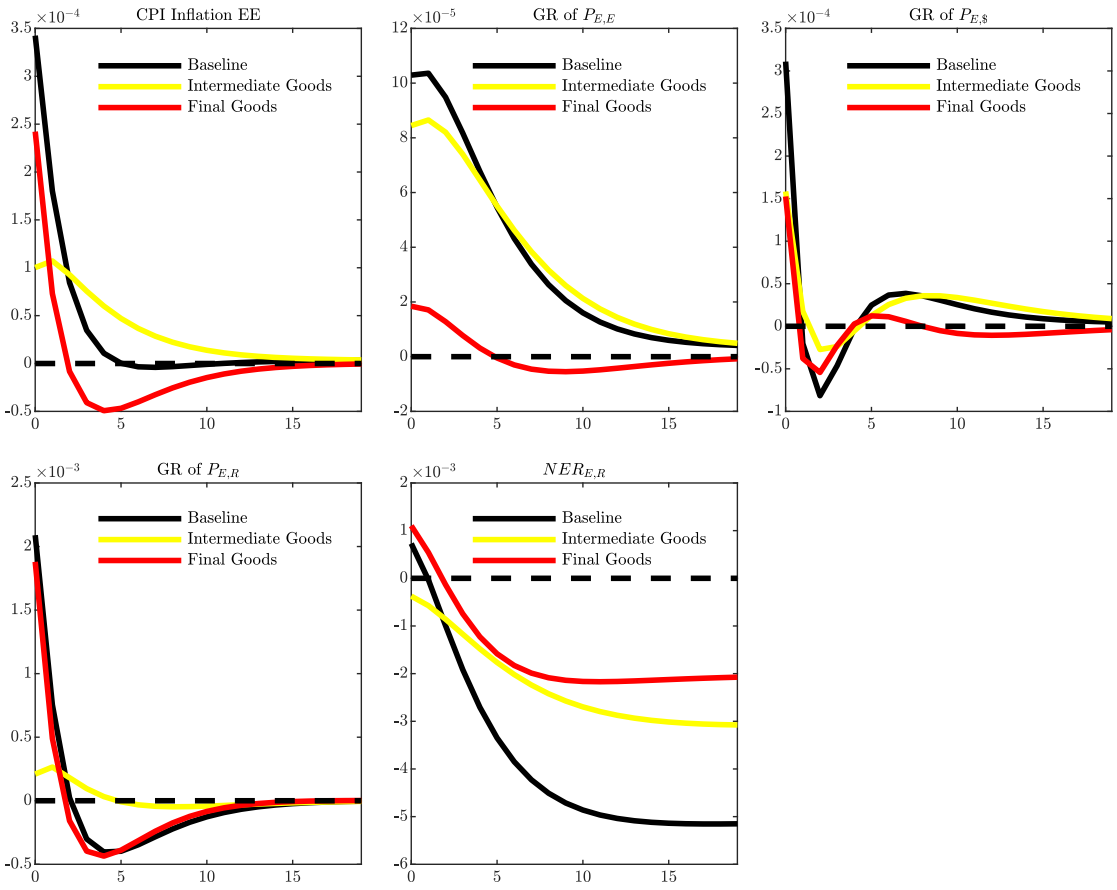


Figure 20: Impulse Responses of CPI Inflation and its sub-components under Producer Currency Pricing to 1 s.d. Global Inflation Shock across different types of goods. The panels contain the following variables (a) CPI Inflation in EE (b) Growth rate of retail price of final goods produced and consumed in the EE (c) Growth rate of retail price of final goods produced in the US for the EE (d) Growth rate of retail price of final goods produced in the ROW for the EE (e) Nominal Exchange Rate between the ROW and the EE.

Under LCP, the largest increase in π_e is observed in the baseline scenario, followed by the intermediate goods model scenario, and lastly by the final goods model scenario (see

Figure 19 (a)). This is mainly due to the largest increase in $P_{E,E}$ under the intermediate goods model relative to the other two model scenarios (see Figure 19 (b)). Under the intermediate goods model, wages in EE fall by less relative to the baseline model, while the cost of composite intermediate goods is only marginally lower than under the baseline model, together, these effects lead to a greater increase in the retail price of the final goods produced and consumed in EE.

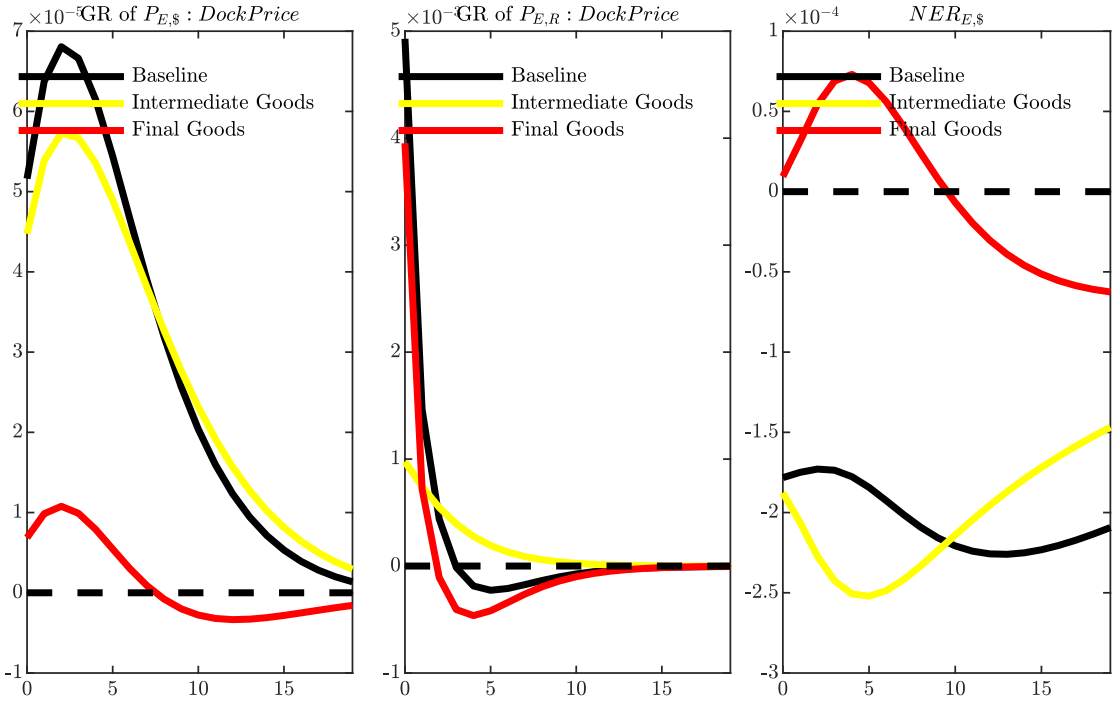


Figure 21: Impulse Responses of Dock Prices and Exchange Rates under Dominant Currency Pricing to 1 s.d. Global Inflation Shock across different types of goods. The panels contain the following variables (a) Growth rate of dock price of final produced in the US for the EE ($P_{E,\$}$: $DockPrice$) (b) Growth rate of dock price of final produced in the ROW for the EE ($P_{E,R}$: $DockPrice$) (c) Nominal Exchange rate between the US and the EE.

The ordering of CPI inflation across the model scenarios is identical under DCP and PCP, with π_e increasing the most under the baseline model scenario, followed by the final goods model scenario and the intermediate goods model scenario (see Figure 18 (a)). The lowest increase in the CPI inflation in EE for the intermediate goods model under DCP is due to the largest fall in $P_{E,\$}$ and the smallest increase in $P_{E,R}$ across all three model scenarios (see Figures 18 (c) and (d)). Under PCP, however, the lowest increase in CPI inflation in EE is due only to the smallest increase in $P_{E,R}$ across all

three model scenarios.

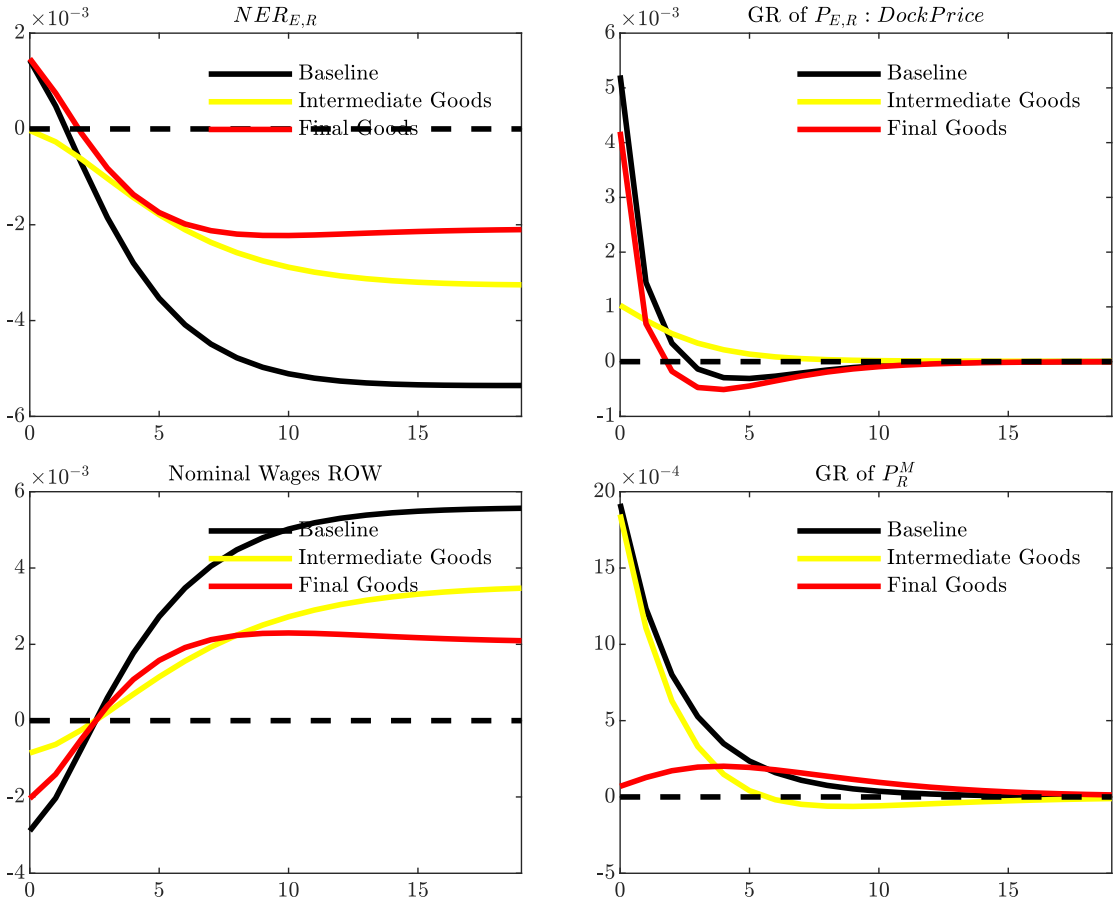


Figure 22: Impulse Responses of Variables of interest under Local Currency Pricing to 1 s.d. Global Inflation Shock across different types of goods. The panels contain the following variables (a) Nominal Exchange Rate between the ROW and the EE ($NER_{E,R}$) (b) Growth rate of dock price of final produced in the ROW for the EE ($P_{E,R}$: $DockPrice$) (c) Nominal Wages in ROW (d) Growth rate of cost of composite intermediate goods in the ROW (P_R^M).

The largest fall in $P_{E,\$}$ and the smallest increase in $P_{E,R}$ for the intermediate goods model under DCP are due to the relevant nominal exchange rate channel. Both $P_{E,\$}^{imp}$ and $P_{E,R}^{imp}$ increase following the global inflation shock (see Figure 21 (a) and (b)). However, the EE's currency simultaneously appreciates the most against the US currency under the intermediate goods model. This exerts downward pressure on the retail price of final goods produced in both the US and the ROW for the EE, leading to a fall in $P_{E,\$}$ and the lowest increase in $P_{E,R}$.

Under PCP, the smallest increase in $P_{E,R}$ for the intermediate goods model stems from

the relevant nominal exchange rate channel. The EE's currency appreciates against the ROW currency under the intermediate goods model, in contrast to the depreciation of the EE's currency under the other two models. The appreciation of the EE's currency under the intermediate goods model exerts downward pressure on $P_{E,R}$, as retailers in the EE are required to pay less, in the EE's currency, to import goods from the ROW.

The transmission of the global inflation shock via the relevant nominal exchange rate channel can also be seen in $P^{imp}_{E,R}$ for the baseline and intermediate goods model scenarios under LCP. The cost of composite intermediate goods in the ROW increases by a similar magnitude on impact under both the baseline and intermediate goods model scenarios, whereas wages in the ROW fall more under the baseline scenario. Even so, $P^{imp}_{E,R}$ increases by less under the intermediate goods model (see Figure 22 (b)). This highlights the downward price pressure stemming from the appreciation of the EE's currency against the ROW currency under the intermediate goods model.

5.4 Trade Openness

Trade openness is another factor that affects ERPT, with the feedback between the two variables being described as multifaceted (Ha et al., 2020). The literature provides evidence of an increase in trade openness being associated with both higher and lower ERPT. The idea behind higher ERPT due to greater trade openness works through the overall composition of foreign goods in domestic markets. The greater the share of foreign goods, the more domestic prices will be affected by movements in exchange rates. Conversely, lower ERPT with greater trade openness is linked to an increase in competition among firms that cater to domestic markets. An increase in the number of competitors results in greater absorption of exchange rate movements by adjusting the markups instead of changing retail prices.

This section focuses on the impact of trade openness on CPI inflation in EE. Across each of the three currency invoicing regimes, the analysis focuses on comparing the two additional scenarios of increased imports from the US and from the ROW with the baseline scenario. The value of the parameter governing the consumption of final goods in EE is increased by 10 points from each of the two remaining blocs for the construction of the two additional scenarios. An analysis focused on the increase in exports from EE to these blocs can be found in the appendix. This exercise allows for an examination of

the impact of trade openness on CPI inflation in EE across different country blocs and currency invoicing regimes, and for both types of trade openness, imports and exports.

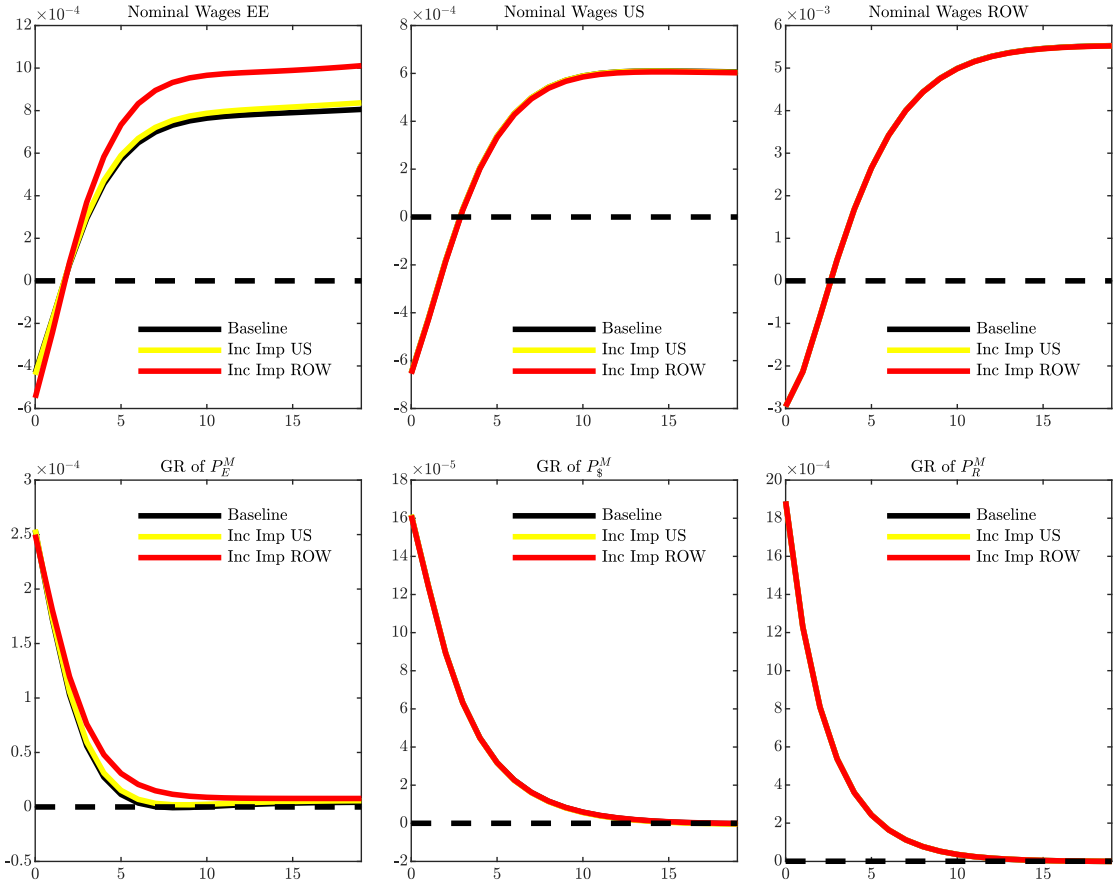


Figure 23: Impulse Responses of Factors of Production under Dominant Currency Pricing to 1 s.d. Global Inflation Shock across the baseline scenario, and increased imports from the US and the ROW. The panels contain the following variables (a) Nominal Wages in EE (b) Nominal Wages in US (c) Nominal Wages in ROW (d) Growth rate of cost of composite intermediate goods in the EE (P_E^M) (e) Growth rate of cost of composite intermediate goods in the US ($P_\$^M$) (f) Growth rate of cost of composite intermediate goods in the ROW (P_R^M).

Opposing price pressures emanating from the factors of production is evident across all model scenarios and currency invoicing regimes, including the model scenarios where exports are increased from the US and the ROW. The price pressures from the two factors of production are identical to those observed in previous sections (see Fifure 23). The cost of composite intermediate goods exerts upward price pressure, while wages exert downward price pressure across all three blocs. However, unique to the analysis of this subsection, the cost of composite intermediate goods and wages in the US and

the ROW follow identical magnitudes and trajectories across all model scenarios and currency invoicing regimes. This makes it possible to isolate the impact of the relevant nominal exchange rate channel on the dock and retail prices of goods, which is the primary factor responsible for the differential impact on $P_{E,\$}$ across different model scenarios.

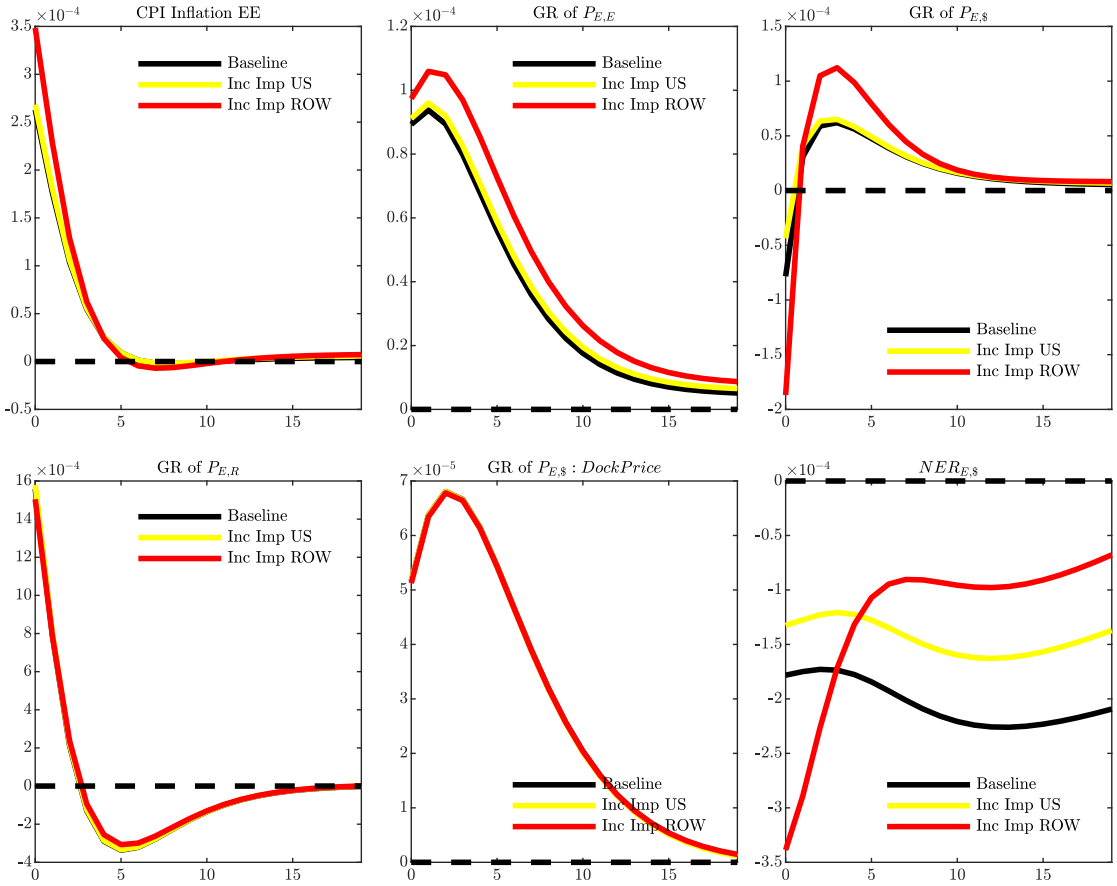


Figure 24: Impulse Responses of CPI Inflation and its sub-components under Dominant Currency Pricing to 1 s.d. Global Inflation Shock across the baseline scenario, and increased imports from the US and the ROW. The panels contain the following variables (a) CPI Inflation in EE (b) Growth rate of retail price of final produced and consumed in the EE ($P_{E,E}$) (c) Growth rate of retail price of final produced in the US for the EE ($P_{E,\$}$) (d) Growth rate of retail price of final produced in the ROW for the EE ($P_{E,R}$) (e) Growth rate of dock price of final produced in the US for the EE ($P_{E,\$}$: $DockPrice$) (f) Nominal Exchange Rate between the US and the EE ($NER_{E,\$}$)

Under DCP, the dock price of the final goods produced in the US for the EE is identical across all model scenarios, owing to identical factors of production and the absence of any relevant nominal exchange rate channel (see Figure 24 (e)). However, at the retail

price level, the relevant nominal exchange rate for the aforementioned goods changes, with the EE's currency appreciating against the US currency. This exerts downward price pressure on the goods produced in the US for the EE. $P_{E,\$}$ falls the most under the model scenario that witnesses the largest appreciation of the EE's currency against the US currency (see Figure (c) and (f)). Similar observations hold for $P^{imp}_{E,\$}$ and $P_{E,\$}$ under PCP, with the only difference being that the EE's currency depreciates against the US currency (see Figure 16 (c) and (e)).

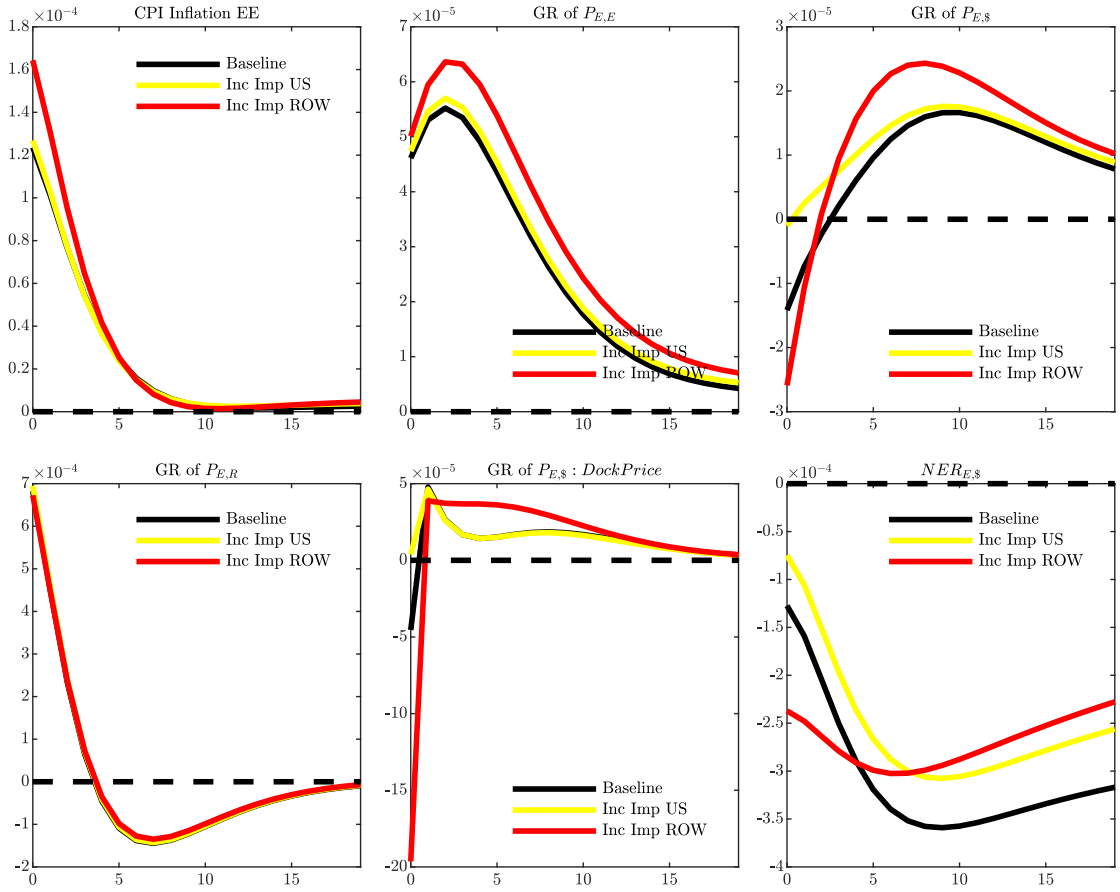


Figure 25: Impulse Responses of CPI Inflation and its sub-components under Local Currency Pricing to 1 s.d. Global Inflation Shock across the baseline scenario, and increased imports from the US and the ROW. The panels contain the following variables (a) CPI Inflation in EE (b) Growth rate of retail price of final produced and consumed in the EE ($P_{E,E}$) (c) Growth rate of retail price of final produced in the US for the EE ($P_{E,\$}$) (d) Growth rate of retail price of final produced in the ROW for the EE ($P_{E,R}$) (e) Growth rate of dock price of final produced in the US for the EE ($P_{E,\$}$: $DockPrice$) (f) Nominal Exchange Rate between the US and the EE ($NER_{E,\$}$)

Under LCP, the relevant nominal exchange rate channel is active for the dock price

of the final goods produced in the US for the EE. Since the factors of production are identical across all model scenarios, in the absence of the relevant nominal exchange rate channel, $P^{imp}_{E,\$}$ would also be identical. Thus, any variations in $P^{imp}_{E,\$}$ can be attributed to the indirect channel of the global inflation shock. $NER_{E,\$}$ is the relevant nominal exchange rate for $P^{imp}_{E,\$}$. The EE's currency appreciates the most against the US currency under the model scenario of increased from the ROW, followed by the model scenario of increased imports from the US, and lastly the baseline scenario. The ordering of the fall in $P^{imp}_{E,\$}$ follows the same pattern, with the largest fall recorded for the model scenario that exhibits the highest appreciation of the EE's currency against the US currency (see Figure 25 (e) and (f)).

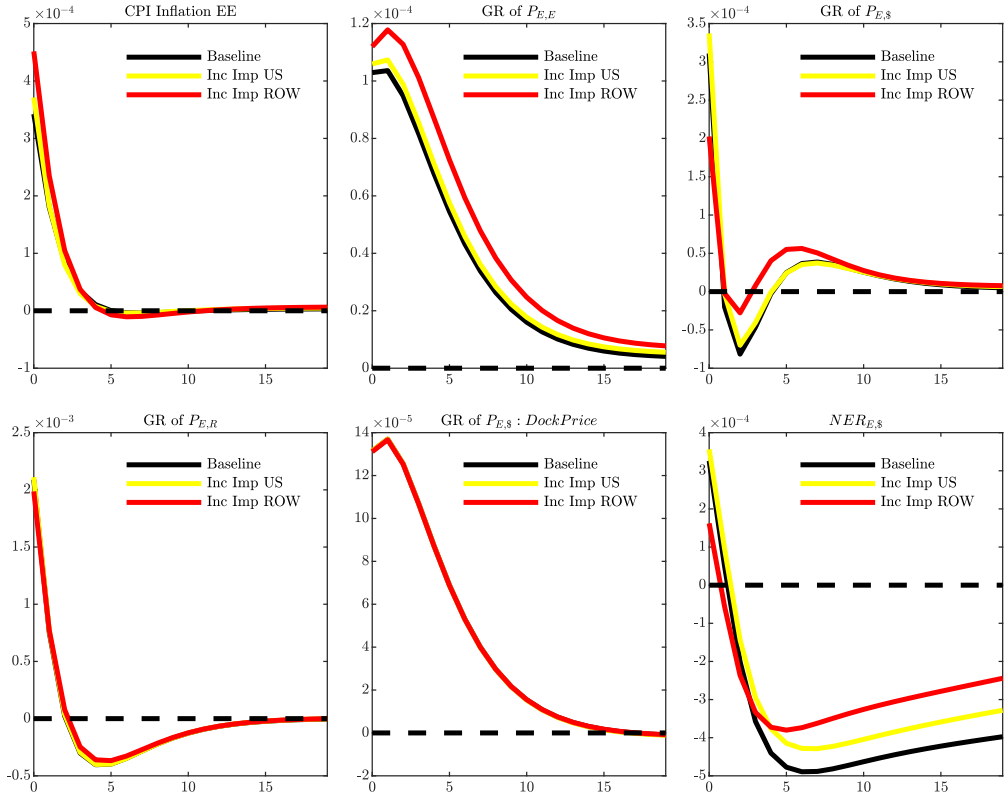


Figure 26: Impulse Responses of CPI Inflation and its sub-components under Producer Currency Pricing to 1 s.d. Global Inflation Shock across the baseline scenario, and increased imports from the US and the ROW. The panels contain the following variables (a) CPI Inflation in EE (b) Growth rate of retail price of final produced and consumed in the EE ($P_{E,E}$) (c) Growth rate of retail price of final produced in the US for the EE ($P_{E,\$}$) (d) Growth rate of retail price of final produced in the ROW for the EE ($P_{E,R}$) (e) Growth rate of dock price of final produced in the US for the EE ($P_{E,\$} : DockPrice$) (f) Nominal Exchange Rate between the US and the EE ($NER_{E,\$}$)

6 Conclusion

The primary goal of this paper is to study the transmission of a global inflation shock to CPI inflation in EE under different currency invoicing regimes. It focuses on both the direct and indirect channels of transmission of a global inflation shock. The direct channel measures the impact on the retail price of final goods arising from the increase in cost pressures stemming from a global inflation shock, while the indirect channel captures the role of the relevant nominal exchange rate in either amplifying or dampening the direct channel of the global inflation shock through changes in the exchange rate.

The two major results that address these questions, and are observed across all the model scenarios analyzed in this paper, are the opposite price pressures emanating from the factors of production and the impact of the indirect channel of the global inflation shock on CPI inflation and its sub-components. Both results hold under the baseline scenario, under different exchange rate pass-through assumptions, across different types of goods, and under greater trade openness. The first overarching result highlights the upward price pressure stemming from the cost of composite intermediate goods, while a downward price pressure is exerted owing to a fall in wages. The second overarching result provides evidence for the existence of an indirect channel of transmission of a global inflation shock, which either amplifies or dampens the effect of the global inflation shock through changes in the relevant nominal exchange rate. The impact of the indirect channel differs based on the choice of the currency invoicing regime.

In addition to the aforementioned major results, this paper also identifies several section-specific findings. The subsection on exchange rate pass-through highlights the importance of differential ERPT under DCP and PCP in producing a differential impact on the variables of interest following a global inflation shock. This is evident from the indistinguishable IRFs of CPI inflation and its sub-components under DCP and PCP following a global inflation shock, when the degree of ERPT is identical between the two currency invoicing regimes. It also finds evidence for the importance of ERPT in sustaining the indirect channel of the global inflation shock. This can be validated by looking at the results of Scenario 5 under equal ERPT, where ERPT is assumed to be zero, the two overarching results identified in the baseline scenario do not hold.

The subsection on the types of goods, namely intermediate and final goods, provides evidence for a differential impact on the variables of interest following a global inflation

shock, depending on the type of good under consideration. Lastly, the subsection on trade openness finds that greater trade openness leads to higher CPI inflation in EE. It also finds that an increase in imports into the EE from the US and the ROW has no impact on the factors of production in those economies across all model scenarios and currency invoicing regimes.

The paper is subject to certain limitations arising from its modelling assumptions. First, the three-bloc open economy model abstracts from capital as a factor of production and does not account for financial frictions. Second, it models the global inflation shock as a cost-push shock in the ROW, however, in practice, other types of shocks may also give rise to global inflationary pressures. Third, the scope of the paper is limited to the impact of the global inflation shock on CPI inflation in EE and its sub-components. The structure of the model would, however, permit the study of broader and more intricate relationships between other variables following a global inflation shock, which remains an avenue for future research.

The findings of this paper carry several implications for policymakers in emerging economies. First, the existence and direction of the indirect channel of the global inflation shock are contingent on the currency invoicing regime and the degree of ERPT. Policymakers should therefore monitor both these aspects in order to assess the true impact of the indirect channel of the global inflation shock's transmission. Second, the paper finds that wages exert downward pressure on π_E following a global inflation shock. Policies designed to leverage this channel may therefore be employed to counteract inflationary pressures, although their broader macroeconomic implications warrant further investigation.

Third, the paper highlights the differential impact on CPI inflation and its sub-components depending on the type of goods affected by the shock. This facilitates the design of targeted policies suited for specific scenarios. Fourth, greater trade openness is associated with higher CPI inflation in EE. While trade integration carries well-documented benefits, it also entails greater exposure to external inflationary pressures. A comprehensive assessment of the benefits and costs associated with greater trade openness is therefore warranted before pursuing deeper trade integration. Taken together, these results highlight the importance of currency invoicing, the degree of exchange rate pass-through, and trade structure as channels through which global inflation shocks are transmitted

to domestic prices in EE.

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A Empirical Analysis

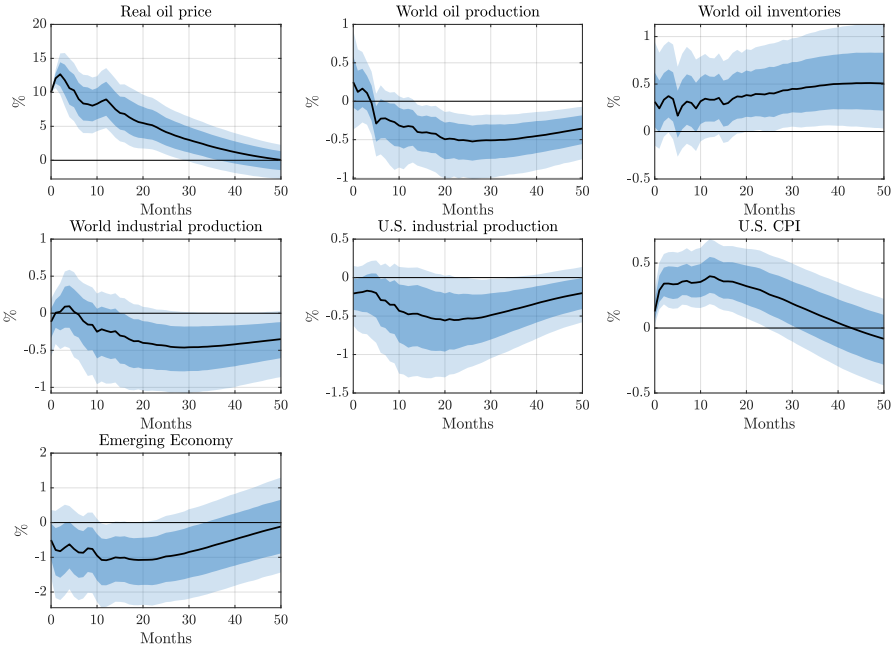


Figure 27: Impulse Responses to a Global Inflation Shock based on VAR analysis of the extended sample. The shock corresponds to a 10% increase in the real oil price. The panel in the third row is the exchange rate between US dollar and basket of emerging economies, where decline in the exchange rate means depreciation of US dollar against the emerging economies' currency

The VAR analysis and local projections are also conducted for an extended sample through M12 2024. For the VAR analysis, the direction and trajectory of all variables remain the same, with only a slight difference in magnitude (see Figure 27). The US currency experiences a slightly prolonged depreciation against emerging economies' currencies in the extended sample. However, the IRFs based on local projections for the extended sample provide different responses across variables. The response of world oil inventories becomes erratic, while world industrial production remains negative throughout the periods considered. US CPI inflation declines and becomes negative, and the US dollar starts to appreciate after around 35 periods. The appreciation of the US dollar is followed by a fall in US CPI inflation, suggesting that a strong dollar causes CPI inflation to fall in the US (see Figure 28).

The IRFs of the VAR analysis and local projections are based on the period from M12 1974 to M12 2017. The calculation of world oil inventories follows the same method as in

Känzig (2021), and is based on OECD Crude Oil Inventories, OECD Petroleum Stocks and U.S. Crude Oil and Petroleum Stocks.

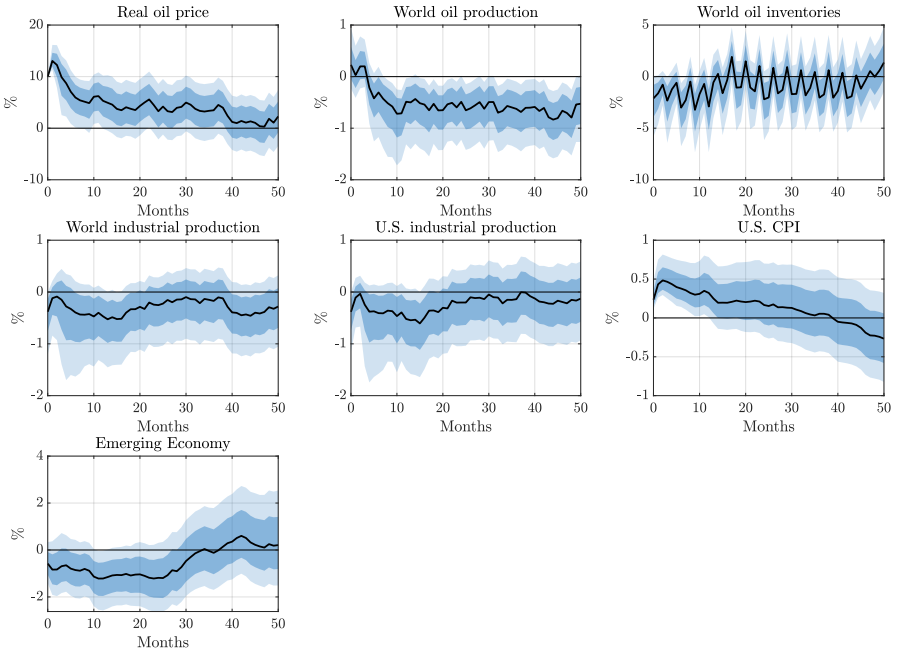


Figure 28: Impulse Responses to a Global Inflation Shock based on Local Projections of the extended sample. The shock corresponds to a 10% increase in the real oil price. The panel in the third row is the exchange rate between US dollar and basket of emerging economies, where decline in the exchange rate means depreciation of US dollar against the emerging economies' currency

The data used for the empirical analysis are as follows:

Table 6: Data Source

Variable	**Data Source**	**Time Period**
GDP based price index of G20 countries	OECD Stat	Q1 1999 to Q4 2024
US Dollar Index vis-a-vis Emerging Economies	FRED	M12 1974 to M12 2024
Oil Price	FRED	M12 1974 to M12 2024
World Oil Production	Datastream	M12 1974 to M12 2024
OECD Crude Oil Inventories	Datastream	M12 1974 to M12 2024
OECD Petroleum Stocks	Datastream	M12 1974 to M12 2024
U.S. Crude Oil and Petroleum Stocks	Datastream	M12 1974 to M12 2024
World Industrial Production	Christin Baumeister's webpage	M12 1974 to M12 2024
U.S. Industrial Production	FRED	M12 1974 to M12 2024
U.S CPI	FRED	M12 1974 to M12 2024

B Model Equations

B.1 Households

The households in country i maximize their utility subject to their budget constraint. This can be done by solving the following Lagrangian:

$$\mathcal{L} = E_t\big[\sum_{t=0}^{\infty} \beta_E^t[log(C_{i,t}) + \varrho_i log(1 - H_{i,t}) + \sum_{j=[E,\$,R]} log(\eta_{i,j,t})$$
$$V(\frac{B_{i,j,t}}{P_{i.t}} NER_{i,j,t})] + \lambda_{i,t}[w_{i,t}H_{i,t} + \sum_{j=[E,\$,R]} RER_{i,j,t} r_{j,t-1} b_{i,j,t-1} \frac{1}{\pi_{j,t-1,t}}$$
$$-[\sum_{j=[E,\$,R]\forall i \neq j} \frac{\kappa_{i,j}}{2} RER_{i,j,t}(b_{i,j,t} - b_{i,j}^{ss})^2] - C_{i,t} - \sum_{j=[E,\$,R]} RER_{i,j,t} b_{i,j,t}]\big]$$

The first-order conditions with respect to $C_{i,t}$, $b_{i,j,t-1}$, $H_{i,t}$ yield the following equations:

$$C_{i,t} : \lambda_{i,t} = \frac{1}{C_{i,t}}$$

$$H_{i,t} : -w_{i,t}\lambda_{i,t} = \frac{-\varrho_i}{1 - H_{i,t}}$$

$$b_{i,j,t-1} : \lambda_{i,t}(1 + \kappa_{i,j}(b_{i,j,t} - b_{i,j}^{ss})) = log(n_{i,j,t}) V'(\frac{B_{i,j,t}}{P_{i.t}} NER_{i,j,t})$$
$$+\beta_i r_{j,t} \lambda_{i,t+1} \Delta NER_{i,j,t,t+1} \frac{1}{\pi_{E,t,t+1}}$$

In order to satisfy the conditions outlined in the main text about $V(.)$, the following functional form is assumed:

$$V(.) = -\lambda_i^{ss} e^{-RER_{i,j,t} b_{i,j,t}}$$

B.2 Final Consumption Goods

Households consume $C_{i,t}$, which is a combination of domestic and imported goods and is defined as follows:

$$C_{i,t} = [\sum_{j=[E,\$,R]} (\omega_{i,j})^{\frac{1}{\rho_i}} (C_{i,j,t})^{\frac{\rho_i - 1}{\rho_i}}]^{\frac{\rho_i}{\rho_i - 1}}$$

where,

$$\sum_{j=[E,\$,R]} \omega_{i,j} = 1$$

Households seek to minimize total expenditure given the total consumption demand and CES preferences. This leads to the following Lagrangian problem:

$$\mathcal{L} = \sum_{j=[E,\$,R]} P_{i,j,t}(q) C_{i,j,t}(q) + P_{i,t}(C_{i,t} - [\sum_{j=[E,\$,R]} (\omega_{i,j})^{\frac{1}{\rho_i}} (C_{i,j,t})^{\frac{\rho_i - 1}{\rho_i}}]^{\frac{\rho_i}{\rho_i - 1}})]$$

First order condition with respect to $C_{i,j,t}$ yields,

$$C_{i,j,t} = \omega_{i,j} C_{i,t} (\frac{P_{i,j,t}}{P_{i,t}})^{-\rho_i}$$

Substituting this back in the equation for consumption yields,

$$1 = [\sum_{j=[E,\$,R]} \omega_{i,j}(\frac{P_{i,j,t}}{P_{i,t}})^{1-\rho_i}]^{\frac{\rho_i}{\rho_i-1}} \tag{B.1}$$

B.3 Firms

B.3.1 Wholesalers

The wholesale firms follow the following production function:

$$\sum_{j=[E,\$,R]} X_{i,j,t}(q) + \sum_{j=[E,\$,R]} X^M_{i,j,t}(q) = A_{i,t}H^{1-\alpha_i}_{i,t}(q)M^{\alpha_i}_{i,t}(q)$$

and are subject to the following maximization problem:

$$w_{i,t}H_{i,t}(q) + p^M_{i,t}M_{i,t}(q) + mc_{i,t}[\sum_{j=[E,\$,R]} X_{i,j,t}(q) + \sum_{j=[E,\$,R]} X^M_{i,j,t}(q) - A_{i,t}H^{1-\alpha_i}_{i,t}(q)M^{\alpha_i}_{i,t}(q)]$$

In a symmetric equilibrium, firms follow the same price, the same inputs, and the same output (Landi (2021)), this provides the following first-order conditions with respect to $H_{i,t}$ and $M_{i,t}$:

$$H_{i,t} : w_{i,t} = (1-\alpha_i)mc_{i,t}A_{i,t}(\frac{h_i}{m_i})^{-\alpha_i}$$

$$M_{i,t} : p^M_{E,t} = \alpha_i mc_{i,t}A_{i,t}(\frac{h_i}{m_i})^{1-\alpha_i}$$

The wholesale firms are monopolistic and determine their optimal price by solving the following Calvo problem:

$$E_t \sum_{k=0}^{\infty} \Lambda_{j,t,t+k} (\xi_{i,j}^{imp})^k [\frac{P_{i,j,t}(i)}{P_{j,t+k}} - MC_{j,t+k}] X_{i,j,t+k}(q)$$

Solving the Calvo function provides the following function:

$$p_{i,j}^* = \frac{\theta_{i,j}}{\theta_{i,j} - 1} \frac{J_{i,j,t}^{imp}}{JJ_{i,j,t}^{imp}}$$

where, for goods that are exported and are invoiced in US dollar, following equations hold

$$J_{i,j,t} = mc_{j,t} x_{i,j,t} + \xi_{i,j} \beta_j \frac{\lambda_{j,t+1}}{\lambda_{j,t}} (\frac{q_{j,\$,t+1}}{q_{j,\$,t}} \frac{p_{i,j,t+1}^{imp}}{p_{i,j,t}^{imp}} \pi_{j,t,t+1})^{\theta_{i,j}^{imp}} J_{i,j,t+1}$$

$$JJ_{i,j,t} = x_{i,j,t} + \xi_{i,j} \beta_j \frac{\lambda_{j,t+1}}{\lambda_{j,t}} (\frac{q_{j,\$,t+1}}{q_{j,\$,t}} \frac{p_{i,j,t+1}^{imp}}{p_{i,j,t}^{imp}} \pi_{j,t,t+1})^{\theta_{i,j}^{imp}} \frac{1}{\pi_{j,t,t+1}} JJ_{i,j,t+1}$$

While for the goods sold domestically, following equations hold:

$$p_{i,j}^* = \frac{\theta_{i,j}}{\theta_{i,j} - 1} \frac{J_{i,j,t}}{JJ_{i,j,t}}$$

$$J_{i,i,t} = mc_{i,t} C_{i,i,t} + \xi_{i,i} \beta_i \frac{\lambda_{i,t+1}}{\lambda_{i,t}} (\frac{p_{i,i,t+1}}{p_{i,i,t}} \pi_{i,t,t+1})^{\theta_{i,i}} J_{i,i,t+1}$$

$$JJ_{i,i,t} = C_{i,i,t} + \xi_{i,i}\beta_i \frac{\lambda_{i,t+1}}{\lambda_{i,t}} (\frac{p_{i,i,t+1}}{p_{i,i,t}} \pi_{i,t,t+1})^{\theta_{i,i}} \frac{1}{\pi_{i,t,t+1}} J_{i,i,t+1}$$

Similar equations hold for the intermediate goods firm.

B.3.2 Production Firm

The composite intermediate goods combined by the production firm is defined as follows:

$$M_{i,t} = [\sum_{j=[E,\$,R]} (m_{i,j})^{\frac{1}{z_i}} (X^M_{i,j,t})^{\frac{z_i-1}{z_i}}]^{\frac{z_i}{z_i-1}}$$

Similar to the final consumption good, the production firm is subject to the following minimization problem:

$$\mathcal{L} = \sum_{j=[E,\$,R]} P^M_{i,j,t}(r) X^M_{i,j,t}(r) + P^M_{i,t}(M_{i,t} - [\sum_{j=[E,\$,R]} (m_{i,j})^{\frac{1}{x_i}} (X^M_{i,j,t})^{\frac{x_i-1}{x_i}}]^{\frac{x_i}{x_i-1}})]$$

The first order condition w.r.t $X^M_{i,j,t}$ yields the following equation:

$$X^M_{i,j,t} = m_{i,j} M_{i,t} (\frac{p^M_{i,j,t}}{p^M_{i,t}})^{-x_i}$$

Substituting this back in the equation for composite intermediate goods provide the following equation:

$$\frac{P^M_{i,t}}{P_{i,t}} = [\sum_{j=[E,\$,R]} m_{i,j} (\frac{P^M_{i,j,t}}{P_{i,t}})^{1-z_i}]^{\frac{1}{1-z_i}}$$

B.3.3 Retailers

The goods assembled by retailers produced in other countries are defined as follows:

$$C_{i,j,t} = [\int_0^1 (X_{i,j,t}^{imp}(r))^{\frac{\theta_{i,j}-1}{\theta_{i,j}}}]^{\frac{\theta_{i,j}}{\theta_{i,j}-1}}$$

Retailers are monopolistic and control the exchange rate pass through by solving the following dynamic problem:

$$E_t \sum_{k=0}^{\infty} \Lambda_{i,t,t+k} (\xi_{i,j})^k [\frac{P_{i,j,t}(i)}{P_{i,t+k}} - \frac{NER_{i,j,t+k} P_{i,j,t+k}^{imp}}{P_{i,t+k}}] X_{i,j,t+k}^{imp}(r)$$

This yields to the following optimal price:

$$p_{i,j}^* = \frac{\theta_{i,j}}{\theta_{i,j}-1} \frac{J_{i,j,t}}{JJ_{i,j,t}}$$

where,

$$J_{i,j,t} = C_{i,j,t} p_{i,j,t}^{imp} + \xi_{i,j} \beta_i \frac{\lambda_{i,t+1}}{\lambda_{i,t}} (\frac{p_{i,j,t+1}}{p_{i,j,t}} \pi_{i,t,t+1})^{\theta_{i,j}} J_{i,j,t+1}$$

$$JJ_{i,j,t} = C_{i,j,t} + \xi_{i,j} \beta_i \frac{\lambda_{i,t+1}}{\lambda_{i,t}} (\frac{p_{i,j,t+1}}{p_{i,j,t}} \pi_{i,t,t+1})^{\theta_{i,j}} \frac{1}{\pi_{i,t,t+1}} JJ_{i,j,t+1}$$

B.4 Aggregation

Wholesalers production function under symmetric equilibrium is as follows:

$$\sum_{j=[E,\$,R]} X_{i,j,t}(q) + \sum_{j=[E,\$,R]} X^M_{i,j,t}(q) = A_{i,t} H^{1-\alpha_i}_{i,t} M^{\alpha_i}_{i,t}$$

From final goods firm and exporters we know that,

$$X_{i,i,t}(q) = (\frac{P_{i,i,t}(q)}{P_{i,i,t}})^{\theta_{i,i}} C_{i,i,t}$$

$$X_{i,j,t}(q) = (\frac{P^{imp}_{i,j,t}(q)}{NER_{j,\$,t} P_{i,j,t}})^{-\theta^{imp}_{i,j}} X^{imp}_{i,j,t}$$

Similar equations hold for the intermediate goods, and substituting them back into the wholesalers production functions, yields the following equation:

$$\begin{aligned} A_{i,t}(H_{i,t})^{1-\alpha_i}(K_{i,t-1})^{\alpha_i} = C_{i,i,t} PD_{i,i,t} + \sum_{j=[E,\$,R] \forall i \neq j} PD^{imp}_{j,i,t} X^{imp}_{j,i,t} \\ + X_{i,i,t} PD^M_{i,i,t} + \sum_{j=[E,\$,R] \forall i \neq j} PD^{MI}_{j,i,t} X^M_{j,i,t} \end{aligned} \tag{B.2}$$

where,

$$PD_{i,i,t} = \xi_{i,i} PD_{i,i,t-1} (\frac{p_{i,i,t}}{p_{i,i,t-1}} \pi_{i,t-1,t})^{\theta_{i,i}} + (1 - \xi_{i,i})(\frac{p^*_{i,i}}{p_{i,i,t}})^{-\theta_{i,i}}$$

$$PD^{imp}_{j,i,t} = \xi^{imp}_{j,i} PD^{imp}_{j,i,t-1} (\frac{RER_{i,\$,t}}{RER_{i,\$,t-1}} \pi_{i,t-1,t} \frac{p^{imp}_{j,i,t}}{p^{imp}_{j,i,t-1}})^{\theta^{imp}_{j,i}} + (1 - \xi^{imp}_{j,i})(\frac{p^{imp*}_{j,i}}{RER_{i,\$,t} p^{imp}_{j,i,t}})^{\theta^{imp}_{j,i}}$$

$$PD^{M}_{i,i,t} = \xi^{M}_{i,i} PD^{M}_{i,i,t-1} (\frac{p^{M}_{i,i,t}}{p^{M}_{i,i,t-1}} \pi_{i,t-1,t})^{\theta^{M}_{i,i}} + (1 - \xi^{M}_{i,i})(\frac{p^{M*}_{i,i}}{p^{M}_{i,i,t}})^{-\theta^{M}_{i,i}}$$

$$PD^{MI}_{j,i,t} = \xi^{MI}_{j,i} PD^{MI}_{j,i,t-1} (\frac{RER_{i,\$,t}}{RER_{i,\$,t-1}} \pi_{i,t-1,t} \frac{p^{MI}_{j,i,t}}{p^{MI}_{j,i,t-1}})^{\theta^{MI}_{j,i}} + (1 - \xi^{MI}_{j,i})(\frac{p^{MI*}_{j,i}}{RER_{i,\$,t} p^{MI}_{j,i,t}})^{-\theta^{MI}_{j,i}}$$

C Calibration

The share of domestic and imported intermediate and final goods is calculated using the Asian Development Bank's multiregional input-output tables. These data are available on an annual basis since 2007 for 62 countries and the rest of the world across 35 sectors. For the purpose of this analysis, emerging economies that form part of the MSCI Emerging Markets Index and for which data are available are considered. These include Brazil, China, Czech Republic, Greece, Hungary, India, Indonesia, Republic of Korea, Malaysia, Mexico, Poland, Philippines, Taiwan, Thailand, and Turkey.

Table 7: Share of Intermediate Goods Consumed

Country	**Domestically Produced**	**Produced in US**	**Produced in ROW**
Brazil	86.93	2.1	10.97
China	93.3	0.53	6.16
Czech Republic	68.12	0.75	31.13
Greece	74.21	1.39	24.4
Hungary	56.93	1.48	41.59
India	84.9	1.24	13.86
Indonesia	80.73	0.94	18.33
Malaysia	71.69	2.45	25.87
Mexico	74.12	13.17	12.71
Philippines	75.9	2.24	21.86
Poland	73.95	0.75	25.31
South Korea	81.75	2.1	16.14
Taiwan	72.89	2.39	24.72
Thailand	76.34	1.68	21.98
Turkey	79.48	1.05	19.47

Table 8: Value of Calibrated Parameters

Variables	Description	Value
$\rho_E, \rho_\$, \rho_R$	Consumption Elasticity of Substitution	1.5
$z_E, z_\$, z_R$	Intermediate Elasticity of Substitution	1.5
$\alpha_E, \alpha_\$, \alpha_R$	Share of Intermediate Goods	0.7
$\beta_E, \beta_\$, \beta_R$	Discount Factor	0.99
$\sigma_E, \sigma_\$, \sigma_R$	Interest Rate Smoothing	0.7
$\theta_{\pi_E}, \theta_{\pi_\$}, \theta_{\pi_R}$	Feedback from Inflation	2
$\theta_{GDP_E}, \theta_{GDP_\$}, \theta_{GDP_R}$	Feedback from Output	0.1
ρ_i	Persistence of AR 1 Process	0.7
$\theta_{i,j}$	Elasticity of Substitution (Final Goods)	6
$\theta_{i,j}^{imp}$	Elasticity of Substitution (Exported Final Goods)	6
$\theta_{i,j}^{M}$	Elasticity of Substitution (Intermediate Goods)	6
$\theta_{i,j}^{MI}$	Elasticity of Substitution (Exported Intermediate Goods)	6
$\xi_{i,i}$	Calvo Parameter (Final Goods)	0.75
$\xi_{i,j}^{imp}$	Calvo Parameter (Exported Final Goods)	0.75
$\xi_{i,i}^{M}$	Calvo Parameter (Intermediate Goods)	0.75
$\xi_{i,j}^{MI}$	Calvo Parameter (Exported Intermediate Goods)	0.75
$D_{E,\$}, D_{E,R}$	Steady State Debt Levels (EE)	-0.05
$D_{\$,E}, D_{\$,R}$	Steady State Debt Levels (US)	0.64
$D_{R,\$}, D_{R,E}$	Steady State Debt Levels (ROW)	0.025
$\kappa_{E,\$}, \kappa_{R,\$}$	Portfolio Adjustment Costs US Bonds	0.001
$\kappa_{E,R}, \kappa_{\$,R}$	Portfolio Adjustment Costs ROW Bonds	0.01
$\kappa_{\$,E}, \kappa_{R,E}$	Portfolio Adjustment Costs EE Bonds	0.01
$\omega_{E,E}$	Proportion of Final Goods produced and consumed in EE	0.76
$\omega_{E,\$}$	Proportion of Final Goods produced in US for EE	0.02
$\omega_{E,R}$	Proportion of Final Goods produced in ROW for EE	0.22
$\omega_{\$,\$}$	Proportion of Final Goods produced and consumed in US	0.786
$\omega_{\$,E}$	Proportion of Final Goods produced in EE for US	0.007
$\omega_{\$,R}$	Proportion of Final Goods produced in ROW for US	0.207
$\omega_{R,R}$	Proportion of Final Goods produced and consumed in ROW	0.977
$\omega_{R,E}$	Proportion of Final Goods produced in EE for ROW	0.0055
$\omega_{R,\$}$	Proportion of Final Goods produced in US for ROW	0.0175
$m_{E,E}$	Proportion of Int Goods produced and consumed in EE	0.77
$m_{E,\$}$	Proportion of Int Goods produced in US for EE	0.02
$m_{E,R}$	Proportion of Int Goods produced in ROW for EE	0.21
$m_{\$,\$}$	Proportion of Int Goods produced and consumed in US	0.886

Table 9: Value of Calibrated Parameters

Variables	Description	Value
$m_{\$,E}$	Proportion of Int Goods produced in EE for US	0.002
$m_{\$,R}$	Proportion of Int Goods produced in ROW for US	0.112
$m_{R,R}$	Proportion of Int Goods produced and consumed in ROW	0.97
$m_{R,E}$	Proportion of Int Goods produced in EE for ROW	0.01
$m_{R,\$}$	Proportion of Intermediate Goods produced in US for ROW	0.02
$\xi_{\$,R}, \xi_{\$,E}$	ERPT parameter for Final Goods imported in US	0.75
$\xi_{E,\$}, \xi_{R,\$}$	ERPT parameter for Final Goods produced in US	0.25
$\xi_{E,R}$	ERPT parameter for Final Goods produced in ROW for EE	0.5
$\xi_{R,E}$	ERPT parameter for Final Goods produced in EE for ROW	0.5
$\xi^M_{\$,R}, \xi^M_{\$,E}$	ERPT parameter for Int Goods imported in US	0.75
$\xi^M_{E,\$}, \xi^M_{R,\$}$	ERPT parameter for Int Goods produced in US	0.25
$\xi^M_{E,R}$	ERPT parameter for Int Goods produced in ROW for EE	0.5
$\xi^M_{R,E}$	ERPT parameter for Int Goods produced in EE for ROW	0.5

C.1 Intermediate Goods

The structure of the multiregional input-output table shows the intermediate goods used by an industry in a country, indicating whether they are produced domestically or imported from any of the other 61 countries and the rest of the world. This allows for the calculation of each industry's intermediate inputs into those produced domestically, those produced in the US, and those produced in the rest of the world. To calculate these shares at the country level, an average is taken across all 35 industries. This is done for each year for all 15 countries mentioned above, and the average across years and countries provides the final value of the intermediate goods share for emerging economies used for the purpose of calibration. In a similar manner, the final shares for the US and the ROW are calculated.

C.2 Final Goods

Similar to the intermediate goods, the multiregional input-output table also provides information on final consumption in a country from each industry across all 61 countries and the rest of the world. The share of final consumption from a given country is calculated by averaging across all industries. The final figures for the share of final consumption goods produced and consumed domestically, final goods produced in the US for EE, and final goods produced in the ROW for EE are calculated by averaging

Table 10: Share of Final Goods Consumed

Country	Domestically Produced	Produced in US	Produced in ROW
Brazil	93.49	0.91	5.59
China	81.65	1.18	17.16
Czech Republic	64.46	0.78	34.75
Greece	71.62	0.83	27.55
Hungary	56.98	0.84	42.18
India	86.4	1.36	12.24
Indonesia	80.05	0.55	19.40
Malaysia	74.87	1.86	23.27
Mexico	73.38	8.92	17.70
Philippines	78.46	1.40	20.14
Poland	73	0.76	26.24
South Korea	71.06	2.54	26.40
Taiwan	73.43	2.05	24.52
Thailand	74.15	1.54	24.31
Turkey	84.14	0.48	15.38

across every year and each of the aforementioned countries. In a similar manner, the final shares for the US and the ROW are calculated.

D Relevant Nominal Exchange Rate Channel in the baseline scenario

D.1 Dominant Currency Pricing

$\mathbf{P_{E,\$}}$: Retail Price

The retail price of the final goods produced in the US declines on impact but rises thereafter, increasing more than its steady-state value. The nominal exchange rate between the US and the EE becomes relevant for the retail price of the final goods, as the invoicing currency of the dock price differs from the currency of the consumer country. The appreciation of EE's currency against the US currency translates into downward price pressure emanating from the nominal exchange rate channel. This yields a decline in the retail price even as the dock price is increasing. Appreciation of EE's currency against the US currency makes the final goods cheaper to import for retailers in the EE, as they are now required to pay a lower amount in EE currency. Thus, the nominal exchange rate channel overpowers the direct impact of the global inflation shock upon impact.

$\mathbf{P_{E,R}}$: Dock Price

Analogous to the price pressures arising from the cost of composite intermediate goods and wages in the EE and the US, the dock price of the final goods produced in the ROW for the EE increases following the global inflation shock. The intensity of the increase in the dock price is strengthened on account of the appreciation of the ROW currency against the US currency. The nominal exchange rate between the US and the ROW becomes relevant because of the difference between the invoicing currency and the currency of the producer country. Appreciation of the ROW's currency against the US currency exerts upward price pressure on the dock price, as exporters earn less in ROW currency for every unit of US currency.

$\mathbf{P_{E,R}}$: Retail Price

The retail price of the final goods produced in the ROW for the EE increases following the global inflation shock and is determined based on its dock price and the nominal exchange rate between the US and the EE. The nominal exchange rate becomes relevant as the invoicing currency differs from the currency of the consumer country. The dock price exerts upward pressure, while the nominal exchange rate channel imposes downward pressure on the retail price of the final goods produced in the ROW for the EE. The downward price pressure of the nominal exchange rate stems from the appreciation of the EE's currency against the US currency, as it becomes cheaper for retailers in the EE to import final goods produced in the ROW.

D.2 Producer Currency Pricing

$\mathbf{P_{E,\$}}$: Retail Price

The retail price of the final goods produced in the US for the EE rises on account of an increase in its dock price and depreciation of EE's currency against the US currency, due to the invoicing currency differing from the currency of the consumer country. This is in contrast to DCP, where the retail price of the final goods produced in the US for the EE falls even as the dock price is increasing, due to the downward pressure stemming from the nominal exchange rate channel.

$\mathbf{P_{E,R}}$: Retail Price

Similar upward price pressure stemming from the relevant nominal exchange rate channel accentuates the retail price of the final goods produced in the ROW for the EE. The

divergence between the invoicing currency and the currency of the consumer country, combined with the depreciation of the consumer country's currency against the invoicing currency, contributes to the observed outcome.

D.3 Local Currency Pricing

$\mathbf{P_{E,\$}}$: Dock Price

Unlike under DCP, the dock price of the final goods produced in the US for the EE falls due to the downward pressure stemming from the nominal exchange rate channel. With the goods invoiced in the currency of the consumer country, the invoicing currency differs from the currency of the producer country. The appreciation of the EE's currency against the US currency after the global inflation shock translates into more US currency for exporters in the US for every unit of EE currency. This allows exporters in the US to lower prices to increase the sale of their goods without a decline in profits.

$\mathbf{P_{E,R}}$: Dock Price

The dock price of the final goods produced in the ROW for the EE increases the most across all three currency invoicing regimes under LCP, mainly due to the divergence between the invoicing currency and the currency of the producer country. Under the nominal exchange rate channel, the EE's currency depreciates against the ROW currency, which makes exporters in the ROW earn less in ROW currency for every unit of EE currency.

E Additional Figures

E.1 Exchange Rate Pass Through

E.1.1 Unequal Exchange Rate Pass Through

Following figures highlights the opposing price pressures emanating from the factors of production for PCP and LCP across the different model scenarios of unequal exchange rate pass through.

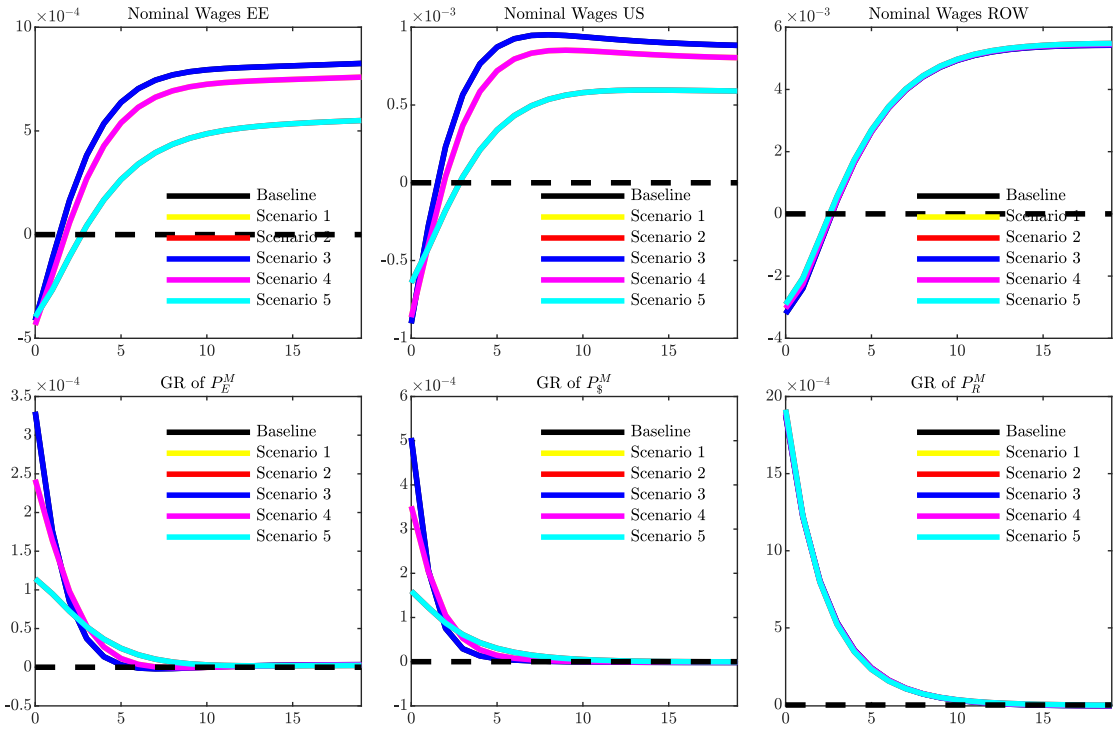


Figure 29: Impulse Responses of Factors of Production under Producer Currency Pricing to 1 s.d. Global Inflation Shock across different model scenarios of unequal exchange rate pass-through. The panels contain the following variables (a) Nominal Wages in EE (b) Nominal Wages in US (c) Nominal Wages in ROW (d) Growth rate of cost of composite intermediate goods in the EE (P_E^M) (e) Growth rate of cost of composite intermediate goods in the US ($P_\$^M$) (f) Growth rate of cost of composite intermediate goods in the ROW (P_R^M).

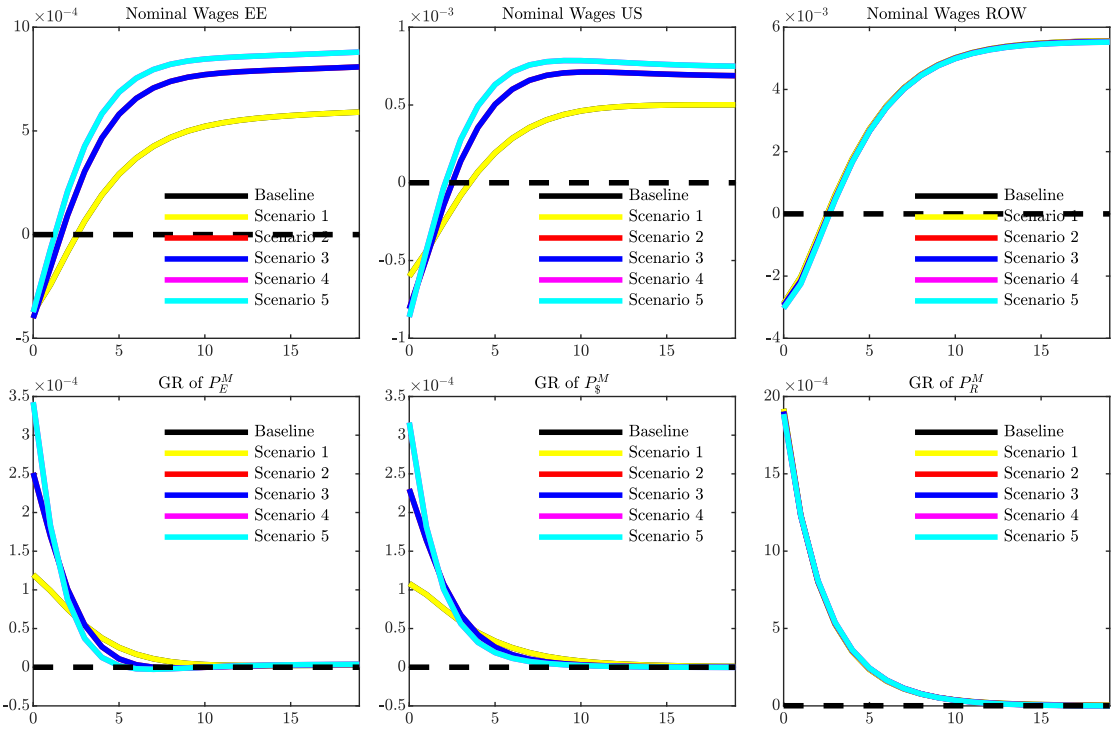


Figure 30: Impulse Responses of Factors of Production under Local Currency Pricing to 1 s.d. Global Inflation Shock across different model scenarios of unequal exchange rate pass-through. The panels contain the following variables (a) Nominal Wages in EE (b) Nominal Wages in US (c) Nominal Wages in ROW (d) Growth rate of cost of composite intermediate goods in the EE (P_E^M) (e) Growth rate of cost of composite intermediate goods in the US ($P_\$^M$) (f) Growth rate of cost of composite intermediate goods in the ROW (P_R^M).

E.1.2 Equal Exchange Rate Pass Through

Following figures highlights the opposing price pressures emanating from the factors of production for PCP and LCP across the different model scenarios of equal exchange rate pass through.

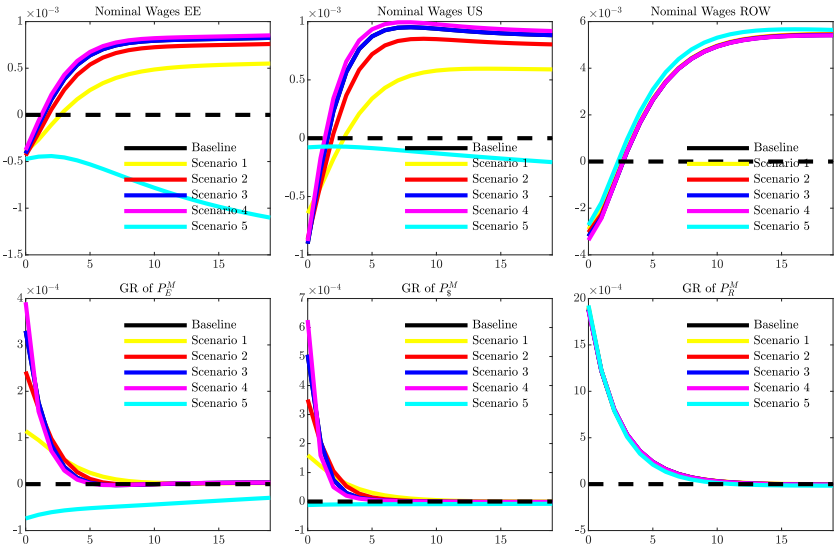


Figure 31: Impulse Responses of Factors of Production under Producer Currency Pricing to 1 s.d. Global Inflation Shock across different model scenarios of equal exchange rate pass-through. The panels contain the following variables (a) Nominal Wages in EE (b) Nominal Wages in US (c) Nominal Wages in ROW (d) Growth rate of cost of composite intermediate goods in the EE (P_E^M) (e) Growth rate of cost of composite intermediate goods in the US ($P_\$^M$) (f) Growth rate of cost of composite intermediate goods in the ROW (P_R^M).

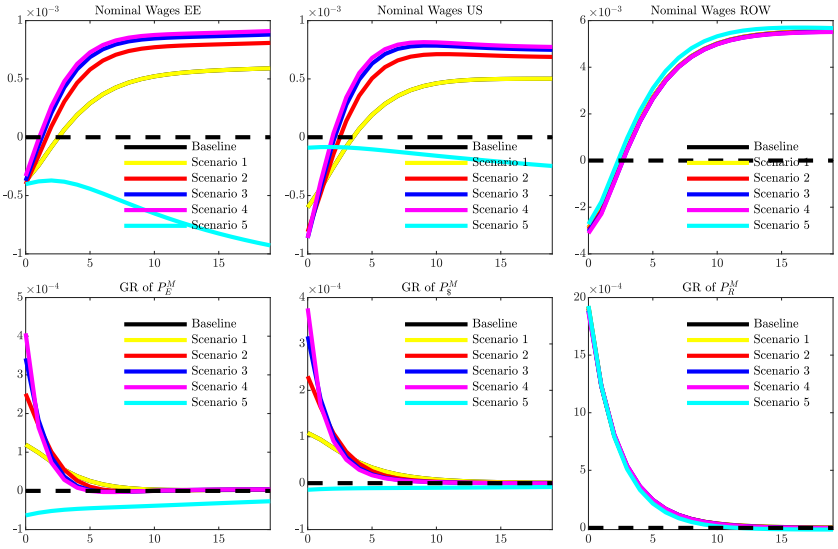


Figure 32: Impulse Responses of Factors of Production under Local Currency Pricing to 1 s.d. Global Inflation Shock across different model scenarios of equal exchange rate pass-through. The panels contain the following variables (a) Nominal Wages in EE (b) Nominal Wages in US (c) Nominal Wages in ROW (d) Growth rate of cost of composite intermediate goods in the EE (P_E^M) (e) Growth rate of cost of composite intermediate goods in the US ($P_\$^M$) (f) Growth rate of cost of composite intermediate goods in the ROW (P_R^M).

E.2 Intermediate and Final Goods Firm

Following figures highlights the opposing price pressures emanating from the factors of production for PCP and LCP across the Intermediate and Final goods.

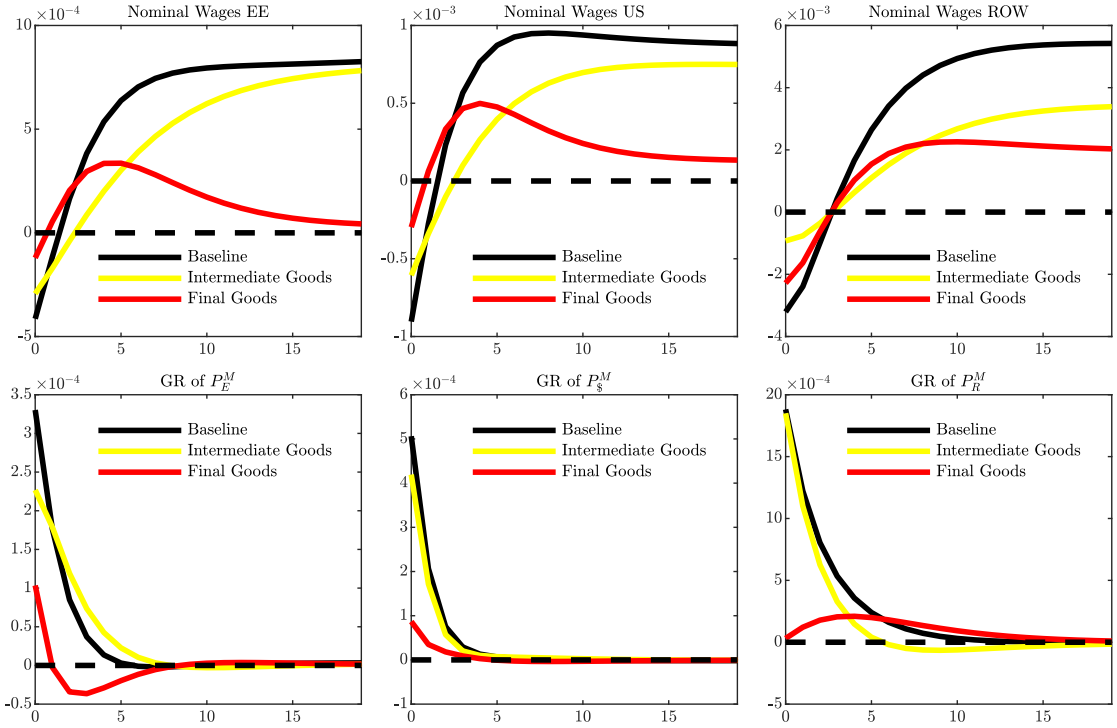


Figure 33: Impulse Responses of Factors of Production under Producer Currency Pricing to 1 s.d. Global Inflation Shock across different types of goods. The panels contain the following variables (a) Nominal Wages in EE (b) Nominal Wages in US (c) Nominal Wages in ROW (d) Growth rate of cost of composite intermediate goods in the EE (P_E^M) (e) Growth rate of cost of composite intermediate goods in the US ($P_\$^M$) (f) Growth rate of cost of composite intermediate goods in the ROW (P_R^M).

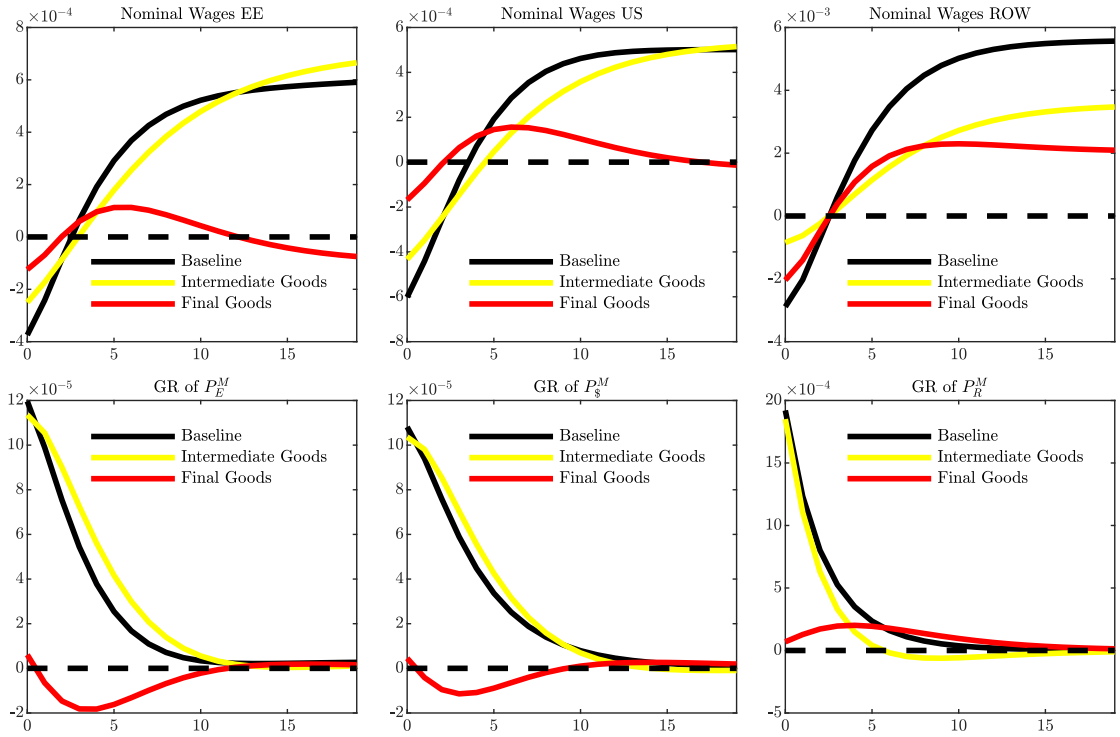


Figure 34: Impulse Responses of Factors of Production under Local Currency Pricing to 1 s.d. Global Inflation Shock across different types of goods. The panels contain the following variables (a) Nominal Wages in EE (b) Nominal Wages in US (c) Nominal Wages in ROW (d) Growth rate of cost of composite intermediate goods in the EE (P_E^M) (e) Growth rate of cost of composite intermediate goods in the US ($P_\$^M$) (f) Growth rate of cost of composite intermediate goods in the ROW (P_R^M).

E.3 Trade Openness

Following figures highlights the opposing price pressures emanating from the factors of production for PCP and LCP across increased imports from the US and the ROW.

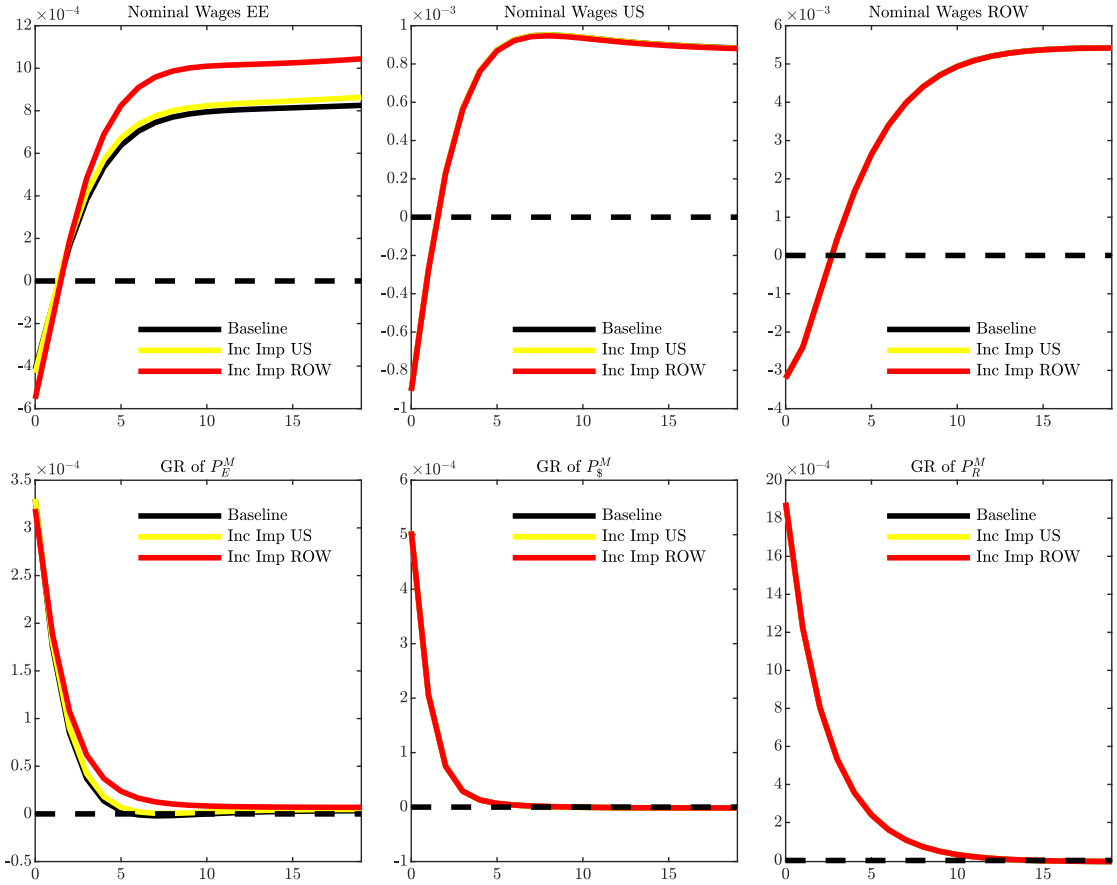


Figure 35: Impulse Responses of Factors of Production under Producer Currency Pricing to 1 s.d. Global Inflation Shock across the baseline scenario, and increased imports from the US and the ROW. The panels contain the following variables (a) Nominal Wages in EE (b) Nominal Wages in US (c) Nominal Wages in ROW (d) Growth rate of cost of composite intermediate goods in the EE (P_E^M) (e) Growth rate of cost of composite intermediate goods in the US ($P_\$^M$) (f) Growth rate of cost of composite intermediate goods in the ROW (P_R^M).

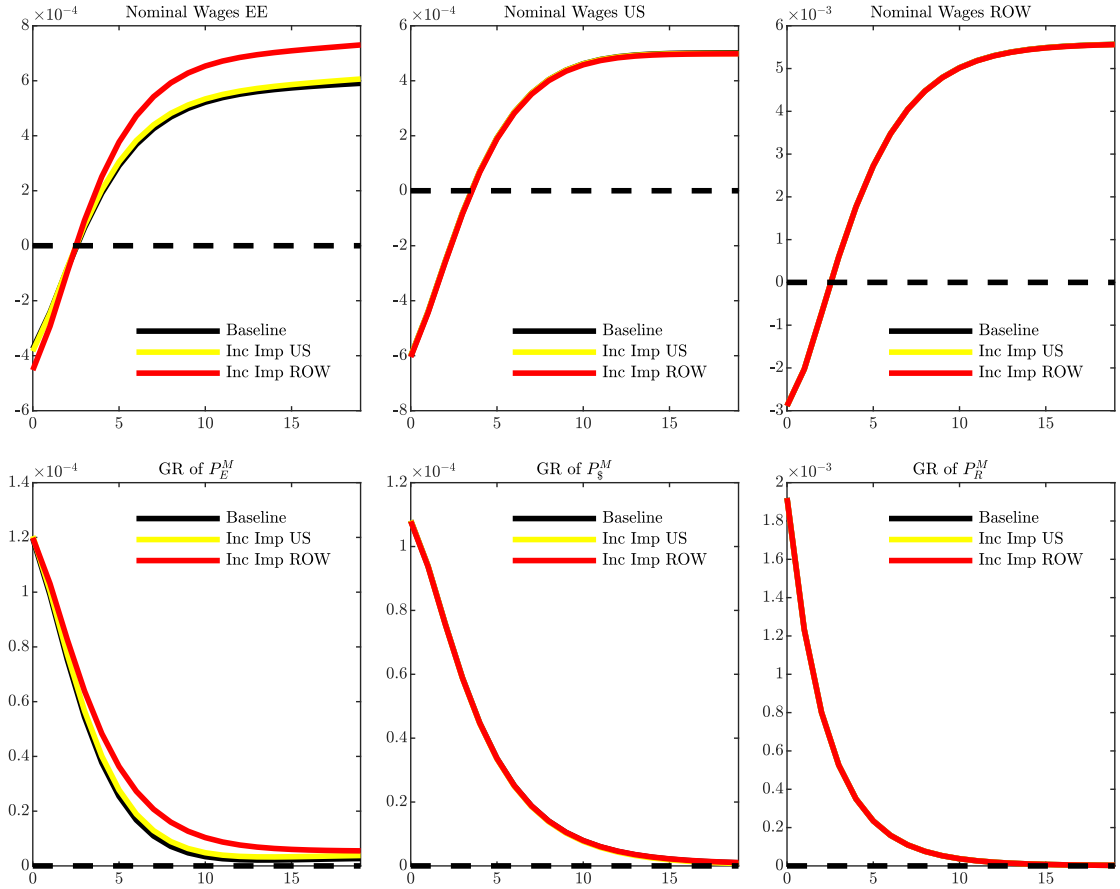


Figure 36: Impulse Responses of Factors of Production under Local Currency Pricing to 1 s.d. Global Inflation Shock across the baseline scenario, and increased imports from the US and the ROW. The panels contain the following variables (a) Nominal Wages in EE (b) Nominal Wages in US (c) Nominal Wages in ROW (d) Growth rate of cost of composite intermediate goods in the EE (P_E^M) (e) Growth rate of cost of composite intermediate goods in the US ($P_\$^M$) (f) Growth rate of cost of composite intermediate goods in the ROW (P_R^M).

F Robustness

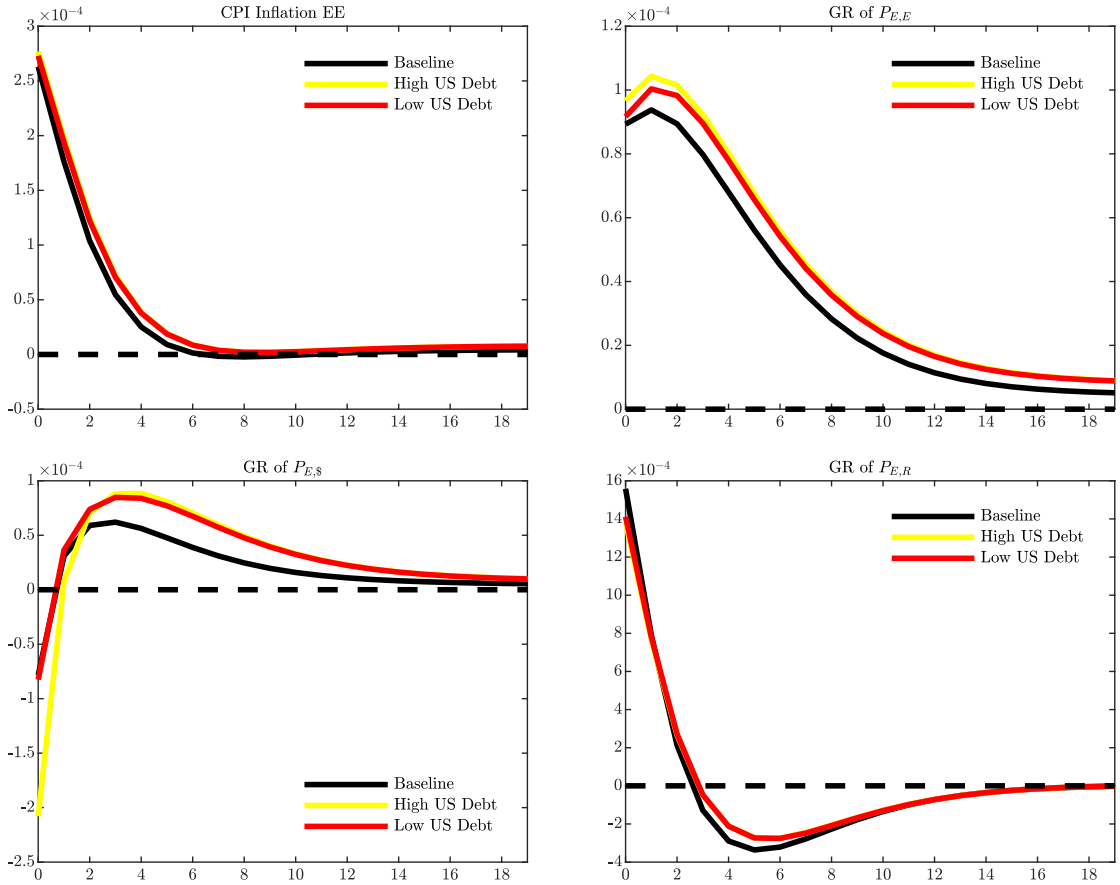


Figure 37: Impulse Responses of CPI Inflation and its sub-components to 1 s.d. Global Inflation Shock under DCP across different scenarios of US Debt. The panels contain the following variables (a) CPI Inflation in EE; (b) Growth Rate of retail price of domestically produced and consumed final goods in EE ($P_{E,E}$); (c) Growth Rate of retail price of final goods produced in the US for the EE ($P_{E,\$}$); (d) Growth Rate of retail price of final goods produced in the ROW for the EE ($P_{E,\$}$)

In order to check the robustness of the results, the calibrated values of the US debt level and portfolio cost associated with holding US bonds are varied under Dominant Currency Pricing. Two additional scenarios are constructed by varying the steady state level of US debt. The US debt parameter is increased to 0.80 in one model scenario, which is referred to as the High US Debt scenario. In the other model scenario, the US debt level is reduced to 0.50. The portfolio cost associated with holding US bonds is assumed to be the lowest in the steady state, in order to highlight the easy availability and marketability of US bonds.

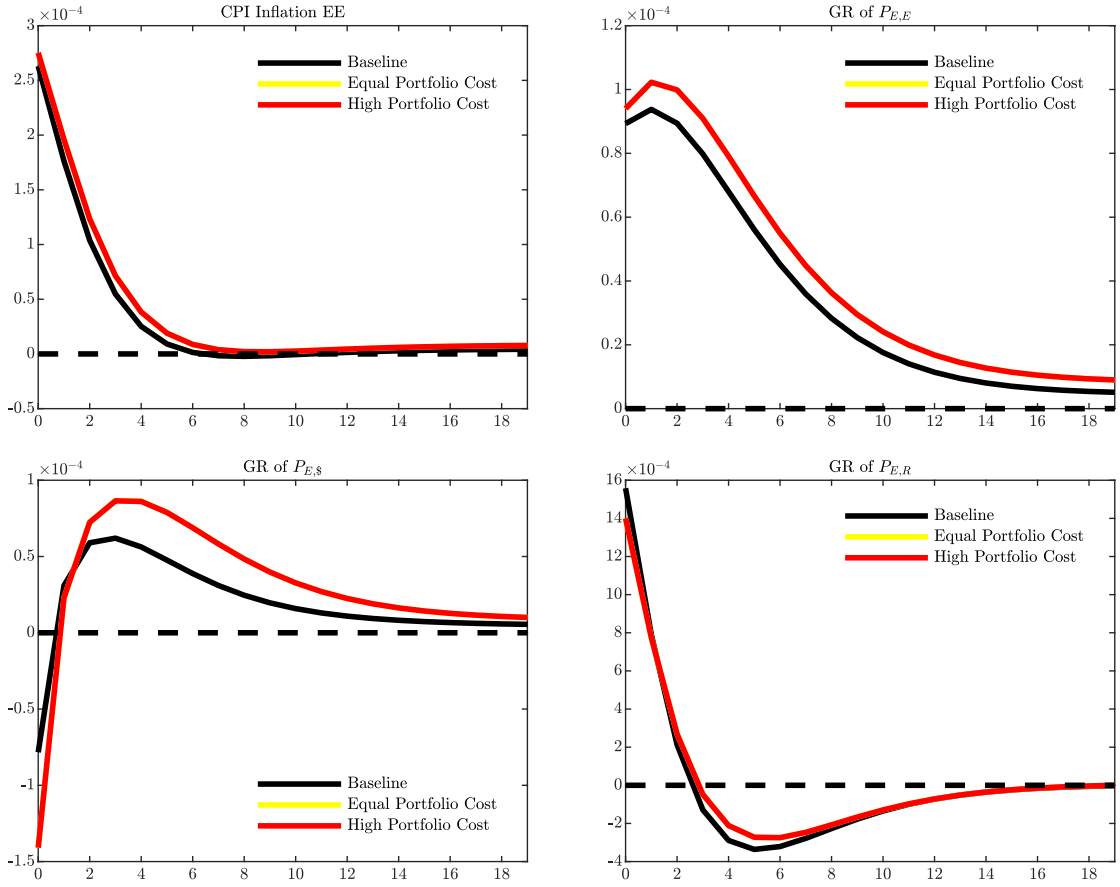


Figure 38: Impulse Responses of CPI Inflation and its sub-components to 1 s.d. Global Inflation Shock under DCP across different scenarios of US bond portfolio cost. The panels contain the following variables (a) CPI Inflation in EE; (b) Growth Rate of retail price of domestically produced and consumed final goods in EE ($P_{E,E}$); (c) Growth Rate of retail price of final goods produced in the US for the EE ($P_{E,\$}$); (d) Growth Rate of retail price of final goods produced in the ROW for the EE ($P_{E,\$}$)

Two additional model scenarios are also constructed for the portfolio cost associated with US bonds. In one of the model scenarios, the portfolio costs are assumed to be equal across all types of bond holdings, and hence this model scenario is referred to as the Equal Portfolio Cost scenario. In the other model scenario, the portfolio costs associated with US bonds are assumed to be higher than all other portfolio costs,and is therefore referred to as the High Portfolio Cost scenario.

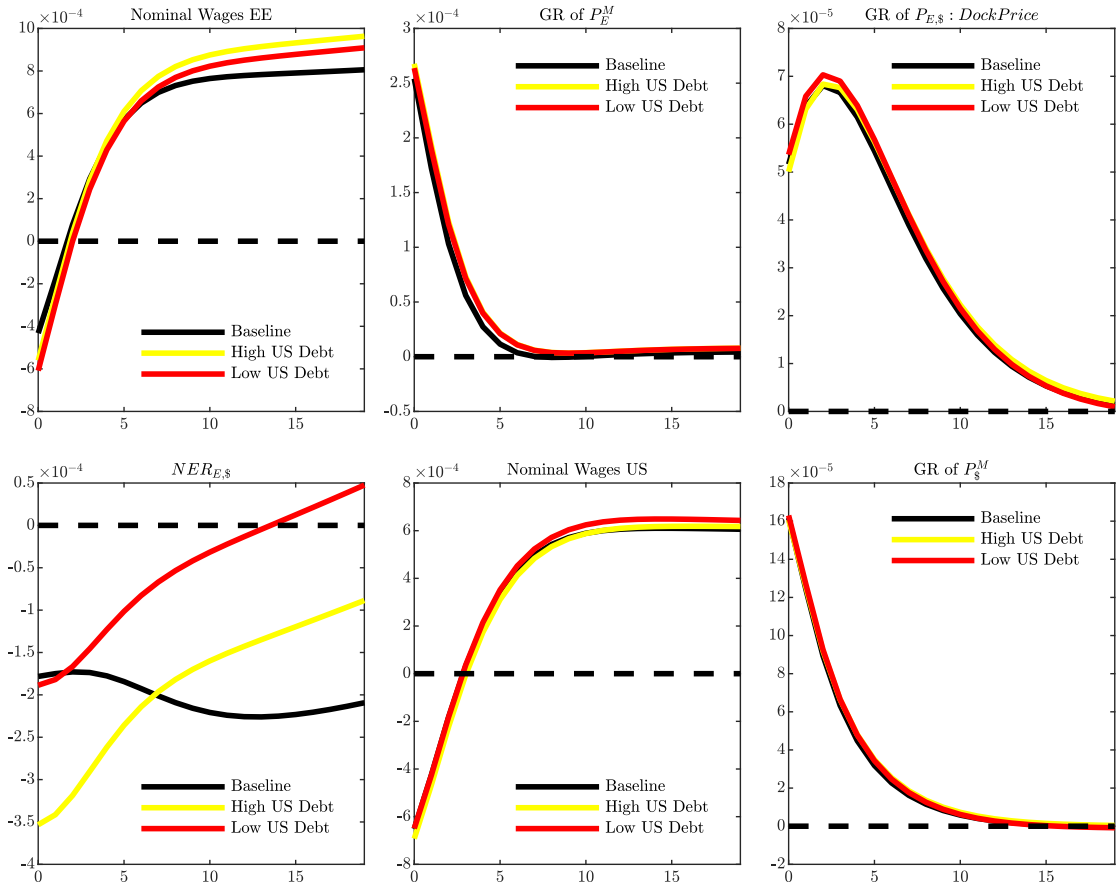


Figure 39: Impulse Responses of CPI Inflation and its sub-components to 1 s.d. Global Inflation Shock under DCP across different scenarios of US Debt. The panels contain the following variables (a) Nominal Wages in EE; (b) Growth Rate of cost of composite intermediate goods in EE (P_E^M); (c) Growth rate of dock price of final goods produced in the US for EE ($P_{E,\$}$: $DockPrice$) (d) Nominal Exchange Rate between the US and the EE ($NER_{E,\$}$); (e) Nominal Wages in US (f) Growth Rate of cost of composite intermediate goods in US ($P_\$^M$)

The direction and trajectory of CPI inflation in EE and its sub-components for the additional model scenarios under US debt and portfolio costs are identical to the baseline scenario. However, the magnitude across the model scenarios differs slightly, thus highlighting the importance of these parameters. The two overarching results identified in the baseline scenario also hold for all the additional model scenarios considered. The factors of production across all three blocs exert opposing price pressures, with the cost of composite intermediate goods exerting upward price pressure, while wages exert downward price pressure.

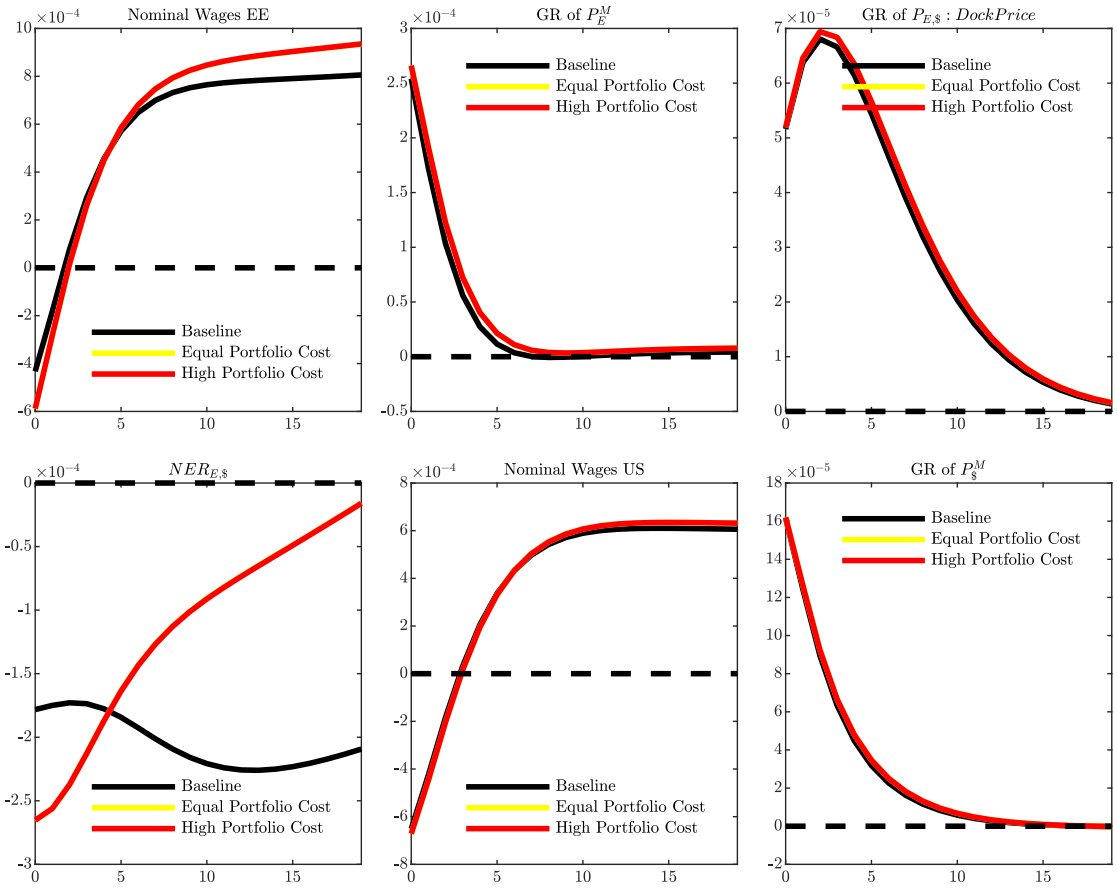


Figure 40: Impulse Responses of CPI Inflation and its sub-components to 1 s.d. Global Inflation Shock under under DCP across different scenarios of US bond portfolio cost. The panels contain the following variables (a) Nominal Wages in EE; (b) Growth Rate of cost of composite intermediate goods in EE (P_E^M); (c) Growth rate of dock price of final goods produced in the US for EE ($P_{E,\$}$: $DockPrice$) (d) Nominal Exchange Rate between the US and the EE ($NER_{E,\$}$); (e) Nominal Wages in US (f) Growth Rate of cost of composite intermediate goods in US ($P_\$^M$)

The relevant nominal exchange rate channel is also functional, and its effect can be seen on the retail price of the final goods produced in the US for EE $P_{E,\$}$. Even when the dock price of the final goods produced in the US for EE $P_{E,\$}$ increases on impact, $P_{E,\$}$ falls due to the appreciation of EE's currency against the US currency. This is due to the downward pressure on the retail price caused by the appreciation of EE's currency, as the retailers in EE are now required to pay less in EE's currency to acquire these goods.

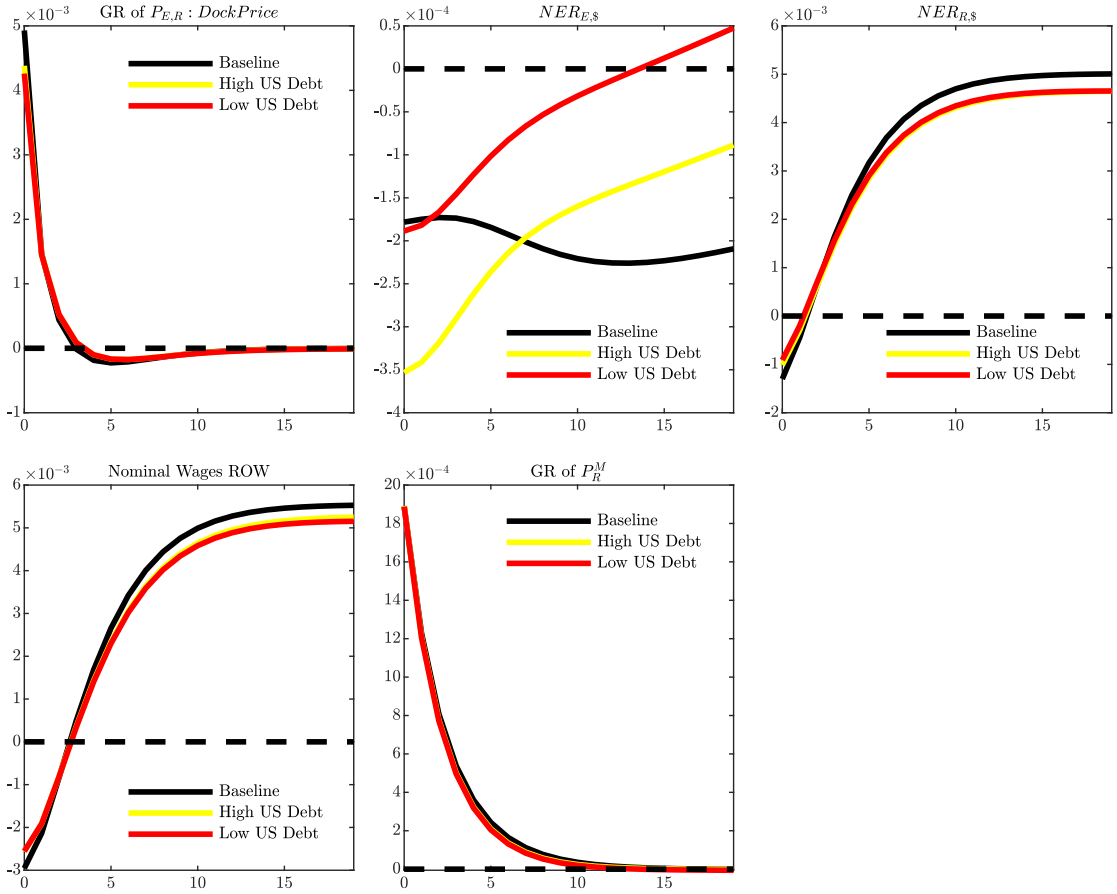


Figure 41: Impulse Responses of CPI Inflation and its sub-components to 1 s.d. Global Inflation Shock under DCP across different scenarios of US Debt. The panels contain the following variables (a)Growth rate of dock price of final goods produced in the ROW for EE ($P_{E,R}$: $DockPrice$); (b) Nominal Exchange Rate between the US and the EE ($NER_{E,\$}$); (c) Nominal Exchange Rate between the ROW and the EE ($NER_{E,R}$) (d) Nominal Exchange Rate between the US and the ROW ($NER_{R,\$}$) ; (e) Nominal Wages in ROW (f) Growth Rate of cost of composite intermediate goods in ROW (P_R^M)

Across all the additional model scenarios, the major difference in the magnitude and trajectory relative to the baseline scenarios occurs in the nominal exchange rate. Under both the US debt and portfolio cost scenarios, EE's currency less in the baseline scenario. For the portfolio cost model scenarios, a higher portfolio cost is associated with greater appreciation of EE's currency as it becomes costly to hold US bonds. Under the US debt model scenarios, when the steady state level of US debt is increased, EE's currency appreciates the most against the US currency. However, when the steady state level of US debt is lower than in the baseline scenario, EE's appreciation is greater than in the baseline scenario, at least on impact. This suggests a nonlinear impact of the US debt

level on the nominal exchange rate between the US and EE.

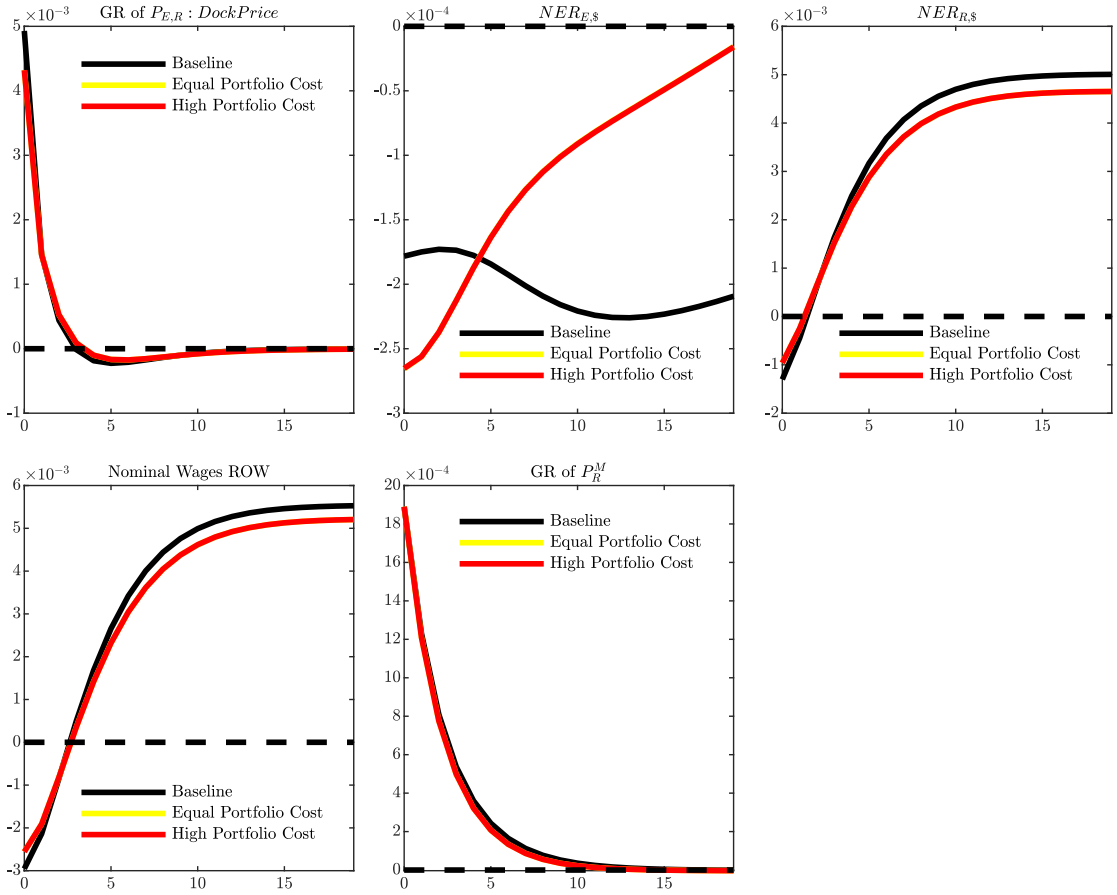


Figure 42: Impulse Responses of CPI Inflation and its sub-components to 1 s.d. Global Inflation Shock under under DCP across different scenarios of US bond portfolio cost. The panels contain the following variables (a)Growth rate of dock price of final goods produced in the ROW for EE ($P_{E,R}$: $DockPrice$); (b) Nominal Exchange Rate between the US and the EE ($NER_{E,\$}$); (c) Nominal Exchange Rate between the ROW and the EE ($NER_{E,R}$) (d) Nominal Exchange Rate between the US and the ROW ($NER_{R,\$}$) ; (e) Nominal Wages in ROW (f) Growth Rate of cost of composite intermediate goods in ROW (P_R^M)